

How Long Do Markets Need to Fully React to Monetary Policy Announcements?

Paul L. Tran*

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Abstract

This paper shows that financial markets need more time than the standard 30 minutes to fully react to monetary policy announcements. I systematically estimate these optimal window lengths using a neural network text analysis method. I find that the required time increases with asset underlying maturity (reaching 50–60 minutes for maturities at least two quarters ahead) and FOMC statement characteristics: complexity, novelty, and dissents. This timing difference has economic consequences: optimal windows correct for the attenuated impact of monetary policy shocks about forward guidance on interest rates, break-even inflation, and equity prices, and yield more precise macroeconomic responses.

Keywords: Event window studies, FOMC statements, monetary policy shocks, neural networks, natural language processing, text analysis

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*Office of Macroeconomic Analysis, US Department of the Treasury (incoming). Department of Economics, The University of Texas at Austin. 2225 Speedway, BRB 2.128, C3100, Austin, TX 78712, USA. Email: pltran@utexas.edu. Website: <https://paulletran.com/>. I am indebted to my committee—Oli Coibion, Chris Boehm, Saroj Bhattarai, and Amy Handlan—for their invaluable guidance and support. For their helpful comments and suggestions, I also wish to thank Andi Mueller, Andres Drenik, Stefano Eusepi, Oliver Pfauti, Andrew Chang, as well as seminar participants from the University of Texas at Austin, the 2025 Texas Macro Job Candidate Conference, and the other institutions where I presented this paper. I am grateful to Chris Boehm and Niklas Kroner for sharing data with me. I also appreciate the Empirical Macroeconomics Policy Center of Texas for their financial support to cover some of the computational costs behind my paper. Finally, I thank Trieu Tran for her research assistance. Views expressed are my own and do not necessarily represent those of the US Treasury or the US government.

1 Introduction

To establish the right price for a stock the market must have adequate information, but it by no means follows that if the market has this information it will thereupon establish the right price. The market’s evaluation of the same data can vary over a wide range, dependent on bullish enthusiasm, concentrated speculative interest and similar influence, or bearish disillusionment. Knowledge is only one ingredient on arriving at a stock’s proper price. The other ingredient, fully as important is sound judgement.

Graham (1974)

Following the August 1999 Federal Open Market Committee (FOMC) announcement, the S&P 500 Index did not instantaneously settle at a new value. Instead, as shown in Figure 1, its price swung in both positive and negative directions for over an hour. This example highlights a key question: how much time do financial markets need to fully react to monetary policy announcements? Applied macroeconomists often measure these price changes within narrow windows around news events such as monetary policy announcements. Small windows avoid contamination from unrelated news, but assume that the market’s first response is its final response. On the other hand, wide windows allow more time for the market to react, but risk mixing the impact of monetary policy announcements with other information. Despite these trade-offs, the monetary policy literature has often defaulted to an “ad-hoc” 30-minute window. This “one-size-fits-all” approach is problematic because it assumes all markets take the same amount of time to react to monetary policy news, even though markets for longer-maturity assets might need more time to understand its more “abstract” information. If the chosen window is wrong, measurements of monetary policy surprises and shocks can become attenuated and less relevant.

I propose a systematic method that pins down these “optimal” event windows. First, I train a neural network for text analysis to approximate the underlying relationship between the text of monetary policy announcements and the market price reaction. This process creates a “text-based signal”, which represents the network’s predictions of the market price response based only on the FOMC statements, word-for-word. Second, I have the network learn this relationship and generate this signal for different event window lengths (e.g., a

10-minute window, 20-minute window, and so on). The “optimal” window is defined as where the network makes the most accurate predictions. This “Goldilocks” window captures the exact time where the market has fully reacted to the announcement, resolving a classic econometric trade-off: it expands the horizon to capture delayed processing, but avoids lengthening it so much that the price becomes overwhelmed by unrelated background noise.¹

My method yields four key findings. First, financial markets require more time to fully react to monetary policy announcements than the 30 minutes typically assumed. I find that regardless of contract maturity, financial markets for futures and equities fully react within a window starting 10 minutes before the announcement and *ending at least 30 minutes* after.

Second, the optimal window length varies systematically with underlying asset maturity. As summarised by Figure 2, the estimated horizon rises from 40 minutes for short-horizon assets to 50–60 minutes for maturities at least two quarters ahead. Importantly, the estimated optimal window allows internal processes behind market delays to conclude, thereby capturing the complete resulting market reaction. Furthermore, because the neural network inputs are restricted solely to the statement text, subsequent press conferences either accelerate market processing via verbal clarification or introduce unmodelled information the network cannot interpret, both of which ultimately penalise its approximation quality at longer horizons. As a result, the estimated optimal windows represent a conservative lower bound of true market processing time. Relatedly, statements with greater complexity, less similarity to previous texts, and the presence of dissents require longer windows on average.

Third, the correlation between monetary policy surprises from optimal versus 30-minute windows decreases as asset maturity increases. While surprises for current and next FOMC meetings are similar, the correlation for long-maturity assets (e.g., fifteen years out) declines by up to 10%. These differences primarily alter the forward guidance component of monetary policy shocks, creating larger peaks and troughs or “shifts in importance” in composition.

Fourth, using the optimal window increases the estimated impact of monetary policy

¹This definition of an optimal horizon under a noise-minimisation criterion is conceptually aligned with the relative exogeneity framework of Casini and McCloskey (2025) regarding dominant announcement shocks.

shocks about forward guidance on interest rates, break-even inflation, and equity prices. Furthermore, the estimated responses of macroeconomic variables also become more precise. This suggests that previously documented effects of monetary policy may be attenuated and imprecise due to the use of suboptimal window lengths.

This paper is not the first to investigate the appropriate event window length. Prior studies offer varying horizons depending on the empirical context, ranging from several days (Chang and Chen, 1989) to a single day (Das and King, 2021) or several hours (Hillmer and Yu, 1979; Boehm and Kroner, 2026), while others find minimal variation across choice lengths (Elenev et al., 2024). Most recently, Casini and McCloskey (2025) argues that the window should expand until the announcement shock no longer dominates background noise. My paper bridges and advances these approaches by introducing an entirely new source of information. While existing studies infer the optimal window using only observed asset price movements, my method combines these price dynamics with a “text-based signal” from a neural network that reads the FOMC statement itself. Centring the estimation around the text allows me to systematically pin down the exact horizon where the text-driven price reaction is fully captured and noise is minimised on average. I further contribute to this literature by detailing the stark empirical effects of window choice on measured monetary policy shocks and surprises, particularly regarding forward guidance.

This paper also joins a growing literature using text analysis to study the effects of central bank communication on expectations and the macroeconomy. While traditional economic methods relied on dictionary-based word counts to classify text (e.g., Gentzkow, Kelly, and Taddy, 2019; Husted, Rogers, and Sun, 2020; Aruoba and Drechsel, 2024; Acosta, 2023; and others), they frequently missed the nuances of context. In contrast, modern neural networks capture the full context and interdependencies of words. Recent studies have deployed these advanced tools to read FOMC statements and speeches, directly constructing new measures of monetary policy shocks and financial market reactions (e.g., Handlan, 2022b; Doh, Song, and SK Yang, 2023; Bianchi, Ludvigson, and Ma, 2024; Piller, Schranz, and Schwaller, 2025;

and others). I contribute to this new literature from a unique angle: instead of using neural networks to directly create policy shocks, I use them to first resolve the event window timing problem. By applying the network to determine the optimal window length where it can best understand the relationship between text and price reactions, my method allows for the construction of more precise and relevant monetary policy surprises and shocks.

The remainder of this paper is structured as follows. Section 2 provides a framework of the dynamics of asset market prices in response to news. Despite simplifying assumptions imposed, the framework motivates the necessity of my text-analysis neural network approach. Section 3 describes the input and output data of the text analysis method as well as the data used to investigate the effects of event window choice on monetary policy shock impacts. Section 4 details the process of systematically estimating the optimal event window lengths and the neural network approach behind it all. Section 5 presents the estimated length of time that financial markets need to fully react to monetary policy announcements. Section 6 details the effects of event window choice on monetary policy surprises and shocks. Section 7 details how the complexity, similarity, and presence of dissents within the FOMC statements can affect the optimal window lengths. Finally, Section 8 concludes.

2 Motivating Framework

This section presents a motivating framework to show why the shortest or longest event windows are not always optimal. The framework illustrates how conflicting noise components can distort asset price dynamics following a news release. For simplicity, this framework imposes strong linearity and orthogonality assumptions on the noise components. These assumptions are purely illustrative, serving only to show how noise affects asset prices and, in turn, the optimal window length. My systematic estimation method does not impose these assumptions because the true composition of market price responses is unknown. Therefore, this framework and its related simulations serve to motivate the natural language processing

approach used to extract a signal from the text of the news releases.²

Consider the price of one asset market at time t , P_t . Assume that the news is released at the beginning of the time period, $t = 0$. As shown in Equation 1, the asset price is the sum of three components:

$$\{P_t = P_t^f + \varepsilon_t^c + \varepsilon_t^n | t \geq 0\}, \quad (1)$$

where P_t^f is the fundamental price component, ε_t^c is cognitive noise, and ε_t^n is unrelated news.

P_t^f represents what asset price should be set by the market due to *only* the news release, meaning it is assumed that $P_t^f = P^f \in \mathbb{R}$.

Cognitive noise represents the collective “market noise” from participants’ limited capacity to process information. Immediately after a news release, factors like algorithmic trading, over-reactions, under-reactions, or liquidity constraints mean market participants may have incomplete information or differing beliefs about the news. As time progresses, processing costs fall and investors learn from each other’s price responses, leading to a convergence towards the fundamental price component. In other words, cognitive noise and its error “decay” over time, eventually converging to zero. The framework reflects this behaviour by representing cognitive noise as the following modified AR(1) process:

$$\varepsilon_t^c = \rho_c \varepsilon_{t-1}^c + e^{-\mathcal{D}t} \nu_t^c,$$

where coefficient $|\rho_c| < 1$, decay term $\mathcal{D} \in \mathbb{R}^+$, and random noise ν_t^c is normally distributed with mean zero and variance σ_c^2 . This setup is consistent with standard rational expectations only if cognitive noise is identically zero, as the persistence parameter ρ_c otherwise introduces systematic, predictable deviations from fundamentals. To model these informational frictions and bounded rationality, I ensure the cognitive noise process decays over time by assuming the first two conditions and an initial variance at $t = 0$ equal to σ_c^2 . This departure from full information is necessary to motivate the study of optimal window lengths, as asset prices

²Details behind the simulation process can be found in Appendix A.

would react instantaneously if market participants possessed unbounded cognitive capacity.

The effect of unrelated news on the asset price is modelled as a random walk:

$$\varepsilon_t^n = \varepsilon_{t-1}^n + \nu_t^n,$$

where random noise ν_t^n is normally distributed with mean zero and variance σ_n^2 . Similar to cognitive noise, the variance of the unrelated news process at time $t = 0$ is equal to zero.

2.1 The Effects of Noise on the Variance of Asset Prices

The expression of P_t as Equation 1 allows for the derivation of the variance of the asset price for all time $t \geq 0$ through iterative substitution.

At time $t = 0$ when news is released, one can express the variance of the asset price as $\text{Var}(P_0) = \text{Var}(\varepsilon_0^c) + \text{Var}(\varepsilon_0^n) = \sigma_c^2$. Similarly, the expression at time $t = 1$ is $\text{Var}(P_1) = \text{Var}(\varepsilon_1^c) + \text{Var}(\varepsilon_1^n) = \sigma_c^2(\rho_c^2 + e^{-2\mathcal{D}}) + \sigma_n^2$. Iterating this process yields the following expression:

$$\begin{aligned} \text{Var}(P_t|t \geq 0) &= \left[\sum_{i=0}^t \rho_c^{2(t-i)} e^{-2\mathcal{D}i} \right] \sigma_c^2 + t\sigma_n^2 \\ &= \left[\frac{\rho_c^{2(t+1)} - e^{-2(t+1)\mathcal{D}}}{\rho_c^2 - e^{-2\mathcal{D}}} \right] \sigma_c^2 + t\sigma_n^2, \end{aligned} \quad (2)$$

where $\lim_{t \rightarrow \infty} \left[\frac{\rho_c^{2(t+1)} - e^{-2(t+1)\mathcal{D}}}{\rho_c^2 - e^{-2\mathcal{D}}} \right] = 0$ by the assumptions that $|\rho_c| < 1$ and $\mathcal{D} > 0$.

I define t^{one} as the time where the variance of P_t is minimised. Solving for this time horizon yields the following indirect expression that provides important insight into the factors influencing the appropriate event window length:

$$t^{one} : \mathcal{D} [e^{-2(t+1)\mathcal{D}}] + \ln(\rho_c) \rho_c^{2(t+1)} = \left[\frac{(e^{-2\mathcal{D}} - \rho_c^2)}{2} \right] \frac{\sigma_n^2}{\sigma_c^2}. \quad (3)$$

An interior solution (i.e., $t^{one} > 0$) is mathematically guaranteed if the derivative of Equation 2 is strictly negative at the time of release ($t = 0$), which requires the relative

variance of background noise to satisfy the ceiling $\frac{\sigma_n^2}{\sigma_c^2} < \frac{2[\ln(\rho_c)\rho_c^2 + \mathcal{D}e^{-2\mathcal{D}}]}{e^{-2\mathcal{D}} - \rho_c^2}$. Intuitively, if background noise accumulates too quickly relative to initial cognitive confusion, this condition is violated, and the optimal horizon collapses to the boundary solution $t = 0$.

Two important findings from Equation 3 are that $\frac{\partial t^{one}}{\partial \sigma_n^2} < 0$ and $\frac{\partial t^{one}}{\partial \sigma_c^2} > 0$. In other words, an increased presence of unrelated news (cognitive noise) results in the time where the variance of the asset price is minimised to decrease (increase).³ The dynamics caused by these two components are why strictly using narrow event windows isn't always appropriate.

2.2 Estimator Form

The dynamics described by Equation 3 provide insight into the trade-offs for a *single* news announcement. To apply this framework empirically such that the methodology of this paper is motivated, this concept must be extended to the *average* price reaction across many news releases. Therefore, the goal of this paper is not to find t^{one} for any single event, but to systematically estimate t^* , the time at which the market fully reacts to news announcements *on average*. Formally, consider N news announcements and the price of one asset market. For each news i that is released at time t , the asset price $P_{i,t}$ and its components will respond to the announcement. Therefore, I define t^* as the optimal time horizon that minimises the mean squared error (MSE) between $P_{i,t}$ and $P_{i,t}^f$ for all N news:

$$t^* : \min_t \frac{1}{N} \sum_{i=1}^N \left(P_{i,t} - P_i^f \right)^2 = \min_t \frac{1}{N} \sum_{i=1}^N \left(\varepsilon_{i,t}^c + \varepsilon_{i,t}^n \right)^2 \quad (4)$$

However, assume that the fundamental component is unobservable to the econometrician. Instead, suppose that the econometrician observes a noisy signal of the fundamental component, $s_i = P_i^f + \xi_i$, where $\xi_i \sim \mathcal{N}(0, \sigma_s^2)$. By making the core assumption that the sum of the asset price noise components, $(\varepsilon_t^c + \varepsilon_t^n)$, and the signal noise are *independent*, I

³Because Equation 3 is unable to have t isolated on one side, I numerically verify the dynamics of the t^{one} for various values of σ_c^2 and σ_n^2 in the indirect expression while holding the other parameters constant.

am able to derive the following expression for the minimisation problem of the MSE:

$$\begin{aligned}
t^* : \min_t \frac{1}{N} \sum_{i=1}^N (P_{i,t} - s_i)^2 &= \min_t \frac{1}{N} \sum_{i=1}^N \left(P_i^f + \varepsilon_{i,t}^c + \varepsilon_{i,t}^n - P_i^f - \xi_i \right)^2 \\
\Rightarrow t^* : \min_t \frac{1}{N} \sum_{i=1}^N (P_{i,t} - s_i)^2 &= \min_t \left[\frac{1}{N} \sum_{i=1}^N (\varepsilon_{i,t}^c + \varepsilon_{i,t}^n)^2 + \sigma_s^2 \right]
\end{aligned} \tag{5}$$

This derivation implies that the econometrician can solve for the optimal time horizon, t^* , using only the noisy signal s_i . Asymptotically, the signal’s precision has no effect on finding t^* , but in finite samples common to the monetary policy literature, precision determines estimation feasibility. Equivalently, t^* represents the horizon where the average impact of noise is minimised, meaning the fundamental component P_i^f , and its text-based signal s_i , has the largest share of the observed asset price variance.

2.3 Framework Takeaways

I simulate the asset price process over time for multiple news announcements to illustrate the effects of cognitive and unrelated noise on the optimal time horizon, and to demonstrate why a good signal is needed for this MSE minimisation. The main implications are discussed in the main text, while Appendix A provides full details on the process.

I consider three scenarios for the asset price response: one with cognitive noise, one with unrelated news, and one with both. The simulations show that in all scenarios, the optimal time horizon (t^*) estimated from the noisy signal s_i is essentially equal to the true horizon estimated from the fundamental price P_i^f . A “good” signal, therefore, makes it possible to estimate the time horizon that reflects a full market reaction. However, recall that this framework and its simulations impose strong linearity and orthogonality assumptions purely to demonstrate the effects of noise. In reality, the true compositions and processes of the fundamental and noise components are unknown, yet still influence the overall asset price.

Abstracting from this simple framework, I argue that the text-analysis neural network

approach used in this paper does not need to know the true noise processes to approximate the underlying mapping between monetary policy communications and asset price changes. Crucially, lengthening the event window to extract a “good” signal introduces a classic econometric trade-off: a wider horizon captures the delayed market reaction but hazards introducing endogeneity bias via unrelated background noise. Optimising the window length via a noise-minimisation criterion remains valid, however, if the announcement represents an overwhelming variance shock that dominates baseline market noise across all considered horizons.⁴ In the context of scheduled monetary policy announcements, where the direct source of market movement is explicitly the institutional text, this type of exogeneity is naturally embedded in the data. Because the text disciplines the informational signal, identification remains intact even as the window expands to maximise precision. As discussed in detail in Subsection 4.2, prior studies in the monetary policy literature have shown that this condition is highly credible throughout extended horizons.

3 Data

The sample period for my analysis runs from May 1999 to October 2019. The start date coincides with the FOMC’s adoption of the practice of issuing a statement after every scheduled meeting, regardless of whether policy rates were changed. The end date provides a pre-COVID-19 benchmark for the analysis, thereby avoiding the structural breaks and unprecedented market dynamics associated with the subsequent global pandemic.⁵ As the principal entity determining U.S. monetary policy, the Federal Open Market Committee (FOMC) is closely watched by markets. Of particular importance are the released FOMC statements,

⁴Casini and McCloskey (2025) call this condition relative exogeneity. This is a technical condition which means the policy news is so large that it effectively drowns out all other market noise within that window.

⁵Future versions of this paper could extend the sample period by adapting the systematic estimation to incorporate FOMC statements released after October 2019.

which are the initial and primary source of monetary policy news for market participants.⁶ Because markets are known to dissect and react to these statements word-by-word,⁷ understanding the time it takes for them to fully react is critical for monetary policy event studies. However, the mapping from FOMC statements to asset price reactions is complex, nonparametric, and difficult to capture with traditional text analysis methods. Approximating this underlying relationship motivates the text-analysis neural network approach in this paper. Before detailing the systematic estimation, I provide descriptive information on the FOMC statements (i.e., the inputs) and the financial market asset prices (i.e., the outputs) used in this paper, as summarised in Appendix Table E2.

3.1 Inputs: FOMC Statements

FOMC decisions are announced via press releases following its eight scheduled annual meetings. I exclude statements from unscheduled meetings to ensure that asset price changes are driven strictly by the text content, rather than the meeting itself. Sourcing these announcements from the Federal Reserve Board website yields a final sample of 165 statements.⁸

Text Pre-processing To prepare the FOMC statements for the neural network, I preprocess all 165 statements by standardising them to plain UTF-8 text with uniform spacing, removing URLs, release timestamps, and the list of regional bank request approvals. I also remove the FOMC member voting record from the end of each statement for two main reasons. First, it ensures the neural network, which has a tight constraint on input length, focuses entirely on the core economic narrative (see Subsection 4.3). Second, while voting records can affect stock markets (C Madeira and J Madeira, 2019), they have less impact on interest-rate assets. Given my focus on interest-rate futures, this exclusion should not

⁶Appendix Figure E1 shows U.S. interest spikes on Google Trends for phrases such as “FOMC meeting” and “FOMC statement” during scheduled meeting dates. Further support for financial market reactions coming primarily from the FOMC statements is the fact that the 1st–3rd query results on Google Search direct to the Board of Governors of the Federal Reserve System website.

⁷E.g., [CNBC coverage of the January 2025 FOMC statement text](#).

⁸<https://www.federalreserve.gov/monetarypolicy/fomc.htm>

significantly alter the estimated optimal event windows.⁹

Content and Evolution of Statements A typical FOMC statement discusses macroeconomic conditions, communicates economic expectations, and concludes with policy rate decisions. Following the 2008–2009 Financial Crisis, these statements began incorporating unconventional policies like quantitative easing. This shift increased statement complexity and length, as the Committee relied heavily on forward guidance to manage expectations when constrained by the zero lower bound. As shown in Figure 3, statement length grew rapidly until 2014, and while this trend partially reversed thereafter, statements remain significantly longer than their pre-Crisis counterparts.

The following example is an excerpt of the FOMC statement released on 01 August, 2018:

Information received since the Federal Open Market Committee met in June indicates that the labor market has continued to strengthen and that economic activity has been rising at a strong rate. [...]

The Committee expects that further gradual increases in the target range for the federal funds rate will be consistent with sustained expansion of economic activity, strong labor market conditions, and inflation near the Committee’s symmetric 2 percent objective over the medium term. Risks to the economic outlook appear roughly balanced.

In view of realized and expected labor market conditions and inflation, the Committee decided to maintain the target range for the federal funds rate at 1-3/4 to 2 percent. The stance of monetary policy remains accommodative, thereby supporting strong labor market conditions and a sustained return to 2 percent inflation.

3.2 Outputs: Interest-rate and Equity Futures

The neural network outputs are interest-rate and equity futures prices. Sourced from the Thomson Reuters Tick History database (LSEG), these intraday tick data are necessary to analyse the short event windows common in the high-frequency identification literature. Within these windows, variations in interest-rate futures prices capture changes in expected policy paths in response to FOMC announcements.

I select a range of futures contracts commonly used to construct high-frequency monetary policy surprises (e.g., Kuttner, 2001; Gürkaynak, Sack, and Swanson, 2005; Nakamura and

⁹Nonetheless, future versions of this paper will explore including the voting records as inputs.

Steinsson, 2018; and others). For short-term expectations, I use Federal Funds futures liquid up to three months out ($FF1-FF4$). To capture expectations from four months to one year ahead, I include Eurodollar futures ($EDcm2-EDcm4$) following Acosta, Brennan, and Jacobson (2024). For longer horizons, I use Treasury futures with underlying maturities of approximately two ($TUC1, TUC2$), four ($FVC1, FVC2$), seven ($TYC1, TYC2$), and fifteen years ($USC1, USC2$) (Gürkaynak, Kisacikoğlu, and Wright, 2020). Finally, I include the S&P 500 Index and its E-mini futures ($SPX, ESc1, ESc2$) because equities are central to the monetary policy literature. Because all considered futures have traded outside regular hours since July 2003, they provide the continuous data necessary to test extended window lengths.

Dependent Variable Construction I collect futures contract price levels at 10-minute intervals, from 10 minutes before to 18 hours after an FOMC statement release. The output of interest for the neural network is the log-price difference for each contract, constructed for event windows starting 10 minutes before and ending n minutes after the announcement:

$$DP_{t+n} = \ln \left(\frac{P_{t+n}}{P_{t-10}} \right), \quad (6)$$

The neural network considers price log-differences for event windows up to 70 minutes.¹⁰

3.3 Data to Investigate Impacts of Monetary Policy Shocks

To investigate the effects of event window choice on monetary policy shocks, I use several additional datasets. Daily data for nominal Treasury yields are from Gürkaynak, Sack, and Swanson (2005).¹¹ Daily data for Treasury-inflation-protected-security (TIPS) yields and break-even inflation come from Gürkaynak, Sack, and Wright (2010).¹² Monthly data for industrial production (IP) and the consumer price index (CPI) are from Federal Reserve Economic Data. The monthly excess bond premium (EBP) measure is from Gilchrist and

¹⁰Longer windows are left for future research due to computational and financial constraints.

¹¹Treasuries: <https://www.federalreserve.gov/data/nominal-yield-curve.htm>

¹²TIPS and break-even inflation: <https://www.federalreserve.gov/data/tips-yield-curve-and-inflation-compensation.htm>

Zakrajšek (2012) and sourced from the Federal Reserve Board.¹³

4 Systematically Estimating Optimal Event Windows

Systematically estimating the optimal event window length hinges on approximating the underlying, nonparametric relationship between monetary policy communications and financial market price reactions. Obtaining the text-based signal needed for this estimation requires a precise approximation of this mapping. This precision, in turn, demands a method that can quantify the *full* information content of the FOMC statement, capturing all dimensions of communication that drive market reactions.

Traditional text analysis, such as “fitting predictive models on simple counts of text features” (Gentzkow, B Kelly, and Taddy, 2019), typically fails to capture the full information content of text by ignoring context and word interdependencies. For example, a “bag-of-words” approach equates the phrases “*employment* went up, but *inflation* did not” and “*inflation* went up, but *employment* did not”. Similarly, “n-gram” methods miss information spread across a sentence. A trigram captures “economic growth slowed” but misses the reversal in “economic growth slowed, but should pick up later this year”. Consequently, these popular methods cannot realistically capture the full nuance of FOMC statements.¹⁴

In contrast, text-analysis neural networks *can* capture complex relationships like context and interdependencies. For example, the words “slowed” and “pick up pace” could both be associated with “economic growth” for prediction, even if they are not adjacent. The power of these networks is mathematically justified by the Universal Approximation Theorem (Hornik, Stinchcombe, and White, 1989; and others), which states that any continuous function (such as the underlying mapping from FOMC statements to price reactions) can be approximated by a network with at least one hidden layer to any desired accuracy.¹⁵

¹³https://www.federalreserve.gov/econresdata/notes/feds-notes/2016/files/ebp_csv.csv

¹⁴Babur and Cleophas (2017) find that performance does not increase monotonically with n-gram size.

¹⁵The theorem guarantees existence but does not specify network structure. In practice, deep architectures reduce parameter and computational costs for complex approximations.

Given their success in other social sciences, these methods offer similar benefits for studying monetary policy communication in economics (Gentzkow, B Kelly, and Taddy, 2019).

4.1 Approach: XLNet-Base, a Neural Network for Text Analysis

I employ the XLNet-Base language model from Z Yang et al. (2019), which builds upon the transformer architecture of Vaswani et al. (2017).^{16,17} This model abandons the common “masking” technique, which assumes word independence and can create training discrepancies.¹⁸ Instead, XLNet-Base uses a permutation-based pre-training phase, allowing it to learn context bi-directionally without the weaknesses of masking. This makes XLNet-Base better suited for semantic tasks than unidirectional generative models (e.g., ChatGPT).¹⁹ The model’s versatility makes it ideal for fine-tuning on new regression tasks, such as approximating the underlying relationship between FOMC statements and price reactions. I start with the pre-trained XLNet-Base model and fine-tune it to “transfer” its general language knowledge to this specific task, allowing me to obtain the text-based signal.²⁰

4.2 The Inputs, Outputs, Approach, and Optimal Window Length

Recall from the motivating framework that the optimal event window is where noise is minimised on average, allowing the fundamental price component to have the largest “share” of the price change. The text-based signal is the neural network’s *predicted* price change for a given window, using only the FOMC statement text. It follows that the optimal window length is where the network’s approximation simultaneously has the highest predictive per-

¹⁶While the model architecture could quantitatively influence the results, my core methodology—identifying the optimal window via maximum predictive accuracy—is flexible and easily implemented using alternative approaches capable of mapping text to price changes. Future research could explore the sensitivity of these findings to different architectures or approximation techniques.

¹⁷The open-source, pre-trained neural network has 12 layers, 12 attention heads, 117 million parameters, and a 32,000 token vocabulary. See [Hugging Face](#) for details.

¹⁸E.g., if “target” and “rate” are masked in “...increase its target rate”, their implicit relation is ignored.

¹⁹Generative models are “left-to-right” and predict a word using only preceding words.

²⁰Transfer learning, as shown by Z Yang et al. (2019) and others, reduces data requirements for new tasks while maintaining high accuracy.

formance, because only within this window is noise minimised on average, and produces the most precise signal. This simultaneity drives my systematic estimation procedure:

1. For a given event window length, XLNet-Base regresses futures price changes on the FOMC statement text to assess its predictive performance and signal precision.
2. I repeat this regression for all event windows up to 70 minutes in length.
3. The optimal event window for that asset is the length that yields the network’s best predictive performance.

A valid econometric consideration when expanding the estimation horizon up to 70 minutes is its impact on identification credibility. While traditional high-frequency identification relies on tight windows to exclude potential endogeneity bias, a precision-maximising criterion remains valid if the announcement constitutes an overwhelming variance shock. In the context of scheduled monetary policy announcements, where the institutional text explicitly captures the direct source of market movement, this relative exogeneity is a robust feature of the data, thereby allowing the neural network to safely optimise based on out-of-sample performance. Furthermore, prior studies leveraging massive asset price variance spikes (Boehm and Kroner, 2026) and microstructure frameworks mapping elevated trading volume durations (Jensen, 2025) have shown that the policy statement remains the primary driver of market variation over expanded horizons. Because this institutional signal heavily swamps residual background noise, expanding the horizon does not introduce significant endogeneity bias. Instead, centring the event window selection around the text yields substantial noise reduction to achieve greater precision.

While relative exogeneity justifies searching across wider horizons, the neural network’s restricted input layer simultaneously guards against over-expanding into truly confounded windows. This structural boundary explains why post-statement institutional events, such as the subsequent FOMC press conference, reinforce rather than disrupt my estimation strategy. Because inputs are restricted solely to the statement text, XLNet-Base is blind

to unmodelled information introduced during the press conference. If the event window expands into horizons where unscripted Q&A dynamics or fresh context drive asset prices, the network cannot interpret these movements. This omission acts as a severe out-of-sample predictive penalty, mechanically degrading approximation quality beyond the true processing horizon. Conversely, if the press conference acts as an informational clarification channel that accelerates market reactions to the monetary policy announcement, then this acceleration encourages the network to best approximate at shorter window lengths. Because the network is explicitly aided by text clarification but penalised by unmodelled information, the estimated optimum represents a conservative lower bound of true market processing time.

4.3 Stratified-sampling Cross Validation

To prepare the neural network for systematic estimation, I split the sample five times, with 20 per cent of FOMC statements in each testing subsample. This splitting is stratified by the FOMC’s rate decision, the FOMC Chair, pre-/post-2007, and statement word count to ensure equal distributions. This process yields five distinct training-testing folds such that no statement is shared across testing subsamples.²¹

These splits consist of input-output pairs: the FOMC statements (inputs) and the futures price log-differences (outputs). For each event window length and futures contract, I train XLNet-Base on a training subsample to approximate the underlying relationship. I then test its performance on the held-out subsample. This process is repeated five times, once for each split. The final accuracy for each event window length is the average performance across all five test splits. This method ensures a robust measure of out-of-sample performance and accounts for prediction variance across the splits.²²

I restrict the context window of XLNet-Base (i.e., the number of tokens considered per input) to 512 tokens for each FOMC statement.²³ Given an average statement length of 327

²¹This process is known as stratified 5-fold cross-validation in the machine learning literature.

²²Stratified 5-fold cross-validation minimises differences between the full-sample and subsample distributions, which is crucial for transfer learning on finite samples.

²³For example, “decreasing” and “increasing” are tokenised into sub-word units: {“de”, “in”, “creas”, “ing”}.

words (peaking at 800), this 512-token window provides adequate headroom to capture most statements. The main hyperparameters for fine-tuning are listed in Appendix Table E3.

4.4 Accuracy Metrics

The primary accuracy metric for neural network fine-tuning is the *generalised* R^2 statistic from Hawinkel, Waegeman, and Maere (2024), denoted R_{OOS}^2 . This out-of-sample performance metric is chosen because the conventional R^2 breaks down for non-linear methods. For neural networks, total variance does not neatly decompose into model and residual components, meaning the squared Pearson correlation coefficient fails to equal R^2 .²⁴ Mathematical details are provided in Appendix B.

As is standard in machine learning, model performance is compared to a baseline. This baseline assumes *no* relationship between the FOMC statement text and futures price changes, naively predicting with the in-sample average. Therefore, the R_{OOS}^2 measures the reduction in predictive error achieved by the neural network compared to this naive baseline.

During fine-tuning, XLNet-Base also tracks other metrics: the out-of-sample Pearson correlation between predicted and actual values, the out-of-sample mean absolute error, and the in-sample mean squared error. This last metric is important for verifying that the network is genuinely learning the underlying relationship.

The decisive criterion for systematic estimation is the out-of-sample R^2 *averaged across sample splits* for each event window length and futures contract:

$$\overline{R_{OOS}^2} = \frac{\sum_{i=1}^K R_{OOS,i}^2}{K}, \quad (7)$$

where $K = 5$ is the number of sample splits. This $\overline{R_{OOS}^2}$ statistic measures the neural network’s average performance improvement within a given window, relative to the naive baseline. The optimal event window is therefore the length that yields the largest $\overline{R_{OOS}^2}$.

²⁴Results are robust to using conventional R^2 , though R_{OOS}^2 is the methodologically preferred metric.

This is the point where the network’s predictive power and generalisability is highest, which only occurs when noise impact is minimised and the fundamental reaction is fully captured.

4.5 Fine-tuning the Neural Network for Approximation

Individual neural network parameters are not interpretable like standard parametric coefficients (Athey and Imbens, 2019). As a result, rather than establishing causal effects for specific words, the training process approximates the complex, underlying relationship between FOMC communications and asset prices. These predictions are used to pin down the window size that best reflects the market’s full reaction to monetary policy communication.

Fine-tuning a 110-million-parameter model like XLNet-Base makes overfitting an inevitable issue, which appears as a gap between R_{OOS}^2 and in-sample R^2 . I mitigate this difference by tuning two key hyperparameters: the number of training iterations and the learning rate. Too many iterations cause the network to over-fit the training data, deteriorating out-of-sample accuracy. Too few iterations will cause it to under-fit, failing to learn the underlying relationship. The learning rate governs how quickly the network updates its parameters. A rate that is too high causes the neural network weights to update too dramatically, forgetting prior information and degrading out-of-sample predictions. Conversely, a rate that is too low will also result in underfitting, as the network never fully learns the mapping between FOMC communications and asset price changes. Because I use transfer learning, the network’s initial parameters are already optimised for general English. My goal is therefore to limit parameter updates within and across training iterations in order to preserve this generalisability while adapting the network to the specific task.

Ultimately, training the neural network is a balancing act: XLNet-Base has to meaningfully approximate the underlying relationship without overfitting, which would make the predictions non-generalisable and create a bad signal. I perform this balancing act using a two-step process. First, I set the maximum training iterations to 2040 steps (120 epochs). Second, I perform a “hyperparameter sweep” on the learning rate using Bayesian optimi-

sation.²⁵ During this sweep, accuracy metrics from Subsection 4.4 are tracked. Training stops via early stopping: When out-of-sample accuracy begins to degrade. After a 24-hour sweep for each split, I select the learning rate and training iteration that yield the highest R_{OOS}^2 . Given the potential for volatile out-of-sample predictions on finite samples, I select the model from this iteration only if subsequent iterations show a permanent decline in out-of-sample accuracy, mitigating the inherent risk of overfitting by ensuring the network does not memorise sample noise. I use the Simple Transformers library from Rajapakse, Yates, and Rijke (2024), which simplifies fine-tuning Transformer models like XLNet-Base.

4.6 Addressing Look-ahead Bias

A valid concern for text-analysis neural networks is look-ahead bias. Per Sarkar and Vafa (2024), this bias occurs if a network is pre-trained on future data to predict past values. The risk stems from large, unknown pre-training corpora, which may include future data.²⁶ I argue that this bias is mitigated in my method for two reasons: the known pre-training corpora of XLNet-Base and the specific language composition of FOMC statements.

First, the authors of XLNet-Base mitigate this concern by restricting the pre-training data to *only* BookCorpus and the entire English Wikipedia. This restriction makes it highly unlikely that the network learned any specific information about the intersection of FOMC communications and futures market price reactions during its pre-training. Importantly, Z Yang et al. (2019) showed that this limited pre-training data does not negatively impact the model’s generalisability compared to other models.

Second, look-ahead bias is mitigated because the information content in FOMC statements are temporally anonymised. Similar to Glasserman and Lin (2023), the pre-processed text contains no references to relative times (e.g., t and $t + 1$). Therefore, even if XLNet-Base’s pre-training data included future FOMC statements, the network has no chronological

²⁵The set of considered learning rates is $[1e - 5, 9e - 5]$, assuming a uniform distribution.

²⁶Look-ahead bias is especially prevalent for closed-source large language models (e.g., ChatGPT) whose pre-training data are not publicly accessible.

information to exploit. It cannot “figure out” the temporal order of the statements from the text alone, preventing it from learning from future information.

5 How Long Until Markets Fully React to MP News?

In short, the systematic estimation is searching for the event window length where the neural network’s predictive power is highest. This is the optimal window: the length where, as argued in the motivating framework, noise has a minimal average impact on the asset price change. This window length allows the network to best approximate the underlying relationship, yielding the most precise and generalisable text-based signal.

Figure 4 presents the results for selected federal funds, Eurodollar, and Treasury futures, as well as the S&P 500 Index. Appendix Figures E2–E8 show the results for all futures contracts. In each sub-figure, the horizontal axis shows the end time of the event window lengths (e.g., $t + 20$ implies a 30-minute window ending 20 minutes after release), and the vertical axis is the R_{OOS}^2 metric. The cross points show the $\overline{R_{OOS}^2}$ statistic for each window, with the solid yellow point marking the maximum. Box-and-whisker plots show the distribution of R_{OOS}^2 across the five sample splits for each window length.²⁷

Regardless of futures maturity and asset type, the event window length yielding the largest $\overline{R_{OOS}^2}$ is always at least 40 minutes, ending 30 minutes after the FOMC statement release. Examining other distributional metrics, such as the 25th or 75th percentiles, leads to the same conclusion. Financial markets fully react within a 40-minute event window *at minimum*, implying the popular 30-minute window is too short.²⁸

The optimal event window length increases with the underlying maturity of the futures, a trend summarised in Figure 2. For federal funds futures, the average window is 40 minutes for front-month contracts, rising to 45 minutes for two- and three-month-ahead contracts.

²⁷Standard errors are omitted due to the small sample splits. While increasing K via leave-one-out cross-validation would allow for formal hypothesis testing and confidence intervals, the computational costs are currently prohibitive. Conducting these significance tests remains a valuable direction for future research.

²⁸Results are qualitatively identical using median R_{OOS}^2 ; the optimal window is never 30 minutes.

For Eurodollar futures (two to four quarters ahead), the window is 50 minutes. This 50-minute length also applies to 2- and 5-year Treasury futures, but increases to 60 minutes for 10- and 30-year futures. One possible explanation is that traders of longer-horizon assets are more exposed to “soft” information (e.g., forward guidance), which is harder to interpret (Indriawan, Jiao, and Tse, 2021). This may increase belief and information uncertainty and result in higher risk premia (Piazzesi and Swanson, 2008). Relatedly, Okada (2025) document that longer-duration Treasury bonds are more affected by the FOMC announcement premium. Observed liquidity differences may be a symptom of these optimal event window lengths. For example, Fleming and Piazzesi (2005) find abnormal trading volumes persist longer in longer-maturity Treasury markets after FOMC announcements.

For equities, the S&P 500 Index and its front-month E-mini futures market (*ESc1*) fully react within 50 minutes, while the market for second-month E-mini futures (*ESc2*) takes 60 minutes. The neural network estimates that the average optimal window length is 53.3 minutes. These longer-than-30-minute windows may be because equities are less directly impacted by policy rates, leading to more investor disagreement and misinterpretations about interest rate decisions (Zhang, Kappou, and Urquhart, 2025). The difference between *ESc1* and *ESc2*, despite both sharing the S&P 500 Index’s infinite underlying maturity, may stem from their differing contract expirations. *ESc2* has a more distant expiration and lower liquidity than *ESc1*. This could cause *ESc2* traders to weigh “soft” information more heavily, requiring more time to fully process the announcement.

Overall, the $\overline{R_{OOS}^2}$ broadly exhibits a “hump” shape. The neural network’s predictive performance increases with event window length, peaks at 30–50 minutes after statement release, then declines. This pattern is consistent with the motivating framework. As the window initially lengthens, predictive performance improves as cognitive noise decays and the market’s full reaction is captured. After the peak, performance declines as unrelated news begins to dominate the observed price reaction, reducing the network’s approximation quality and generalisability. Structurally, this empirical decay maps cleanly to the institu-

tional timeline of post-announcement central bank communication. Within my sample, I document that the average duration dedicated to the verbal statement reading and targeted clarification at the start of these press conferences is 10 minutes and 40 seconds. This underpins a text-centric baseline of roughly 40 minutes, aligning perfectly with the empirical optimal window length identified for assets with short underlying maturities. Once the press conference moves deeper into unscripted Q&A dynamics, the introduction of fresh macroeconomic context and unmodelled fundamental updates outside the statement text behaves as a severe predictive penalty. The network’s out-of-sample performance consequently deteriorates as this post-statement noise enters the window, reinforcing the interpretation that my estimated optimal window lengths are a conservative lower bound of how long markets need to fully react to monetary policy announcements.

Importantly, the neural network achieves a positive $\overline{R_{OOS}^2}$ for all event window lengths and assets. In a machine learning context, a consistently positive out-of-sample R^2 against a historical-mean baseline confirms that despite the finite sample of announcements, the regularisation and transfer learning techniques mitigate the risk of the network overfitting to sample noise. In other words, these positive statistics suggest the network can approximate the systematic relationship between FOMC statement text and asset price changes.

Relative to this naive baseline, the network’s generalisability within the optimal window improves by an additional 2–17 percentage points (7.4 p.p. on average) compared to its performance in a conventional 30-minute window.²⁹ This improvement in predictive performance occurs precisely because it is the point where average noise impact is minimised, allowing the neural network to best approximate the underlying relationship.

5.1 Different Event Windows, Different Market Responses

I now compare the market price responses measured within the estimated optimal windows to those from the conventional 30-minute window, as illustrated for selected contracts and

²⁹Appendix Table E4 summarises these results for each futures contract.

the S&P 500 Index in Figure 5.³⁰ The horizontal axis plots the log-price differences within the optimal window against the 30-minute window responses on the vertical axis, with a dashed 45-degree line tracking equivalence on scheduled FOMC meeting dates.

To characterise whether markets under- or over-react to FOMC statements, I regress the price log-differences from the optimal windows on those from a 30-minute window. If the slope coefficient is significantly less than one at the 5% level, the market under-reacts. In each sub-figure, the regression line is red for under-reaction and grey otherwise.

Financial markets for all interest-rate futures under-react to FOMC statements, *ex post*. The regression slope decreases as maturity increases, consistent with the post-FOMC drift documented in Indriawan, Jiao, and Tse (2021) and Brooks, Katz, and Lustig (2023). Theoretically, this prolonged price discovery can reflect sequential learning among investors with heterogeneous technologies (Ai, Bansal, and Han, 2021). Under this channel, interest-rate markets are highly exposed to “soft” forward guidance information, which is costlier to process (Indriawan, Jiao, and Tse, 2021), while slow portfolio adjustments also play a role (Brooks, Katz, and Lustig, 2023). The 16 December 2008 policy surprise provides a clear example, where cutting the target rate to 0–0.25% alongside a commitment to “employ all available tools” likely extended the market’s reaction time. As shown in Figure 6 (top), Eurodollar price changes become increasingly positive as the window expands beyond 30 minutes.

The S&P 500 Index also under-reacts, *ex post*, corroborating Neuhierl and Weber (2024) and Golez, P Kelly, and Matthies (2025) regarding information processing limitations. However, equities also exhibit the most price reversals, often flipping from negative to positive as the event window expands. This behaviour matches macro-finance frameworks where announcements alter the mixture of economically meaningful signals being processed. For instance, the expanded window could capture a delayed “information effect” conveying the central bank’s private assessment (Nakamura and Steinsson, 2018), a gradual market realisation of the Fed’s response to news (Bauer and Swanson, 2023b), or time-varying attention and

³⁰Appendix Figures E9–E15 display the comparisons for all remaining interest-rate and equity futures.

risk premia adjustments as structural uncertainty resolves (Fisher, Martineau, and Sheng, 2022). The 29 April 2015 meeting, shown in Figure 6 (bottom), provides a clear example: the initial negative response reversed and increased until market close as early rate hike expectations dissipated. With more time, cognitive noise subsided, allowing the initial “knee-jerk” reaction to reverse as positive information was digested. This aligns with Bordalo et al. (2024), where stock market overreactions reverse as beliefs correct.

Overall, these findings support the premise that the optimal event window length cannot be simply assumed. Instead, it varies systematically across asset types and underlying maturities. My estimation procedure operates independently of any specific assumption regarding the macro-finance frictions driving these delayed adjustments. Because the overwhelming variance shock of the announcement safely maintains relative exogeneity across these horizons, optimising the window length for minimised noise impact ensures that the complete market reaction is captured, regardless of the underlying mechanisms. As further explored in Section 7, statement characteristics, such as contextual similarity between sequential statements, systematically correlate with these estimated optimal window lengths, reinforcing the interpretation that informational complexity drives the gradual price adjustment.

6 What Happens to Monetary Surprises and Shocks?

Having shown that optimal event windows consistently exceed the standard 30 minutes, this section investigates the subsequent impact on monetary policy surprises and shocks. Constructing these shocks requires a uniform event window length across assets. Standard macro-finance models extracting shocks from the yield curve rely on the property of “invertibility” or “spanning” (Duffee, 2013), a condition well-defined for a temporal cross-section of yields but not for yields measured at varying, asset-specific horizons. Therefore, using the same data and meeting dates from Section 3, I construct and test monetary policy surprises separately for each baseline optimal window identified in Section 5: 40, 50, and 60 minutes.

6.1 Effects of Window Choice on Monetary Policy Surprises

I construct interest-rate surprises for each FOMC meeting date using the standard 30-minute window and the 40-, 50-, and 60-minute optimal windows. Following Kuttner (2001), Gürkaynak, Sack, and Swanson (2005), Nakamura and Steinsson (2018), and others, I use federal funds futures for expectations up to three months out, Eurodollar futures for four months to one year ahead, and Treasury futures for expectations out to fifteen years. Appendix C details the conversion from futures price changes to interest-rate surprises.

Figure 7 plots the correlation between monetary policy surprises from the optimal windows versus the standard 30-minute window. The horizontal axis shows the underlying maturity of the surprises (e.g., “0–1M” for the FOMC meeting happening in the current or next month). The individual lines show the correlations for the 40-, 50-, and 60-minute optimal windows, while the blue-solid line shows their average.

The correlations consistently decline as the event window lengthens. This decline is more pronounced at longer underlying maturities. For example, while surprises for near-term meetings are highly correlated (above 95%) regardless of window choice, the correlation for 15-year-ahead surprises (measured using 50-minute windows) drops by 10%. This demonstrates that as expectations about monetary policy move further into the future, the resulting interest-rate surprises become more sensitive to the choice of event window.

6.2 Effects of Window Choice on Monetary Policy Shock Impacts

I now use the interest-rate surprises as financial instruments to identify exogenous variation in monetary policy, following several popular methods. I include Treasury futures in my instrument set for two reasons. First, recent studies (e.g., Brennan et al., 2024; An, Stedman, and Lusompa, 2025) show longer-maturity instruments help prevent understating policy stimulus during the effective lower bound period.³¹ Second, my Figure 7 results show

³¹For my sample, the effective lower bound is 16 December, 2008, through 16 December, 2015.

that Treasury futures are the most sensitive to window choice.³² For conciseness, I use the *median* of the optimal window lengths found by the neural network: 50 minutes (10 minutes before and 40 minutes after statement release).

Shock Construction I construct monetary policy shocks following the methods of Gürkaynak, Sack, and Swanson (2005) (GSS), Nakamura and Steinsson (2018) (NS), and Jarociński and Karadi (2020) (JK).³³ All three methods use principal component analysis to reduce the set of interest-rate surprises into factors capturing their common variation.^{34,35}

GSS extract two factors by rotating the first two principal components such that the first, “target” (GSS_T), drives changes in the current federal funds rate, while the second, “path” (GSS_P), does not. GSS_T is scaled one-for-one with the current-meeting surprise, whereas GSS_P is scaled to have an equal impact on the four-quarter Eurodollar surprise.

NS use only the first principal component, which Bauer and Swanson (2023a) explain is essentially a weighted average of the GSS target and path factors. This component is scaled one-for-one with the daily change in the one-year nominal Treasury yield (NS_{MP}).

JK scale the first principal component to match the standard deviation of the four-quarter Eurodollar surprise. To disentangle policy from central bank information effects, they impose sign restrictions based on the shock’s co-movement with equity returns.³⁶ A negative stock price co-movement identifies a monetary policy shock (JK_{MP}), while a positive co-movement identifies a central bank information shock (JK_{CBI}).

For easy interpretation and comparison, I re-scale all shock series to be one-for-one with the daily change in the zero-coupon, nominal one-year Treasury yield.³⁷ Appendix Table E6 provides summary statistics for each shock series. Because these methods (GSS, NS, and JK) jointly account for the key characteristics of monetary policy, such as its multi-dimensional

³²All exercises are robust to using the authors’ original instrument sets, as shown in Appendix D.

³³See Appendix Table E5 for a summary of the shocks.

³⁴Principal component analysis projects data onto new dimensions based on variance. The first principal component captures the largest common variation, the second captures the next largest, and so on.

³⁵While JK originally use implied interest-rate surprises from *FF4*, the first principal component of multiple surprises is used in their modern update: https://github.com/marekjarocinski/jkshocks_update_fed

³⁶Stock market changes are measured within the same event window as the surprises.

³⁷Robustness checks in Appendix D show this scaling choice does not affect the main findings.

nature, its impact beyond the immediate horizon, and information effects, any observed differences in the resulting shocks can be attributed directly to the choice of event window.

Visual Shock Differences I plot the shock series from both event windows across FOMC meeting dates in Figure 8, where the black-solid line represents the 30-minute window shocks and the red-dotted line depicts the optimal window shocks. Regardless of construction method, the optimal window shocks frequently display larger peaks and troughs in magnitude than their 30-minute counterparts.³⁸ For example, at the critical December 2008 and March 2009 FOMC meetings, the optimal window shocks are 3–6 basis points larger, suggesting that the 30-minute window fails to capture the full impact of these statements.

Another notable episode is the August 2011 FOMC meeting, where the statement first included explicit, date-based forward guidance (Crump, Eusepi, and Moench, 2013). On this date, the optimal window is associated with a more negative shock for both GSS_P and NS_{MP} . When applying the JK sign restrictions, the negative central bank information shock (JK_{CBI}) increased in magnitude by 3.6 basis points, while the policy shock (JK_{MP}) was unchanged. NP_{MP} and GSS_P also increased by similar magnitudes. Because the federal funds rate was already at the zero lower bound, this new language on the future path of rates could be interpreted as an information shock about weaker-than-anticipated economic conditions. The optimal window may be capturing the full extent of this negative information shock, which took longer than 30 minutes for the market to fully process.

The event window choice can also cause “shifts in importance” between shock components. A clear example is the January 2010 FOMC meeting. Using a 30-minute window, both the JK_{MP} policy shock and the JK_{CBI} information shock had similar magnitudes. When using a 50-minute window, however, this interpretation changes: the information shock becomes the dominant factor, more than doubling the value of JK_{CBI} when measured within the 30-minute window, while JK_{MP} is essentially zero.

This shift aligns with the statement’s content. The FOMC held rates steady but used

³⁸Plots for GSS_T are omitted. These shocks are identical because markets react instantly to rate changes.

forward guidance to signal moderate expected growth, liquidity promotion, and an intention to hold “low levels of the federal funds rate for an extended period”. Likely needing more time to process these components, the market updated its prospects upwards, consistent with a positive JK_{CBI} shock and the S&P 500’s steady rise over time.

Overall, when comparing the JK shocks, the central bank information shock more frequently increases in magnitude with the optimal window than the policy shock JK_{MP} . While JK_{MP} also sees increases, the differences due to window choice are relatively smaller. The larger JK_{CBI} differences likely reflect a mechanical effect stemming from the sign restriction method combined with the larger S&P 500 price changes observed in the optimal window.

Responses of Interest Rates I now examine how the event window choice affects the impact of monetary policy shocks on nominal and real interest rates. I use the daily change in Treasury yields and TIPS yields as the dependent variables, calculated using end-of-day yields for the day of and the day before the FOMC announcement. Because the shocks (constructed with 30-minute or optimal windows) are measured *within* this daily window, the timing ensures that the yield changes do not affect the shocks.

The regression specification is as follows:

$$y^{i,j} = \beta_0^{i,j,k,l} + \beta_1^{i,j,k,l}(Shock)^{k,l} + \varepsilon^{i,j,k,l}, \quad (8)$$

where $y^{i,j}$ is the daily change in Treasury or TIPS yields (i) at a specific horizon (j). The regressor, $(Shock)^{k,l}$, is the monetary policy shock (k) constructed using either window (l).

Regression results for the responses of nominal and real interest rates are presented in Table 1 and Table 2, respectively. Each estimate comes from a separate OLS regression specified in Equation 8. The tables compare the coefficients (β_1) from shocks constructed using the 30-minute window against those constructed using the median optimal window (50 minutes). The bolded columns (10–12 in Table 1, 8–10 in Table 2) show the difference between these coefficients, isolating the impact of the event window choice.

Expanding the event window from 30 minutes to the optimal length, monetary policy shocks that capture forward guidance (GSS_P , NS_{MP} , JK_{MP}) have larger effects on both nominal and real interest rates. For example, the impact of JK_{MP} on 2-, 5-, and 10-year real yields is 23–26 basis points larger, and the estimates retain their 1% statistical significance.

Conversely, the “target” factor GSS_T has a smaller impact on nominal rates across all horizons in the optimal window, though its impact on real yields is unclear. Only the 2-year nominal yield estimates are statistically significant under both windows. This pattern likely reflects the inclusion of the effective lower bound period, when monetary policy stimulus was driven by forward guidance rather than changes to the target rate.

The main takeaway from these results is that using too short of an event window prevents markets from fully processing the “soft” information in FOMC announcements, particularly regarding the future path of interest rates. As a result, the measured impact of the forward guidance component of monetary policy shocks on financial variables is attenuated.

These findings support the interpretation suggested by Gürkaynak, Sack, and Swanson (2005): while markets can process simple target rate changes almost instantly, the more complex information about the economic outlook and future policy path requires more time to be fully priced in by market participants.

Responses of Break-even Inflation I next investigate the impact of monetary policy shocks on break-even inflation, defined as the difference between nominal Treasury and TIPS yields. The regressions follow the same specification as Equation 8, with the daily change in break-even inflation at various maturities as the dependent variable. Table 3 presents the results, comparing the impact of shocks constructed within the 30-minute window (columns 2–4) to those from the optimal window (columns 5–7). The final columns (8–10) show the difference between these corresponding coefficients, isolating the effect of the window choice.

Similar to Nakamura and Steinsson (2018), 30-minute window estimates lack statistical significance. In contrast, the 50-minute optimal window introduces significance at several horizons, including the longest-horizon JK_{MP} and shortest-horizon GSS_P effects. Further-

more, the optimal window consistently makes the estimated impacts of the forward guidance component of monetary policy shocks on break-even inflation *stronger* and more *precise*.

Responses of Equities I next evaluate how shock timing affects equity prices. Following Equation 8, Table 4 models 100 times the S&P 500 and E-mini futures log-price differences. Columns 2–4 and 5–7 compare coefficients from the 30-minute and optimal windows, respectively, while the final columns (8–10) show the percentage change between these estimates.

Expanding the event window from 30 minutes to the optimal length causes the forward guidance component of monetary policy shocks to impact stock prices more negatively. In the optimal window, the negative impact on the S&P 500 Index is 18.25% larger for JK_{MP} , and 11.51% and 1.23% larger for GSS_P and NS_{MP} , respectively. These larger negative impacts are even more pronounced for the second-month S&P 500 E-mini futures contract. Notably, equity responses vary more widely across shock types than interest rates did.

Consistent with the findings for interest rates, the equity response to the GSS “target” factor (GSS_T) also becomes weaker (by 11.99%) in the optimal window. While the estimates remain statistically significant at the 1% level, this weakening suggests the longer window causes a “shift in importance” toward the “path” factor. Intuitively, the optimal window captures the market’s initial, fast reaction to the target rate, but then allows more time to fully process the more complex forward guidance, which dominates the price change.

The effects on the central bank information shock (JK_{CBI}) are more complex. One explanation for the differences relates to the high idiosyncratic volatility of stock prices (relative to yields) combined with the JK sign restriction method. As shown in Subsection 5.1, equities exhibit both under-reaction and frequent, sharp price reversals. Expanding the window to 50 minutes captures more of these reversals. This volatility can lead to different shock decompositions under the sign restriction, amplifying the JK_{CBI} series in both directions and potentially weakening its net, directional impact on prices.

The large equity impact of the JK shocks requires cautious interpretation, as their sign-restriction methodology mechanically ensures co-movement with the S&P 500. However, the

forward guidance component isolated via alternative methods that omit this restriction, such as GSS_P , yields similarly large negative equity impacts, corroborating the overall result.

Responses of the Macroeconomy via Local Projections To study the macroeconomic impact of the shocks, I estimate impulse responses via local projections. Following Gertler and Karadi (2015) and Bauer and Swanson (2023a), the monthly model includes log CPI, log IP, the nominal two-year Treasury yield, and the excess bond premium (EBP) from Gilchrist and Zakrajšek (2012). The EBP proxies the financial sector’s risk-bearing capacity, serving as the key transmission channel to the real economy. Shock series are set to zero in non-FOMC months. Appendix Tables E7 and E8 provide summary statistics.

I estimate the impulse responses using the lag-augmented local projection method of Olea and Plagborg-Møller (2021). This approach is chosen because adding an additional lag to the specification as a control allows for correct inference using only Eicker-White heteroscedasticity-robust standard errors, avoiding the need to correct for serial correlation.

I run a separate regression for each outcome variable on each shock for each event window length (l) at each horizon (h):

$$y_{t+h}^{i,l} = \theta^{i,k,l} (Shock)_t^{k,l} + controls + \eta^{i,k,l} \quad (9)$$

where $y_{t+h}^{i,l}$ is the outcome variable (i) at horizon h , $(Shock)_t^{k,l}$ is the monetary policy shock (k) from window (l), and $controls$ include the requisite lags. The impulse responses are the θ coefficients, plotted over horizon h . For interpretation, all impulse responses are scaled to show the effect of a 100 basis point contractionary policy shock, based on the common re-scaling to the one-year nominal Treasury yield.

Figure 9 compares the impulse responses of CPI to the monetary policy shocks constructed using both event windows.³⁹ At first glance, the point estimates of the impulse responses are qualitatively similar regardless of window choice. While some differences exist (e.g., the decline in CPI from NS_{MP} is more severe at farther horizons in the optimal win-

³⁹Results for the other macroeconomic variables are robust and presented in Appendix Figures E16–E18.

dow), the window choice does not dramatically alter the response path.

The most noticeable difference, however, is in the confidence intervals. The impulse responses derived from the optimal window shocks are consistently more precise than those from the 30-minute window. To formally measure this, for each event window length, I calculate the width of the confidence intervals for each time horizon, variable, and shock.

I then compute the ratio of the optimal window’s confidence interval width to the 30-minute window’s width. A ratio less than one implies a more precise estimate from the optimal window. The average and median of this ratio across all impulse responses are 0.9173 and 0.9410, respectively.⁴⁰ This confirms that constructing monetary policy shocks within the optimal window results in *more precise* macroeconomic impulse responses.

Overall, the varied responses of interest rates, equities, and the macroeconomy support the findings of Brennan et al. (2024) that long-term rates and shock construction methods are important contributors. This paper provides strong evidence that the event window length is another critical dimension that must be considered.

My findings show that the standard 30-minute window is insufficient. It attenuates the measured impact of the forward guidance component of monetary policy shocks on financial variables because markets have not finished reacting. In contrast, constructing shocks within the optimal window reveals these larger impacts and yields more precise macroeconomic estimates. This optimal window represents the length of maximum generalisability, which secures these benefits because the average impact of noise is minimised.

7 Statement Features and Their Effects on Windows

Recall that the systematic estimation is an exercise where the neural network regresses asset price changes on FOMC statement text across various windows. This process can be viewed as “jointly” estimating the optimal event window length and the text-based signal that yield the greatest precision. Unfortunately, the financial and hardware constraints of this joint

⁴⁰These confidence interval width ratios are robust to the choice of monetary policy surprise instruments.

estimation currently limit the analysis to windows ending 1 hour after the statement release.

To circumvent these constraints and provide initial insights, the alternative “one signal” approach follows the motivating framework more closely. It assumes that the predicted price change from the optimal window (i.e., the most precise signal) is *constant* over time, allowing for the estimation of event windows ending more than one hour after statement release.

However, the core assumption of this “one signal” approach is strong. Because the neural network’s weights are sensitive to both inputs and outputs, it is possible that the “jointly” estimated optimal window differs from the one found using this “one signal” approach.⁴¹

The “one signal” approach applies the text-based signal from the “jointly” estimated optimal window to calculate $\overline{R_{OOS}^2}$ beyond the 60-minute constraint. This shortcut reveals how different communication dimensions affect the optimal window length and checks if a longer window yields a larger $\overline{R_{OOS}^2}$, a possibility discussed in Appendix D.

7.1 How FOMC Statement Characteristics Affect Window Lengths

I next examine how specific features of an FOMC statement affect its optimal event window. Using the “one signal” approach, I correlate window lengths with statement complexity, similarity to previous announcements, and dissents. This heterogeneity analysis requires two strong identifying assumptions: that the text-based signal from the “jointly” estimated window is constant, and that it remains relevant across statement subsamples.⁴²

Text Complexity First, I examine how optimal event window lengths vary with statement complexity. I proxy this *semantic* text complexity using the Flesch-Kincaid Grade Level index, a standard readability metric based on sentence and word structure.⁴³ I first split the FOMC statements into two subsamples based on their Flesch-Kincaid Grade Level relative to the median (16.5). I then calculate the MSEs under the “one signal” approach for

⁴¹A rough analogy is how iterative generalised method of moments (GMM) does not always converge.

⁴²Ideally, one would systematically re-estimate the optimal window for large subsamples of each characteristic, but this is computationally infeasible for the current paper.

⁴³Appendix Figure E19 displays the evolution of the Flesch-Kincaid Grade Level for my sample. Descriptive statistics are in Appendix Table E9.

each subsample across all futures maturities.

Table 5 (columns 1-2) summarises this exercise, showing the minimised MSEs and associated event window lengths, averaged across all assets. The results show a clear difference: statements with a “complex” readability level (≥ 16.5) have a longer average optimal event window of 71 minutes, compared to 60 minutes for “simpler” statements (< 16.5).

Interestingly, both subsamples have similar average MSEs. This finding suggests that the price response to “complex” statements is relatively more affected by cognitive noise. Intuitively, complex information generates more disagreement and market volatility (Smales and Apergis, 2017), ultimately requiring more time for markets to fully process the news.⁴⁴

Text Similarity FOMC statements, though relatively concise, feature evolving language and context over time (Acosta and Meade, 2015; Handlan, 2022a). Using the “one signal” approach, I analyse how optimal window lengths change with the *similarity* between consecutive statements, validating my paper’s systematic estimation. Because the neural network approximates the relationship between statement text and price reactions, it captures the informational “level” of the text rather than the “change”. Consequently, the systematic estimation implicitly assumes the market holds “zero” prior expectations of the statement’s content.⁴⁵ Despite this strong assumption, these initial insights show that using the informational level still provides valuable signals for the neural network.

Rather than relying on simple word counts or sparse frequency metrics that ignore word order, I measure semantic similarity using a dense, context-aware deep learning framework. Specifically, I extract text representations using XLNet-Base that has not been fine-tuned to approximate the relationship between monetary policy announcements and market price reactions. Using only the pre-trained neural network ensures the semantic signals are entirely

⁴⁴Understanding how communication complexity impacts information processing—even among sophisticated market participants—remains an open research question. For example, McMahon (2023) show that the Bank of England’s efforts to simplify statements decreased semantic complexity but simultaneously increased motivating complexity over time.

⁴⁵The ideal modification is to have the neural network approximate the underlying relationship using the quantified difference in the information content expressed in the FOMC statement and what financial markets expected prior to release.

out-of-sample and orthogonal to market price reactions. Unlike traditional bag-of-words approaches, this model processes text bi-directionally to incorporate surrounding context and syntax, mapping each complete document into a 768-dimensional embedding vector.

Utilising these dense representations, I compute the degree of alignment between any two statements (A and B) via their *cosine similarity*:

$$S^{A,B} := \cos \theta = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|}, \quad (10)$$

where $\mathbf{A}$ and $\mathbf{B}$ denote the dense contextual vectors for statements A and B. Cosine similarity values approaching one mean the statements share near-identical contextual prose, while lower values indicate a structural shift or informational update in communication.

I focus on sequential statement similarity, $(t, t-1)$, denoted S^1 and plotted in Figure 10.⁴⁶ Figure 10 shows that contextual cosine similarity remains exceptionally high, massed tightly near a median of 0.998. This distribution reflects a significant language standardisation beginning around 2009–2010 as the federal funds rate approached the zero lower bound. As discussed in Handlan (2022a), the FOMC anchored market expectations by standardising its vocabulary during and after the financial crisis in an attempt to calm market volatility. This highly similar prose continued through 2019, past the zero lower bound, reflecting broader efforts to increase policy transparency under Chairs Yellen and Powell.

To isolate true informational shifts, I partition the sample at the 33rd percentile into “different” ($\leq 33^{rd}$) and “similar” ($> 33^{rd}$) statements. Table 5 (columns 3–4) summarises the “one signal” results. Averaged across all assets, “similar” statements require only 52 minutes, while “different” statements expand the optimal window to 81 minutes with similar minimised MSEs. This finding is intuitive: a “different” statement contains a genuine contextual pivot that requires substantially more time for financial markets to discern and fully process. Conversely, “similar” statements introduce negligible updates, allowing markets to react more

⁴⁶Appendix Figure E20 presents the full document similarity matrix as a heat map. Descriptive statistics for S^1 are in Appendix Table E9.

quickly as the overwhelming majority of the content has already been priced in.

Presence of Dissents The final characteristic I consider is the presence of dissents, which are recorded in roughly 40% of the statements in my sample. While not rare, dissents are significant, as they traditionally signal that a Board member finds the majority’s opinion unacceptable (C Madeira and J Madeira, 2019). The very existence of a dissent thus provides additional, complex information for markets to process.

To investigate this, I split the sample into statements with unanimous votes and those with at least one dissent.⁴⁷ Table 5 (columns 5-6) presents the results of the “one signal” approach for these subsamples, averaged across all assets. The results show a large difference: statements with dissents have an average optimal window of 83 minutes, far longer than the 53-minute window for unanimous statements. This suggests the internal disagreement signalled by a dissent prolongs market processing times. While prior work has shown dissents have a nuanced impact (e.g., C Madeira and J Madeira, 2019; Tsang and Z Yang, 2024 find stronger effects on equities than on interest rates), my finding shows that, on average across all asset types, the presence of a dissent is associated with longer market reaction times.

Overall, these findings show that monetary policy communication affects markets along multiple dimensions, not just the “hard” information of the federal funds rate decision. The “soft” information characteristics of the statement, such as contextual complexity, similarity to prior announcements, and the presence of dissents, systematically influence the time financial markets need to fully process the communication, particularly its forward guidance.

Indeed, future research into how these textual dimensions affect market processing times could provide deeper insight into the transmission of monetary policy.

8 Conclusion

The choice of event window length for measuring financial market responses to news has largely remained an “ad-hoc” decision in the empirical monetary policy literature. This pa-

⁴⁷This analysis does not distinguish between the *reasons* for the dissent (e.g., for tighter or easier policy).

per challenges this convention by asking a simple question: how does one systematically choose the appropriate window? I answer this by developing a new methodology that combines observed price dynamics with a text-based signal derived from a neural network. This method allows me to systematically estimate the event window length that best reflects the market’s full reaction to the information content of FOMC statements.

My systematic estimation yields two key findings. First, the common 30-minute window is insufficient. My results show that, on average, markets fully react within an event window *ending at least 30 minutes after release*, and a 30-minute window is never found to be optimal for any asset. Second, the optimal window length is not “one-size-fits-all”. Instead, the optimal duration increases systematically with asset underlying maturity, rising from 40 minutes for short-term assets to 50–60 minutes for assets with maturities of two quarters or more. Relatedly, I find that statements with greater complexity, less similarity, and the presence of dissents are all associated with longer event windows on average.

The implications of these findings are not merely methodological. The choice of event window has a tangible impact on monetary policy surprises and shocks. I find that the correlation between surprises measured with optimal versus 30-minute windows decreases with asset underlying maturity. These discrepancies are economically meaningful, altering the forward guidance component of policy shocks. Consequently, the measured impact of this component on interest rates, break-even inflation, and equity prices becomes larger, suggesting that conventional, shorter windows attenuate the true effects of policy.

The benefits of the optimal window extend to the broader macroeconomy. I document that when monetary policy shocks are constructed using the optimal event window, the estimated impulse responses of macroeconomic variables become more precise.

Ultimately, this paper argues that determining the optimal event window length is an empirical question, not an assumed parameter. The provided methodology offers a systematic approach for this critical step in event window studies. The results indicate that allowing for longer, asset-specific reaction times yields monetary policy surprises that more accurately

reflect the market’s full response. This occurs because the optimal window maximises generalisability and minimises average noise impact.

As the literature expands to study broader forms of central bank communication (e.g., Neuhierl and Weber, 2019; Swanson and Jayawickrema, 2023; Bauer and Swanson, 2023a; and others), this paper’s methodology can be applied to measure the full reaction time to any text-based news event. Doing so offers a path toward constructing more relevant and precise measurements of unanticipated monetary policy.

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Figures and Tables

Market Price Reactions for S&P 500 Index, 24/08/1999

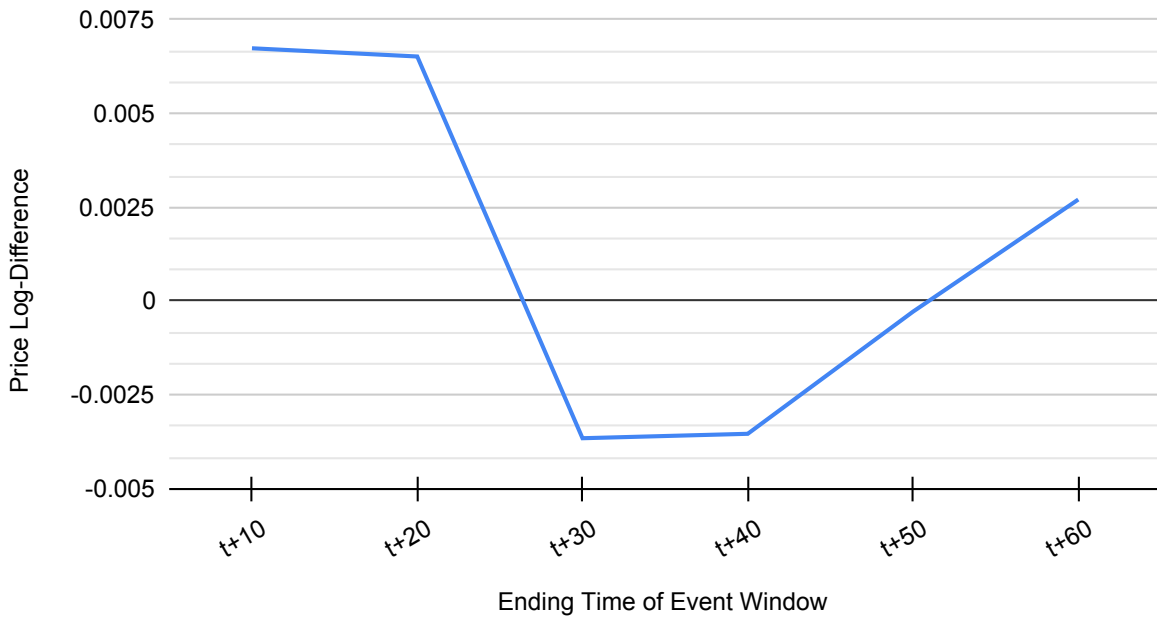


Figure 1: Responses of S&P 500 Index to FOMC Statements

Notes: The price log-differences of the S&P 500 Index for different event window lengths after the release of the 24 August, 1999 FOMC statement is depicted. The horizontal axis of each figure depicts the end time of the considered event window lengths and the vertical axis represents the price-log differences. Every event window starts at 10 minutes before the FOMC statement release.

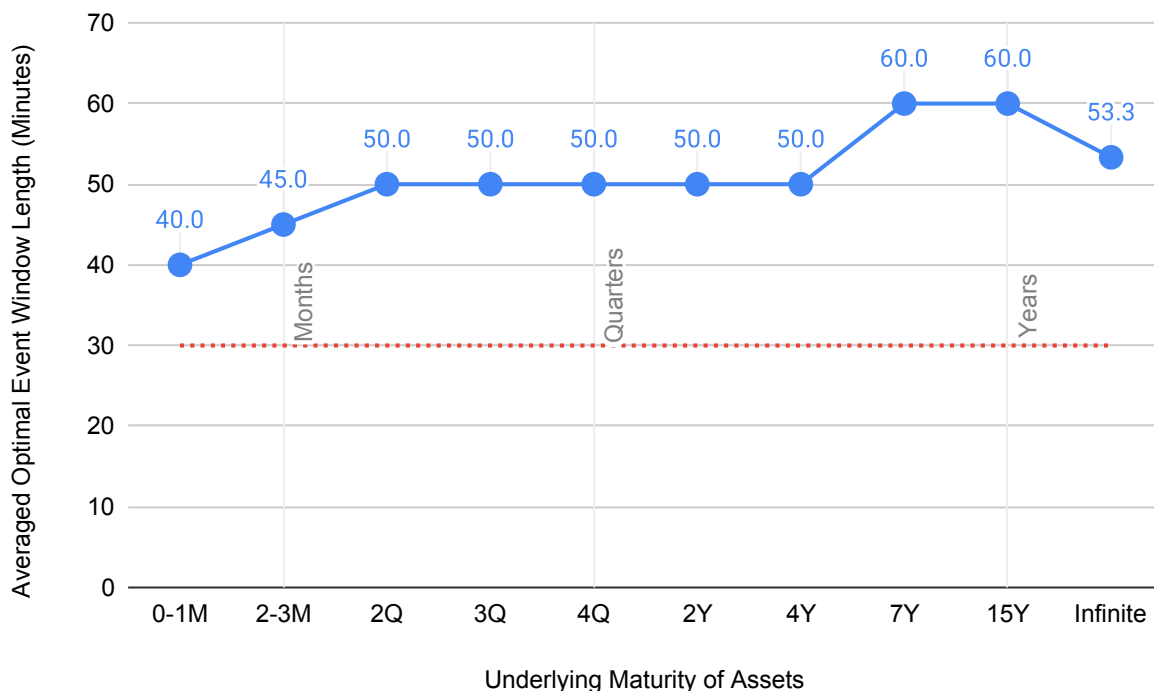


Figure 2: Averaged Optimal Event Window Lengths for Various Asset Underlying Maturities

Notes: The horizontal axis depicts the horizons of all considered futures contracts and equities. The vertical axis depicts the event window lengths in minutes, with all windows starting at 10 minutes before FOMC statement release. The red dotted line depicts the 30-minute window length, common in the literature. The systematically estimated event window lengths are averaged across futures maturities for each asset type. “0–1M” is the average of the event window lengths for front-month and 1-month-ahead Federal Funds futures. “2–3M” is the average of the event window lengths for 2-month and 3-month-ahead Federal Funds futures. “Infinite” is the underlying maturity of the S&P 500 Index and considered E-mini futures contracts. The underlying maturities for the Treasury futures contracts are approximated by [Gürkaynak, Kisacikoğlu, and Wright \(2020\)](#).

Number of Words in FOMC Statements

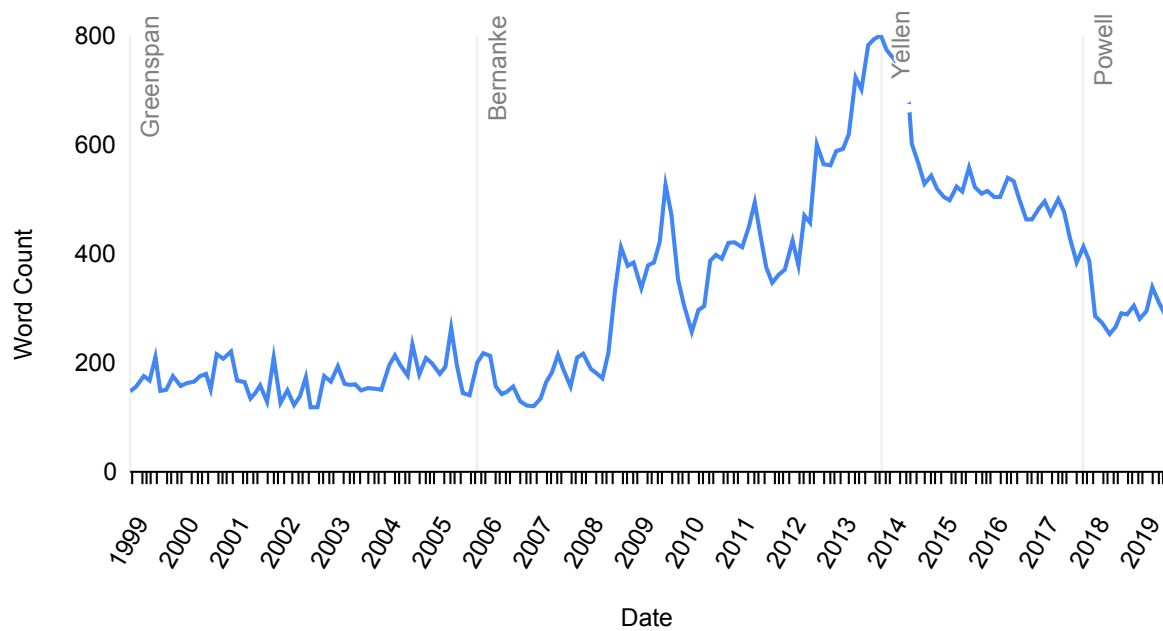
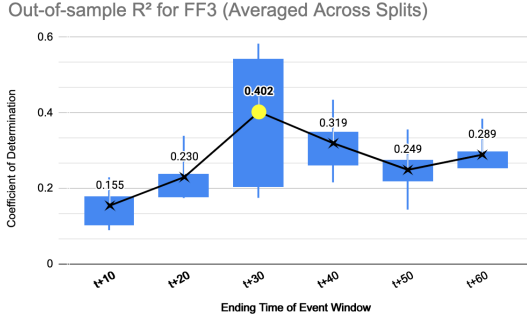
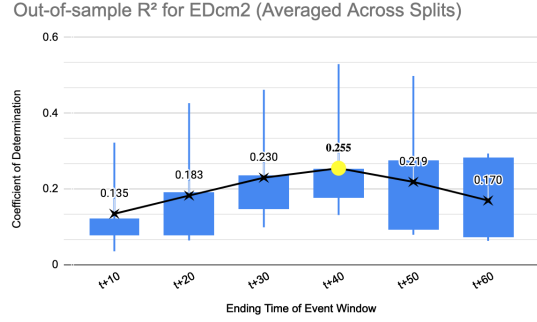


Figure 3: Number of Words in FOMC Statements, May 1999–October 2019

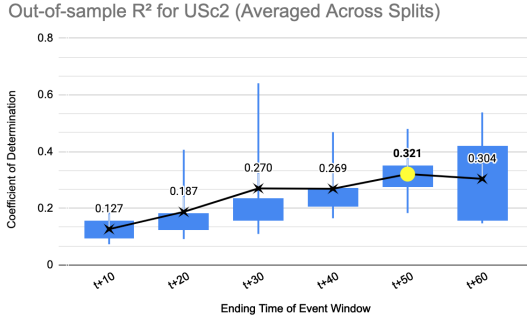
Notes: The above counts are for FOMC statements that have underwent pre-processing, which is explained in Subsection 3.1. From left to right, the vertical grey lines indicate the first FOMC meeting with Greenspan, Bernanke, Yellen, and Powell as Fed Chair.



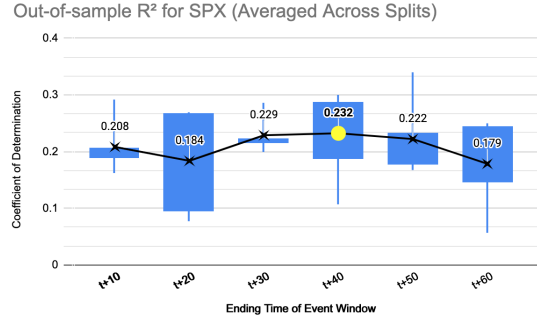
(a) *FF3*



(b) *EDcm2*

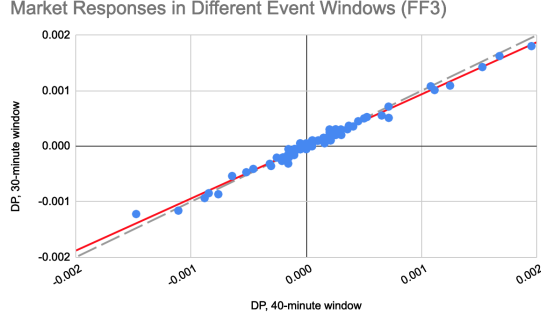


(c) *USc2*

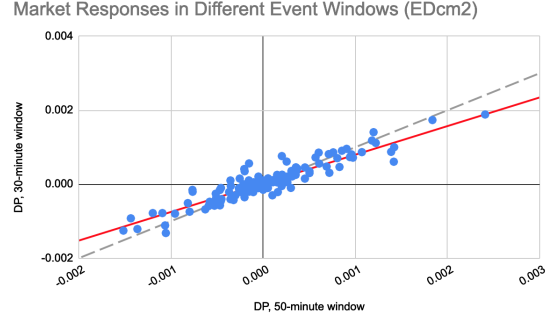


(d) *SPX*

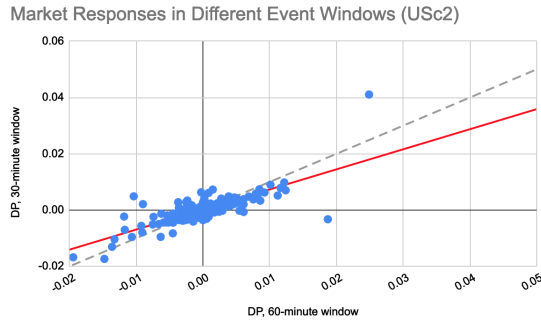
Figure 4: Optimal Event Window Lengths for Interest-Rate Futures and S&P 500 Index Notes: The horizontal axis of each figure depicts the end time of the considered event window lengths and the vertical axis represents the coefficient of determination, R^2_{OOS} . The cross points represent $\overline{R^2_{OOS}}$ for each event window size, where the solid yellow point represents the event window with the largest $\overline{R^2_{OOS}}$. For each event window, box-and-whisker plots are shown surrounding the corresponding averages. $\overline{R^2_{OOS}}$ measures the average reduction in predictive error achieved by the neural network compared to the naive baseline.



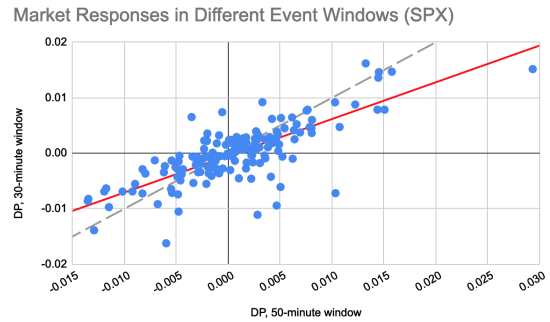
(a) $FF3^{**}$



(b) $EDcm2^{***}$



(c) $USc2^{***}$

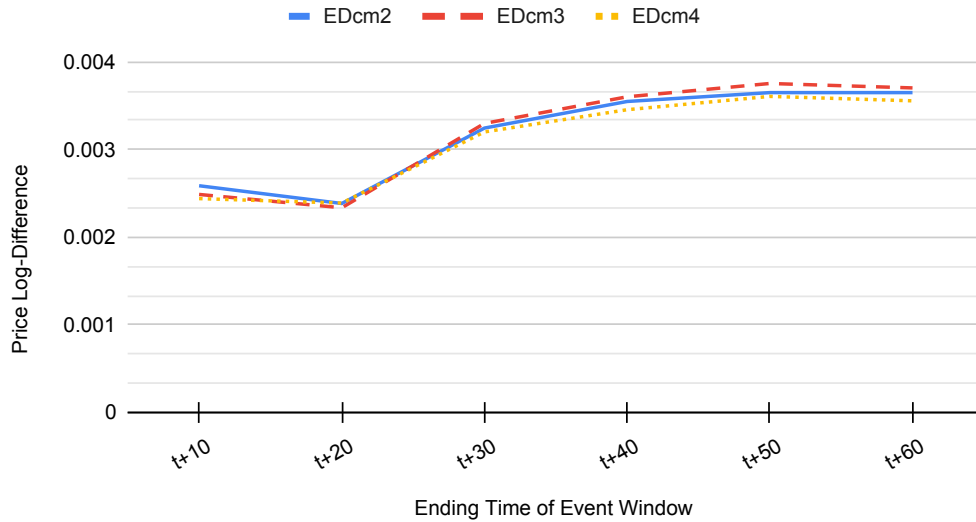


(d) SPX^{***}

Figure 5: Comparing Market Responses in Different Event Windows for Interest-Rate Futures and S&P 500 Index

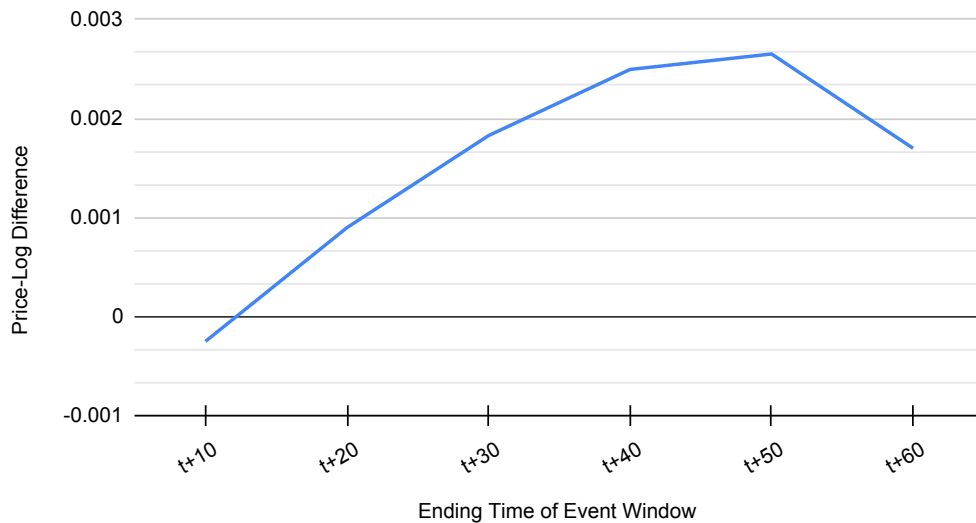
Notes: The horizontal axis depicts the price log-difference within the systematically estimated event window length. The vertical axis represents DP_{t+20} . The 45-degree line is depicted as grey and dashed. The blue dots are market reactions on scheduled FOMC meetings. The price-log differences calculated within the optimal window lengths are regressed on DP_{t+20} through OLS. If the slope coefficient is greater (less) than one and its difference with one is statistically significant for at most $\alpha = 0.05$, then the financial market under-reacts to the information content of FOMC statements on release, ex post, and is depicted in red. ** sig. at the 5% level, *** sig. at the 1% level.

Market Price Reactions for Eurodollar Futures, 16/12/2008



(a) Eurodollar Futures

Market Price Reactions for S&P 500 Index, 29/04/2015



(b) SPX

Figure 6: Responses of Eurodollar Futures and S&P 500 Index to FOMC Statements
 Notes: The top (bottom) sub-figure depicts the price log-differences of Eurodollar futures markets (the S&P 500 Index) for different event window lengths after the release of the December 2008 (April 2015) FOMC statement. The horizontal axis of each figure depicts the end time of the considered event window lengths and the vertical axis represents the price-log differences.

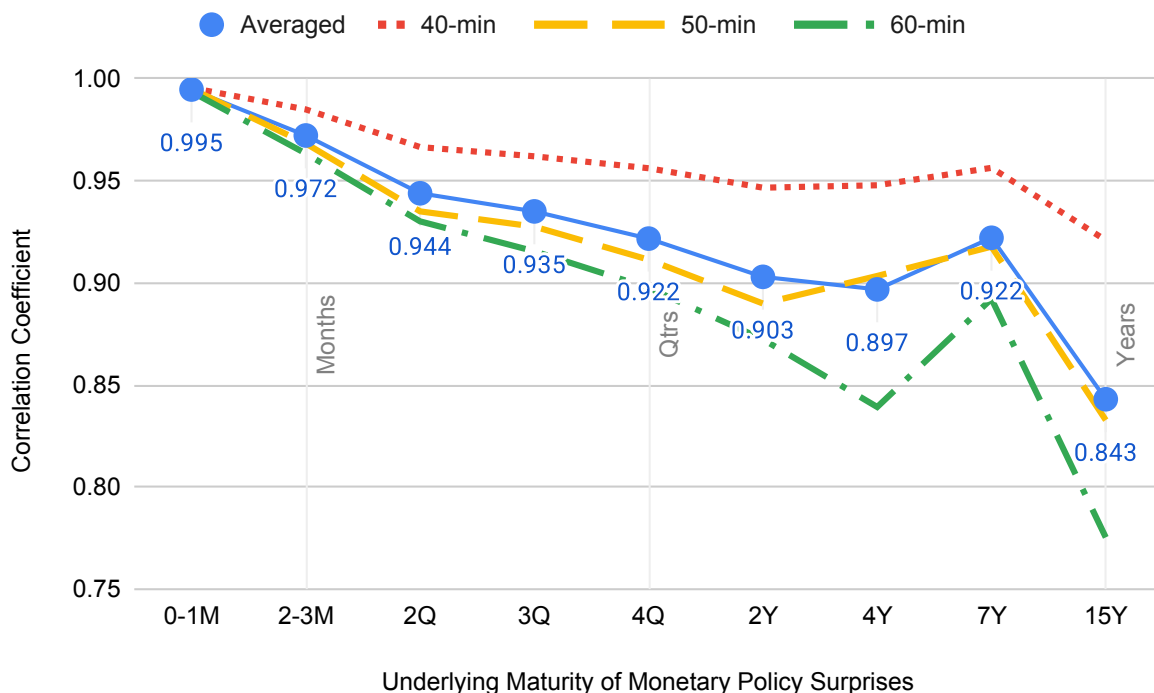


Figure 7: Correlation Between Monetary Policy Surprises Calculated in Optimal v. 30-minute Window Lengths

Notes: The horizontal axis depicts the underlying maturities of interest-rate surprises. The vertical axis represents the Pearson correlation coefficient between the surprises calculated within optimal event window lengths v. 30 minutes. The red-dotted, yellow-dashed, and green-mixed lines represent the correlations calculated for surprises measured within the optimally-found 40, 50, and 60 minutes, respectively. The blue-solid line represents the correlations averaged across the three optimal window lengths. “0–1M” is interest rate surprise calculated using information from front-month and 1-month-ahead Federal Funds futures. “2–3M” represent expectations calculated from 2-month and 3-month-ahead Federal Funds futures. The underlying maturities for the Treasury futures contracts and resulting surprises are approximated by [Gürkaynak, Kisacikoğlu, and Wright \(2020\)](#).

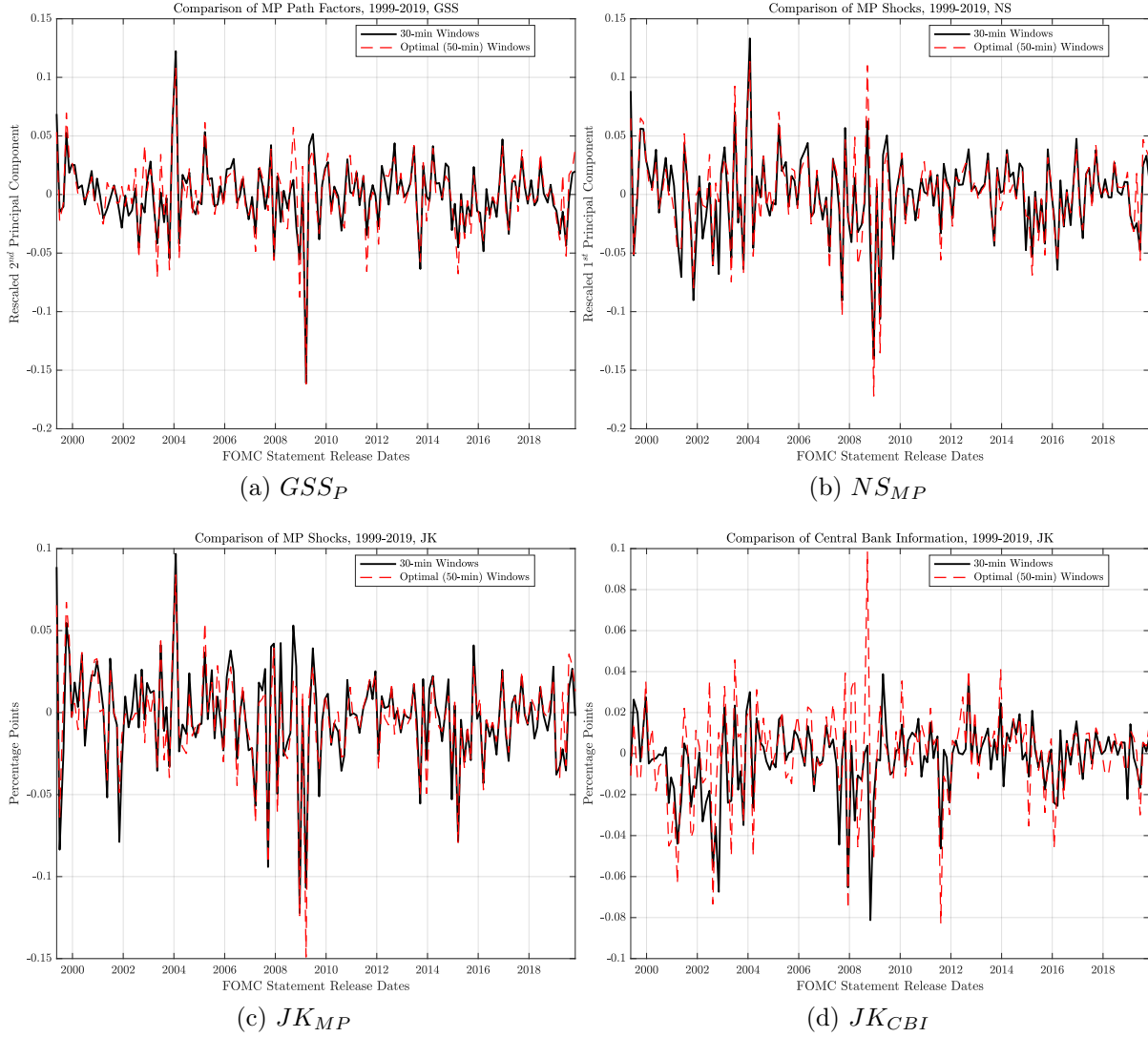


Figure 8: Comparing Monetary Policy Shock Series Derived from Optimal Window Length v. 30 Minutes

Notes: For each sub-figure, the horizontal axis represents FOMC statement release dates. The vertical axis depicts percentage points after rescaling each shock to be one-for-one with the daily change in the zero-coupon, nominal one-year Treasury yield. For all construction methods, the black-solid and red-dotted lines represent the shocks derived from surprises measured within 30 minutes and the median optimal event window length of 50 minutes, respectively.

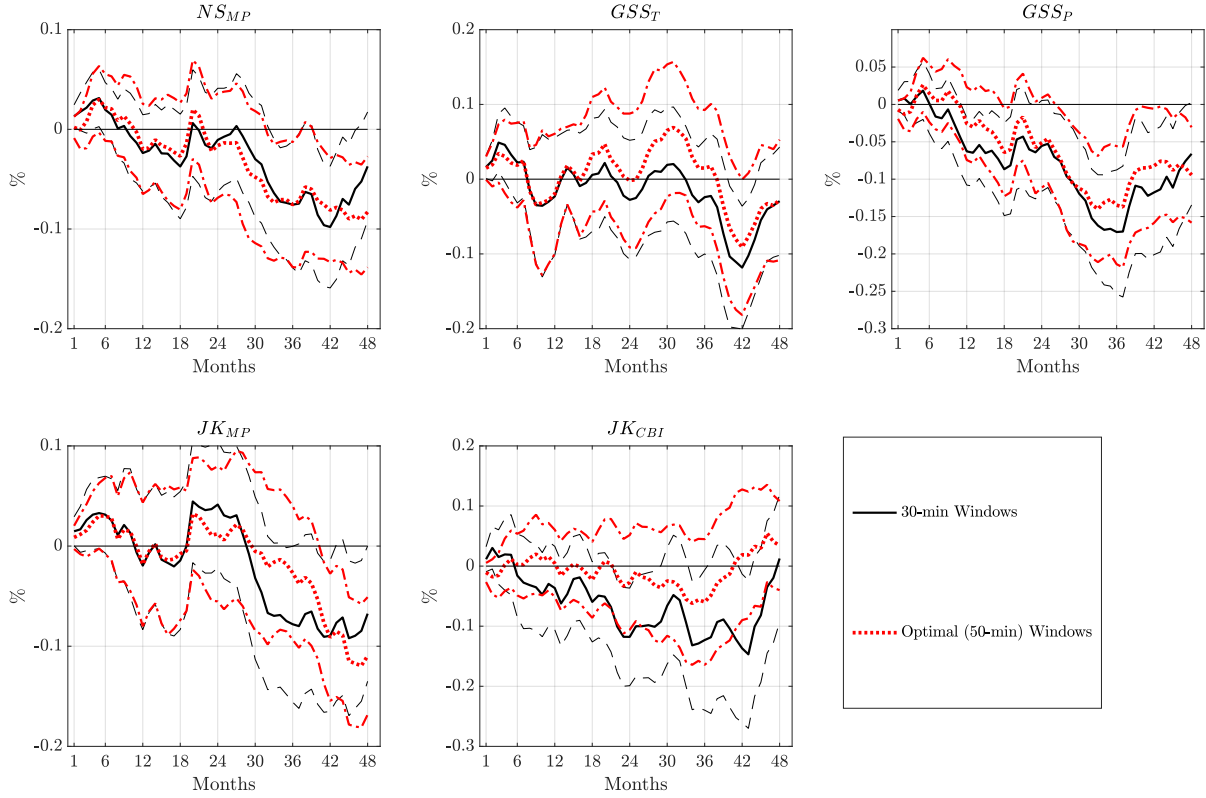


Figure 9: Effects of Event Window Choice on CPI Impulse Responses

Notes: Impulse responses are calculated using the lag-augmented local projection from Olea and Plagborg-Møller (2021). Confidence bands are at the 90% level. Standard errors are Eicker-Huber-White heteroscedasticity-robust standard errors. The responses are to a 100 basis point increase in the monetary policy shock series. The monetary policy shocks are constructed using the full set of monetary policy surprise instruments.

Cosine Similarity of Sequential FOMC Statements

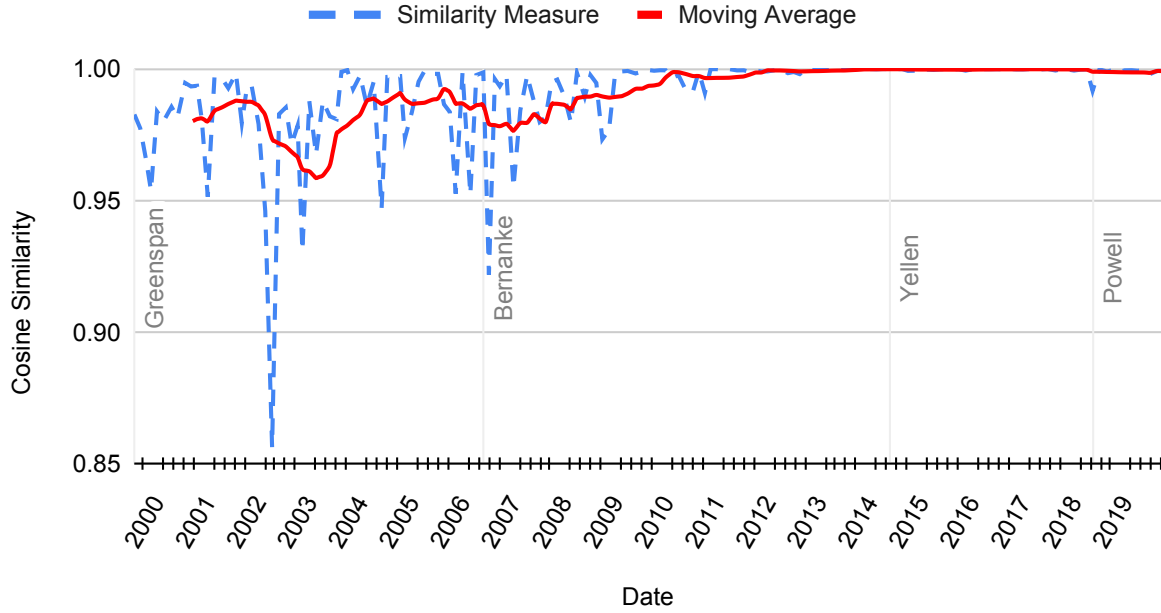


Figure 10: S^1 Cosine Similarity of Sequential FOMC Statements

Notes: S^1 represents the sequential cosine similarity between the dense contextual embedding of an FOMC statement and that of the immediately previous FOMC statement. The underlying representations are extracted using XLNet-Base that has not been fine-tuned to approximate the underlying relationship between monetary policy announcements and market price reactions. Using only the pre-trained neural network ensures the semantic signals are entirely out-of-sample and orthogonal to market price reactions. Cosine similarity values approaching one mean the statements share near-identical contextual prose and policy concepts, while lower values indicate a structural shift or informational update in communication. The moving average in solid red is calculated with a period of 10.

	30-minute Window					Optimal Window					Difference		
	ΔTY_1	ΔTY_2	ΔTY_5	ΔTY_{10}		ΔTY_1	ΔTY_2	ΔTY_5	ΔTY_{10}		ΔTY_2	ΔTY_5	ΔTY_{10}
GSS_T	1.00*** (0.28)	0.82*** (0.34)	0.15 (0.43)	-0.37 (0.41)		1.00*** (0.23)	0.78*** (0.28)	0.08 (0.33)	-0.42 (0.32)		-0.04	-0.07	-0.05
GSS_P	1.00*** (0.09)	1.46*** (0.09)	1.89*** (0.25)	1.64*** (0.36)		1.00*** (0.11)	1.51*** (0.10)	1.93*** (0.21)	1.66*** (0.30)		+0.05	+0.04	+0.02
NS_{MP}	1.00*** (0.07)	1.24*** (0.09)	1.29*** (0.19)	0.95*** (0.25)		1.00*** (0.09)	1.30*** (0.10)	1.39*** (0.18)	1.06*** (0.24)		+0.06	+0.11	+0.11
JK_{MP}	1.00*** (0.11)	1.30*** (0.15)	1.39*** (0.28)	0.99*** (0.36)		1.00*** (0.12)	1.35*** (0.16)	1.52*** (0.32)	1.16*** (0.43)		+0.04	+0.13	+0.17
JK_{CBI}	1.00*** (0.25)	1.04*** (0.30)	1.00*** (0.31)	0.82*** (0.29)		1.00*** (0.23)	1.20*** (0.25)	1.14*** (0.27)	0.85*** (0.26)		+0.16	+0.14	+0.03

Table 1: Differences in Responses of Nominal Interest Rates to Shocks from Event Window Choice

Notes: Each estimate comes from a separate OLS regression. For each regression, the dependent variable is the one-day change in end-of-day Treasury yields, represented by columns. The independent variable is the change in monetary policy shock series over a given event window around the time of scheduled FOMC announcements. Starting from the left, columns 2–5 represent the OLS regressions performed on shock series constructed within 30-minute windows. Columns 6–9 are equivalent, but for shocks constructed within the median optimal window length of 50 minutes. The sample period consists of all scheduled FOMC statement release dates from May 1999 through October 2019, resulting in 165 observations. Newey and West (1987) standard errors are reported in parentheses. A constant is included in all specifications but not reported. Columns 10–12 represent the differences between the coefficients of the regressions using the optimal windows v. those using 30-minute windows. *** sig. at the 1% level.

	30-minute Window			Optimal Window			Difference		
	$\Delta TIPS_2$	$\Delta TIPS_5$	$\Delta TIPS_{10}$	$\Delta TIPS_2$	$\Delta TIPS_5$	$\Delta TIPS_{10}$	$\Delta TIPS_2$	$\Delta TIPS_5$	$\Delta TIPS_{10}$
GSS_T	-0.81 (1.44)	0.02 (0.51)	-0.19 (0.45)	-0.90 (1.57)	0.09 (0.46)	-0.16 (0.37)	-0.09	+0.07	+0.03
GSS_P	2.21*** (0.49)	1.96*** (0.46)	1.74*** (0.44)	2.20*** (0.37)	2.03*** (0.38)	1.75*** (0.37)	-0.00	+0.06	+0.01
NS_{MP}	1.17 (0.73)	1.29*** (0.36)	1.08*** (0.30)	1.31** (0.63)	1.47*** (0.31)	1.20*** (0.28)	+0.14	+0.18	+0.13
JK_{MP}	1.40* (0.83)	1.40*** (0.47)	1.15*** (0.41)	1.66*** (0.63)	1.64*** (0.49)	1.38*** (0.46)	+0.26	+0.24	+0.23
JK_{CBI}	0.51 (0.85)	0.99*** (0.33)	0.85*** (0.25)	0.60 (0.92)	1.13*** (0.33)	0.84*** (0.25)	+0.09	+0.14	-0.01

Table 2: Differences in Responses of Real Interest Rates to Shocks from Event Window Choice

Notes: Each estimate comes from a separate OLS regression. For each regression, the dependent variable is the one-day change in end-of-day TIPS yields, represented by columns. The independent variable is the change in monetary policy shock series over a given event window around the time of scheduled FOMC announcements. Starting from the left, columns 2–4 represent the OLS regressions performed on shock series constructed within 30-minute windows. Columns 5–7 are equivalent, but for shocks constructed within the median optimal window length of 50 minutes. The sample period consists of all scheduled FOMC statement release dates from May 1999 to October 2019, resulting in 165 observations. Newey and West (1987) standard errors are reported in parentheses. A constant is included in all specifications but not reported. Columns 8–10 represent the differences between the coefficients of the regressions using the optimal windows v. those using 30-minute windows. *** sig. at the 1% level.

	30-minute Window			Optimal Window			Difference		
	ΔBEI_2	ΔBEI_5	ΔBEI_{10}	ΔBEI_2	ΔBEI_5	ΔBEI_{10}	ΔBEI_2	ΔBEI_5	ΔBEI_{10}
GSS_T	1.63 (1.46)	0.13 (0.29)	-0.18 (0.17)	1.67 (1.58)	-0.01 (0.29)	-0.26 (0.17)	+0.05	-0.14	-0.08
GSS_P	-0.75 (0.46)	-0.08 (0.24)	-0.10 (0.12)	-0.69* (0.36)	-0.10 (0.23)	-0.09 (0.12)	+0.06	-0.02	+0.01
NS_{MP}	0.07 (0.70)	-0.01 (0.23)	-0.13 (0.12)	-0.01 (0.65)	-0.07 (0.23)	-0.14 (0.12)	-0.08	-0.07	-0.01
JK_{MP}	-0.09 (0.81)	-0.01 (0.29)	-0.17 (0.13)	-0.31 (0.61)	-0.12 (0.25)	-0.22** (0.11)	-0.22	-0.11	-0.05
JK_{CBI}	0.54 (0.76)	0.01 (0.29)	-0.02 (0.20)	0.60 (0.86)	0.01 (0.30)	0.02 (0.23)	+0.07	+0.00	+0.04

Table 3: Differences in Responses of Break-even Inflation to Shocks from Event Window Choice

Notes: Each estimate comes from a separate OLS regression. For each regression, the dependent variable is the one-day change in end-of-day break-even inflation, represented by columns. The independent variable is the change in monetary policy shock series over a given event window around the time of scheduled FOMC announcements. Starting from the left, columns 2–4 represent the OLS regressions performed on shock series constructed within 30-minute windows. Columns 5–7 are equivalent, but for shocks constructed within the median optimal window length of 50 minutes. The sample period consists of all scheduled FOMC statement release dates from May 1999 to October 2019, resulting in 165 observations. Newey and West (1987) standard errors are reported in parentheses. A constant is included in all specifications but not reported. Columns 8–10 represent the differences between the coefficients of the regressions using the optimal windows v. those using 30-minute windows. * sig. at the 10% level. ** sig. at the 5% level. *** sig. at the 1% level.

	30-minute Window			Optimal Window			Percentage Difference		
	DP_{SPX}	DP_{ESc1}	DP_{ESc2}	DP_{SPX}	DP_{ESc1}	DP_{ESc2}	DP_{SPX}	DP_{ESc1}	DP_{ESc2}
GSS_T	-8.40** (2.71)	-8.83*** (2.68)	-7.25*** (2.78)	-7.39** (3.10)	-7.43** (3.15)	-7.34** (3.12)	-11.99%	-15.92%	+1.25%
GSS_P	-6.14** (1.81)	-6.27*** (1.83)	-6.12*** (1.76)	-6.85** (2.88)	-6.96** (2.91)	-7.63*** (2.81)	+11.51%	+11.00%	+24.61%
NS_{MP}	-6.92** (1.32)	-7.15*** (1.37)	-6.51*** (1.31)	-7.00*** (1.85)	-7.10*** (1.89)	-7.55*** (1.84)	+1.23%	-1.00%	+16.00%
JK_{MP}	-14.76*** (0.81)	-15.08*** (0.91)	-13.73*** (0.94)	-17.46*** (1.04)	-17.77*** (1.08)	-17.30*** (1.06)	+18.25%	+17.88%	+26.00%
JK_{CBI}	15.19*** (2.29)	13.84*** (2.35)	15.18*** (2.39)	14.08*** (2.11)	14.44*** (2.14)	12.12*** (2.07)	-7.36%	-4.90%	-12.43%

Table 4: Differences in Responses of Stock Prices to Shocks from Event Window Choice

Notes: Each estimate comes from a separate OLS regression. For each regression, the dependent variable is the price log-difference of the S&P 500 Index or E-mini futures, represented by columns. The independent variable is the change in monetary policy shock series over a given event window around the time of scheduled FOMC announcements. Starting from the left, columns 2–4 represent the OLS regressions performed on shock series constructed within 30-minute windows. Columns 5–7 are equivalent, but for shocks constructed within the median optimal window length of 50 minutes. The sample period consists of all scheduled FOMC statement release dates from May 1999 to October 2019, resulting in 165 observations. Newey and West (1987) standard errors are reported in parentheses. A constant is included in all specifications but not reported. Columns 8–10 represent the percentage differences between the coefficients of the regressions using the optimal windows v. those using 30-minute windows, where positive (negative) values represent a stronger (weaker) effect in the same direction. *** sig. at the 1% level.

Metric	Simple	Complicated	Different	Similar	Unity	Dissents
<i>Minimised MSE</i> Average	1.25e-5	1.06e-5	9.99e-6	9.66e-6	9.57e-6	1.43e-5
<i>Event Window Length (Minutes)</i> Average	60	71	81	52	53	83

Table 5: Average MSEs and Event Window Lengths Using the “One Signal” Approach, Conditioned by FOMC Statement Complexity, Similarity, and Presence of Dissents

Notes: The complexity of FOMC statements is measured by the Flesch-Kincaid Grade Level, defined as: $0.39 \times \text{average sentence length} + 11.8 \times \text{average number of syllables per word} - 15.59$, and displayed in the first two columns. Changes in FOMC statements are measured using a pairwise-statement cosine similarity measure and displayed in the third and fourth columns from the left. The event window lengths are displayed in minutes. “Simple” statements have grade levels up to 16.5. “Complicated” statements have grade levels above 16.5. “Different” are sequential statements with a cosine similarity of less than to 0.994. “Similar” are sequential statements with a cosine similarity of more than 0.994. “Unity” statements are those without votes of dissent. “Dissents” are statements with recorded dissent votes. For all futures contracts, event window lengths considered as outliers under the “one signal” approach are set equal to the median of the sub-set window lengths in order to lessen their effects.

A Motivating Framework Simulations

This appendix details the simulation for the motivating framework in Section 2. I simulate the asset price process for multiple news announcements to illustrate how cognitive noise and unrelated news affect the optimal time horizon. The simulation also demonstrates why a “good” signal is necessary for this estimation, providing a rationale for the paper’s methodology.

The simulation assumes news is released at time $t = 0$. The fundamental price, $P_{i,t}^f$, jumps to a random value $P_i^f \in [-100, 100]$ and remains there. Simultaneously, the cognitive noise component jumps to a random value in $[-100, 100]$, while the unrelated news component is zero. The econometrician observes a noisy signal, s_i , with a normally distributed error (mean zero, finite variance). I simulate this process for $N = 10,000$ news announcements up to $t = 100$. For each simulation, I calculate both the true MSE (using P_i^f) and the estimated MSE (using s_i). The true optimal time (t^*) and the estimated optimal time ($\hat{t}$) are the points that minimise these respective MSEs, per Equation 4.

A.1 Simulation Results

I consider three scenarios for the asset price response: (1) high cognitive noise and low unrelated news, (2) low cognitive noise and high unrelated news, and (3) high levels of both noise components. The simulation parameters for these three scenarios are listed in the topmost rows of Appendix Table E1.

The bottom rows of Appendix Table E1 show the results. A market with high cognitive noise and low unrelated news (left column) has a longer optimal time horizon (t^*). Conversely, a market with high unrelated news and low cognitive noise (middle column) has a very small t^* . These results are consistent with Equation 3. Intuitively, a longer reaction process requires a longer optimal time horizon. In contrast, when markets react quickly but unrelated news accumulates, a short window is optimal. The rightmost column shows the most realistic scenario, which includes both cognitive and unrelated noise. Here, the optimal time horizon lies between the two extremes.

Importantly, the simulations show minimal differences between the true (t^*) and estimated ($\hat{t}$) time horizons, even when using only the noisy signal s_i . This demonstrates that a “good” signal allows for a precise estimation of the market’s full reaction time.

This motivating framework, however, relies on strong linearity and orthogonality assumptions to isolate the effects of noise. In reality, the true compositions and processes of the fundamental and noise components are unknown, yet they still drive the overall asset price.

Abstracting from this simple framework, I argue the paper’s neural network approach can extract a “good” signal by approximating the mapping between communications and price changes, regardless of the true, unobserved noise processes.

B The Primary Accuracy Metric for Judging XLNet

As mentioned in Subsection 4.4, the primary metric for neural network fine-tuning is a generalised R^2 statistic from Hawinkel, Waegeman, and Maere (2024):

$$R_{OOS}^2 = 1 - \frac{\widehat{MSE}}{\widehat{MST}} = 1 - \frac{T^{-1} \sum_{i=1}^T (y_i - \hat{y}_i)^2}{\frac{T+1}{T(T-1)} \sum_{i=1}^T (y_i - \overline{y_{IS}})^2}, \quad (\text{B1})$$

where $\widehat{MSE}$ is the neural network’s out-of-sample mean squared error; $\widehat{MST}$ is the out-of-sample MSE from a naive model that predicts using only the in-sample mean ($\overline{y_{IS}}$); and T is the testing sample size. While the network’s explicit objective function is to minimise $\widehat{MSE}$, this action simultaneously maximises R_{OOS}^2 .

This metric is chosen for two main reasons. First, the conventional R^2 (i.e., the proportion of variance explained by a model) breaks down for non-linear methods like neural networks. For such models, the total variance no longer neatly decomposes into model and residual components, meaning the squared Pearson correlation coefficient does not equal R^2 . Second, the R_{OOS}^2 formula is specifically designed to assess out-of-sample performance, not in-sample fit.⁴⁸

As is standard in machine learning, model performance is compared to a baseline. This baseline assumes *no* relationship between the FOMC statement text and futures price changes, naively predicting with the in-sample average. Therefore, the R_{OOS}^2 measures the reduction in predictive error achieved by the neural network compared to this naive baseline.

During fine-tuning, XLNet-Base also tracks other metrics: the out-of-sample Pearson correlation between predicted and actual values, the out-of-sample mean absolute error, and the in-sample mean squared error. This last metric is important for verifying that the network is genuinely learning the underlying relationship.

The decisive criterion for systematic estimation is the out-of-sample R^2 *averaged across sample splits* for each event window length and futures contract:

$$\overline{R_{OOS}^2} = \frac{\sum_{i=1}^K R_{OOS}^2}{K}, \quad (\text{B2})$$

where $K = 5$ is the number of sample splits. This $\overline{R_{OOS}^2}$ statistic measures the neural network’s average performance improvement within a given window, relative to the naive baseline. The optimal event window is therefore the length that yields the largest $\overline{R_{OOS}^2}$. This is the point where the network’s predictive power and generalisability is highest, which only occurs when the impact of noise is minimised and the fundamental reaction is fully captured.

⁴⁸Using a conventional R^2 yields similar optimal window lengths and network quality, but R_{OOS}^2 is a more appropriate metric.

C Construction of High-Frequency Monetary Surprises

C.1 Financial Data Overview and Price Selection

The intraday data for interest-rate and equity futures (May 1999 to October 2019) is from the Thomson Reuters Tick History database via LSEG. Table E2 provides an overview of this data. For each futures contract, I use a minutely price series. The remainder of this appendix details the construction of interest-rate surprises from these contracts. Following the literature, I use federal funds futures for expectations up to three months, Eurodollar futures for two to four quarters, and Treasury futures for expectations out to fifteen years.

As discussed in Subsection 3.2, I collect prices at 10-minute intervals, from 10 minutes before to 18 hours after an FOMC statement release. Because trades do not occur at every exact interval, I apply two rules. For the pre-announcement price ($t - 10$), I use the last recorded trade *at or before* that timestamp. For all post-announcement intervals (e.g., $t + 10, t + 20$), I use the first recorded trade *at or after* that interval's timestamp. This method assumes the price is constant between recorded trades.

C.2 Surprises from Federal Funds Futures

A federal funds rate futures (FFF) contract's settlement price is based on the average effective federal funds rate over its expiry month. The contract pays out 100 minus this average rate. Therefore, the market's expected average federal funds rate is implied by its price. Let $p_{\tau,t}^{FFj}$ be the price of the $(j - 1)$ month-ahead FFF at time t for FOMC meeting τ . The expected average federal funds rate is then calculated as $100 - p_{\tau,t}^{FFj}$.

C.2.1 Surprises from Federal Funds Futures: Current Meeting

I calculate the “current meeting” monetary policy surprise ($mp1$) for each FOMC announcement τ at time $t + n$ as:

$$mp1_{\tau,t+n} = \frac{m}{m-d} (FF1_{\tau,t+n} - FF1_{\tau,t-10}), \quad (B3)$$

where $FF1$ is the implied federal funds rate from the current-month contract ($100 - p_{\tau,t}^{FF1}$), d is the day of the announcement, and m is the number of days in the month.

A special case applies if the FOMC announcement occurs in the last seven days of the month ($m - d + 1 \leq 7$). To avoid distortion from the large $\frac{m}{m-d}$ scaling factor, the surprise is calculated as the simple change in the next month's futures contract: $mp1_{\tau,t+n} = FF2_{\tau,t+n} - FF2_{\tau,t-10}$.

C.2.2 Surprises from Federal Funds Futures: Next Meeting

The surprise for the next scheduled FOMC meeting ($mp2$) is more complex, as it must isolate the new information for that specific meeting. First, the relevant futures contract (j) is identified, which corresponds to the month of the next scheduled meeting ($j \in \{2, 3, 4\}$).

The surprise is then calculated as:

$$mp2_{\tau,t+n} = \frac{m_2}{m_2 - d_2} \left\{ [FFj_{\tau,t+n} - FFj_{\tau,t-10}] - \frac{d_2}{m_2} mp1_{\tau,t+n} \right\}, \quad (\text{B4})$$

where FFj is the implied rate from the relevant contract, d_2 and m_2 are the day of and total days in the month of the next meeting, and $mp1_{\tau,t+n}$ is the current-meeting surprise.

As with the current-meeting surprise, a special case applies if the next meeting falls in the last seven days of its month ($m_2 - d_2 + 1 \leq 7$). To avoid distortion, the surprise is calculated as the simple change in the following month's contract: $mp2_{\tau,t+n} = FF(j+1)_{\tau,t+n} - FF(j+1)_{\tau,t-10}$.

C.3 Surprises from Eurodollar Futures

Eurodollar contracts are quarterly contracts (expiring in March, June, September, or December) that pay out 100 minus the US Dollar BBA LIBOR interest rate at expiration. Let $p_{\tau,t}^{edj}$ be the price of the j^{th} closest quarterly futures contract at time t on FOMC meeting date τ .

The implied interest rate is $edj_{\tau,t} = 100 - p_{\tau,t}^{edj}$. The monetary policy surprise for this contract, mp_{edj} , is calculated as the simple change in this implied rate within the event window:

$$mp_{edj,\tau,t+n} = edj_{\tau,t+n} - edj_{\tau,t-10}. \quad (\text{B5})$$

C.4 Surprises from Treasury Futures

Treasury futures contracts oblige the seller to deliver Treasury bonds within a specific maturity range at expiration. Let $p_{\tau,t}^{tkj}$ be the price of the j^{th} nearest quarterly k -year Treasury futures contract at time t on FOMC meeting date τ .

The implied yield surprise ($tk_{\tau,t+n}$) for an event window ending at $t+n$ is calculated by dividing the price change by its approximate maturity (l) and flipping the sign:

$$tk_{\tau,t+n} = - \left(p_{\tau,t+n}^{tkj} - p_{\tau,t-10}^{tkj} \right) / l, \quad (\text{C.1})$$

where the approximate maturities, $l \in \{2, 4, 7, 15\}$, are from Gürkaynak, Kisacikoglu, and Wright (2020). Following Gorodnichenko and Ray (2017), if the FOMC meeting occurs during an expiry month (March, June, etc.), I use the next-nearest contract ($j+1$) due to its higher liquidity.

D Robustness Checks

D.1 Original Surprise Instruments for MP Surprises & Shocks

This appendix repeats the exercises from Subsection 6.2, but constructs the monetary policy shocks using the *original* instrument sets from the cited authors.⁴⁹

Visual Shock Differences I plot the shock series constructed from the original instrument sets in Figure E28. The axes and scaling in each sub-figure match the original authors’ specifications.

The results are robust. The visual differences in the forward guidance component of monetary policy shocks caused by the event window choice are similar to those in the main text (Figure 8), which used the full instrument set. This robustness also applies to the JK_{CBI} shock, which continues to show “shifts of importance” in the optimal window, a result of the Jarociński and Karadi (2020) sign restriction method.

Responses of Interest Rates Appendix Tables E10 and E11 present the regression results for nominal and real interest rates, respectively, following the specification in Equation 8. The results are robust. Opening the event window to the optimal 50 minutes yields stronger point estimates (in basis points) for all monetary policy shocks containing forward guidance. Furthermore, using the optimal window also increases the statistical significance of several coefficient estimates, such as the 2-year TIPS yield response to GSS_P or the nominal Treasury yield responses to the JK_{CBI} shock.

Responses of Break-even Inflation Regression results for break-even inflation using the original surprise instruments are in Appendix Table E12. The results are robust to the instrument choice. Similar to the main text, monetary policy shocks constructed within the optimal window have more negative impacts on break-even inflation at all maturities. Furthermore, moving from the 30-minute to the optimal window again increases the statistical significance for JK_{MP} at the furthest horizon, although most estimates are statistically insignificant.

Responses of Equities I next examine the impact of monetary policy shocks on equity prices, again using the original instrument sets. Appendix Table E13 presents analogous results for the S&P 500 Index and its E-mini futures, following the specification in Equation 8, with the dependent variable as 100 times the log-price difference.

Compared to the main results (Subsection 6.2), the effects of the window choice are more mixed for equities. For example, while the optimal window strengthens the impact of GSS_P on the S&P 500 Index by 2.52%, it weakens the impact of NS_{MP} by 3.32%. Although all shocks remain statistically significant and have the expected sign, the overall effects of the window choice are smaller than in the main text. In contrast, the results for the second-month S&P 500 E-mini futures contract remain similar to those in the main text.

The overall results suggest that the original instrument sets, which rely on shorter-maturity surprises, may understate the shock effects. This limitation is particularly relevant as my sample period includes the effective lower bound, where forward guidance is not fully captured by surprises with maturities of only one year. This finding aligns with recent

⁴⁹For consistency with the other shock constructions, I use the next-meeting surprise ($mp2_\tau$) to build the Jarociński and Karadi (2020) shocks, as it represents a similar underlying maturity to their original $FF4$ instrument.

studies (e.g., Brennan et al., 2024; An, Stedman, and Lusompa, 2025; and others) which show that incorporating longer-maturity instruments is necessary to prevent understating monetary policy shocks in this era.

Responses of the Macroeconomy via Local Projections I estimate the macroeconomic impulse responses using the lag-augmented local projection method of Olea and Plagborg-Møller (2021). The model estimates the responses of log consumer price index, log industrial production, the nominal two-year Treasury yield, and the excess bond premium (EBP) to each monetary policy shock, following the specification in Equation 9.

Figures E29–E32 plot the impulse responses. Similar to the main results, the event window choice does not qualitatively change most point estimates, though some exceptions exist (e.g., the EBP response to the NS_{MP} shock becomes statistically significant and positive after 36 months).

Also consistent with the main findings, the confidence intervals of the impulse responses from the optimal window are visibly smaller. To formalise this, I calculate the confidence interval width ratios as discussed in Subsection 6.2. Using the original instrument set, the average and median ratios are 0.9033 and 0.9506, respectively. This confirms that using the optimal event window length yields more precise estimates of the macroeconomic responses, even with the original instrument set.

D.2 Optimal Event Window Lengths Beyond 70 Minutes?

The “one signal” approach also allows for a crude check of whether a larger $\overline{R_{OOS}^2}$ exists beyond the 70-minute window used in the “joint” estimation. Appendix Figures E21–E27 display this exercise, extending the event window out to 18 hours after the FOMC statement release. For most assets, the event window with the largest $\overline{R_{OOS}^2}$ is the same one found via the “joint” approach, supporting the overall validity of the 70-minute estimation limit.

However, for a few contracts (e.g., $FF2$, $FF4$, $TUc1$, $TYc2$, and $USc1$), the “one signal” approach does suggest an optimal window longer than 70 minutes. To confirm these, I calculate the $\overline{R_{OOS}^2}$ using the full “joint” approach for these specific longer windows. This check yields two insights. First, the true optimal window for these contracts is still within 60 minutes.

Second, even though the “one signal” optimum was incorrect, the “joint” estimation for these longer windows (beyond 60 minutes) still yielded a higher $\overline{R_{OOS}^2}$ than the conventional 30-minute window. This suggests the global optimum for some assets may lie somewhere between 60 minutes and the time predicted by the “one signal” approach. Regardless, this exercise confirms that a 30-minute window is never optimal.

Appendix Figures and Tables

Google Trends for FOMC Meeting Terms

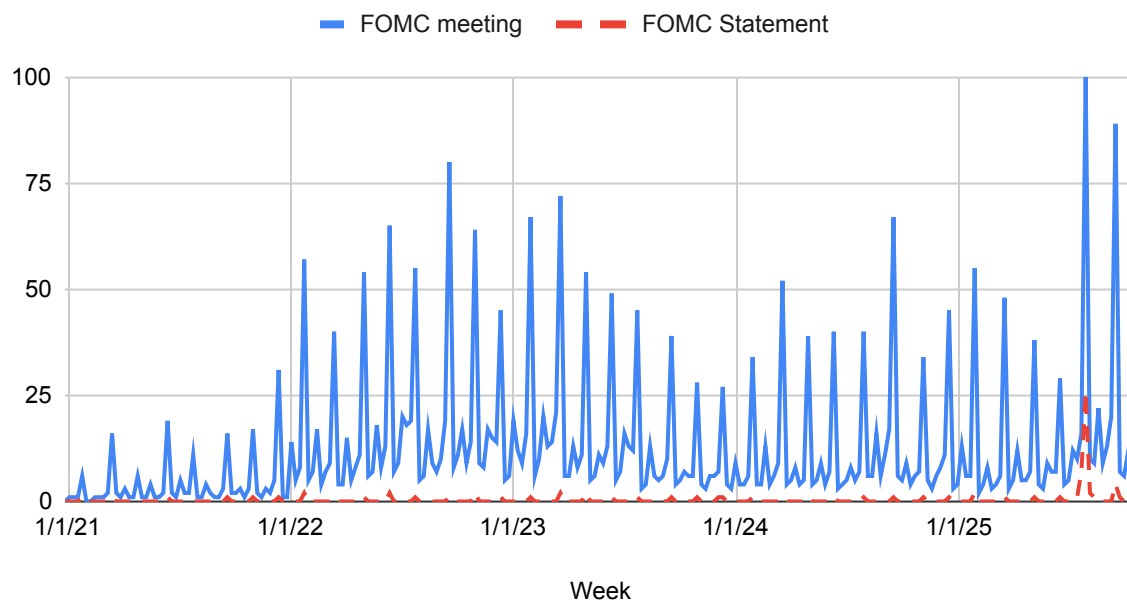
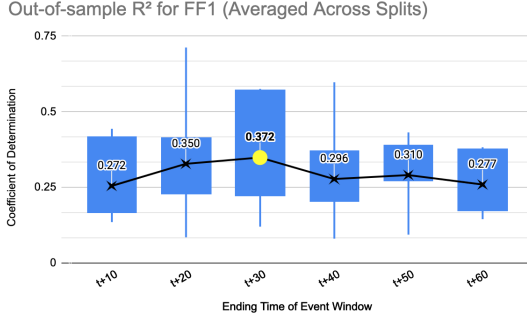
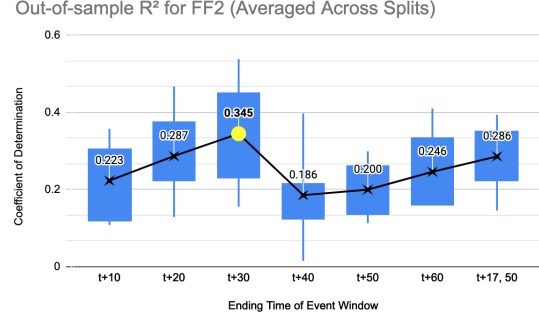


Figure E1: Google Trends Interest About FOMC Statements over Time, January 2021–October 2025

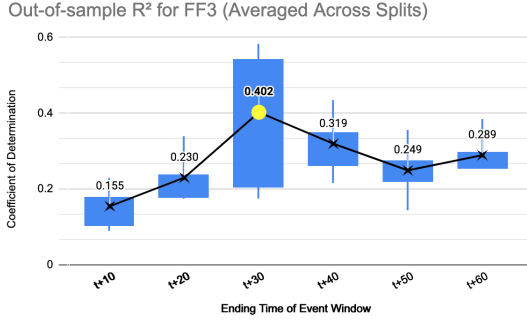
Notes: The blue and solid (red and dashed) line represents the search interest for the phrase “FOMC meeting” (“FOMC statement”) in the U.S.



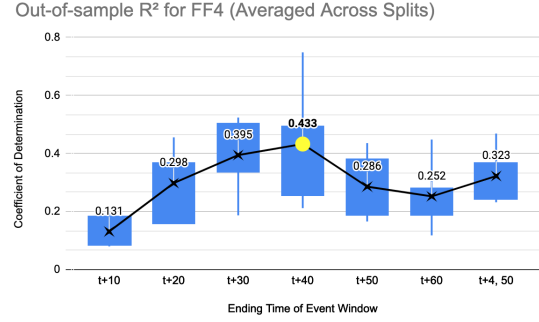
(a) $FF1$



(b) $FF2$

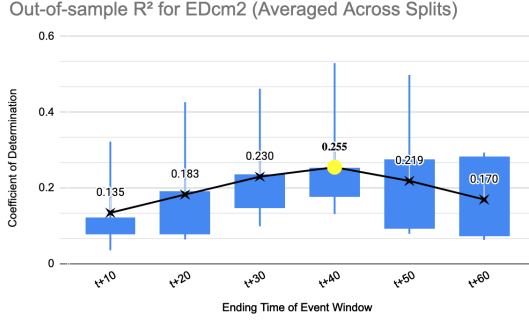


(c) $FF3$

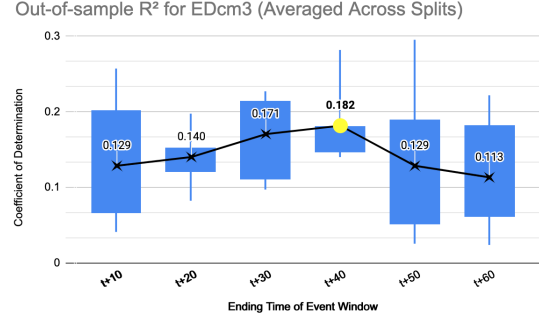


(d) $FF4$

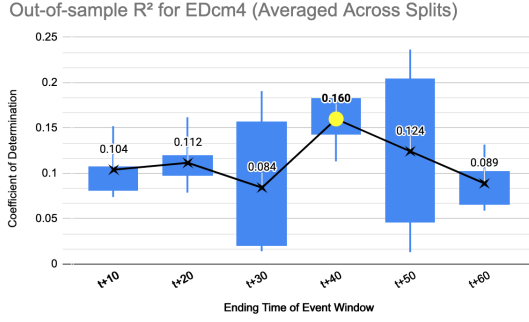
Figure E2: Optimal Event Window Lengths for Federal Funds Futures
Notes: The horizontal axis of each figure depicts the end time of the considered event window lengths and the vertical axis represents the coefficient of determination, R^2_{OOS} . The cross points represent $\overline{R^2_{OOS}}$ for each event window size, where the solid yellow point represents the event window with the largest $\overline{R^2_{OOS}}$. For each event window, box-and-whisker plots are shown surrounding the corresponding averages. $\overline{R^2_{OOS}}$ measures the average reduction in predictive error achieved by the neural network compared to the naive baseline.



(a) $EDcm2$



(b) $EDcm3$

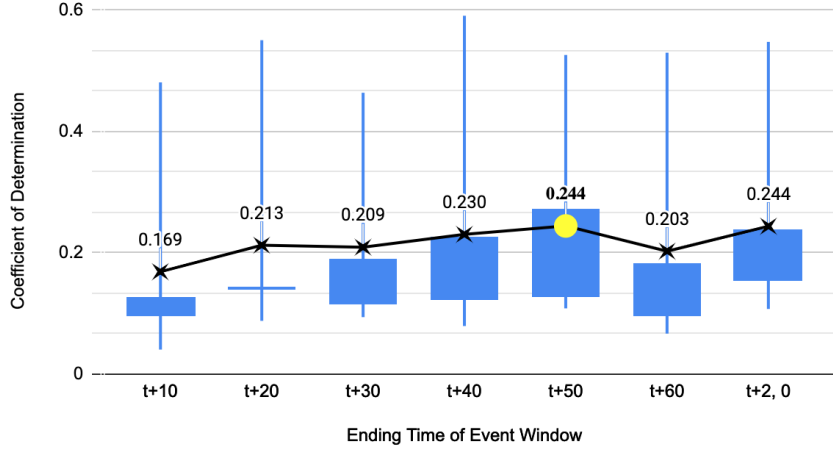


(c) $EDcm4$

Figure E3: Optimal Event Window Lengths for Eurodollar Futures

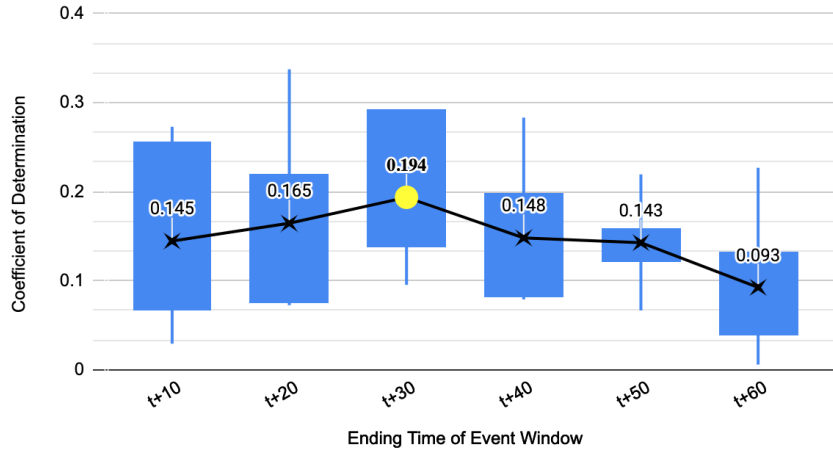
Notes: The horizontal axis of each figure depicts the end time of the considered event window lengths and the vertical axis represents the coefficient of determination, R^2_{OOS} . The cross points represent $\overline{R^2_{OOS}}$ for each event window size, where the solid yellow point represents the event window with the largest $\overline{R^2_{OOS}}$. For each event window, box-and-whisker plots are shown surrounding the corresponding averages. $\overline{R^2_{OOS}}$ measures the average reduction in predictive error achieved by the neural network compared to the naive baseline.

Out-of-sample R^2 for TUC1 (Averaged Across Splits)



(a) *TUC1*

Out-of-sample R^2 for TUC2 (Averaged Across Splits)

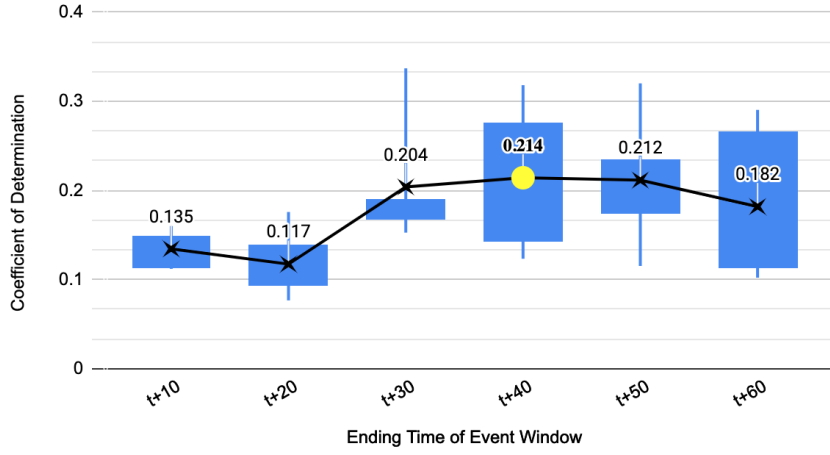


(b) *TUC2*

Figure E4: Optimal Event Window Lengths for 2-Year Treasury Futures

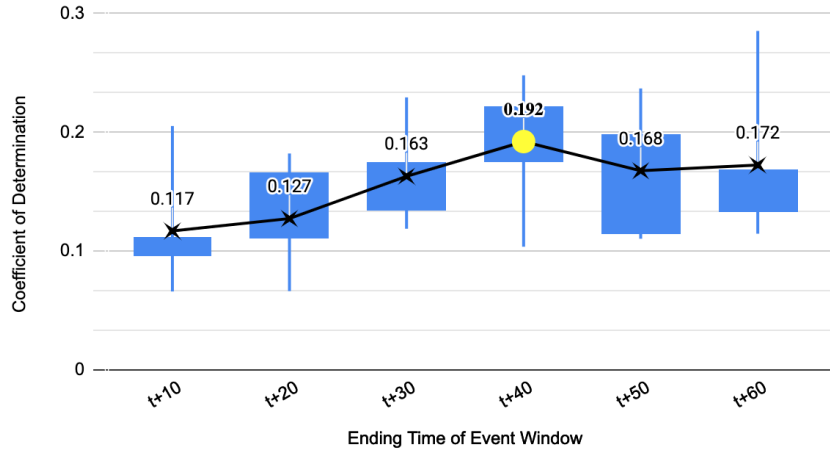
Notes: The horizontal axis of each figure depicts the end time of the considered event window lengths and the vertical axis represents the coefficient of determination, R^2_{OOS} . The cross points represent $\overline{R^2_{OOS}}$ for each event window size, where the solid yellow point represents the event window with the largest $\overline{R^2_{OOS}}$. For each event window, box-and-whisker plots are shown surrounding the corresponding averages. $\overline{R^2_{OOS}}$ measures the average reduction in predictive error achieved by the neural network compared to the naive baseline.

Out-of-sample R^2 for FVc1 (Averaged Across Splits)



(a) $FVc1$

Out-of-sample R^2 for FVc2 (Averaged Across Splits)

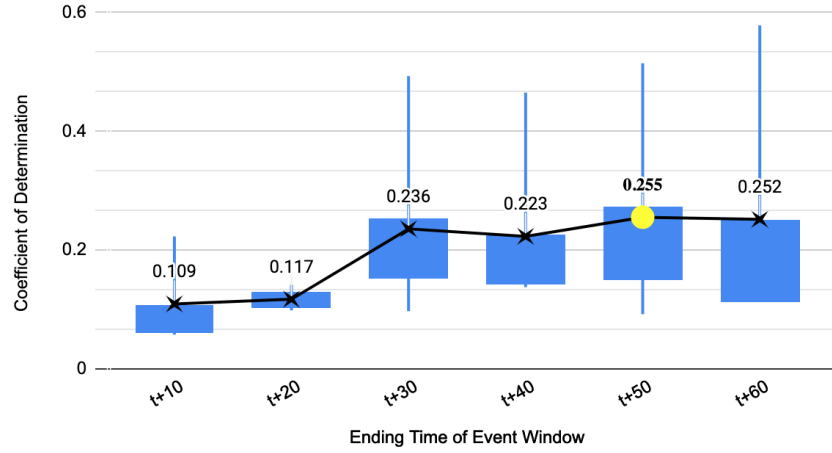


(b) $FVc2$

Figure E5: Optimal Event Window Lengths for 5-Year Treasury Futures

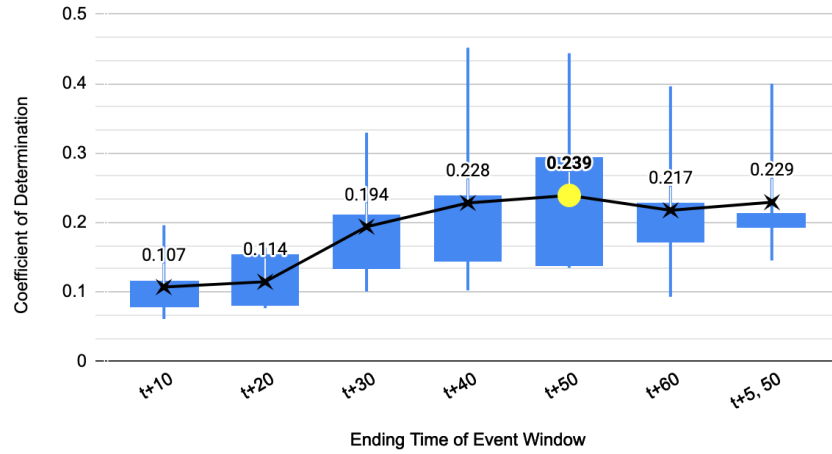
Notes: The horizontal axis of each figure depicts the end time of the considered event window lengths and the vertical axis represents the coefficient of determination, R^2_{OOS} . The cross points represent $\overline{R^2_{OOS}}$ for each event window size, where the solid yellow point represents the event window with the largest $\overline{R^2_{OOS}}$. For each event window, box-and-whisker plots are shown surrounding the corresponding averages. $\overline{R^2_{OOS}}$ measures the average reduction in predictive error achieved by the neural network compared to the naive baseline.

Out-of-sample R^2 for TYc1 (Averaged Across Splits)



(a) *TYc1*

Out-of-sample R^2 for TYc2 (Averaged Across Splits)

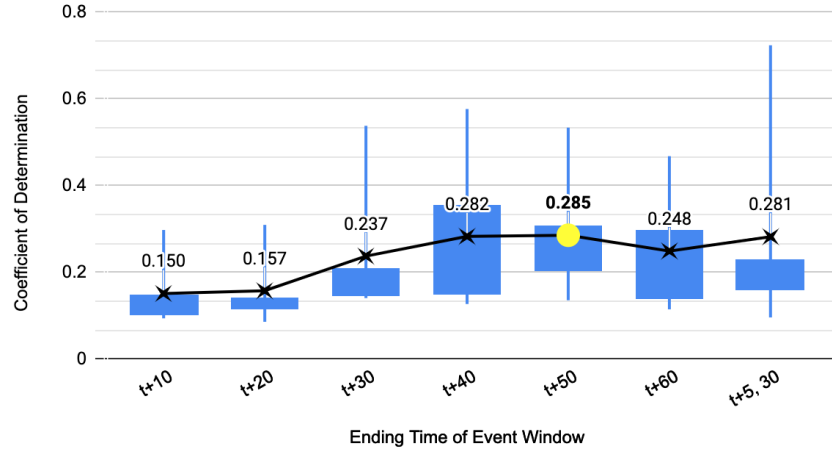


(b) *TYc2*

Figure E6: Optimal Event Window Lengths for 10-Year Treasury Futures

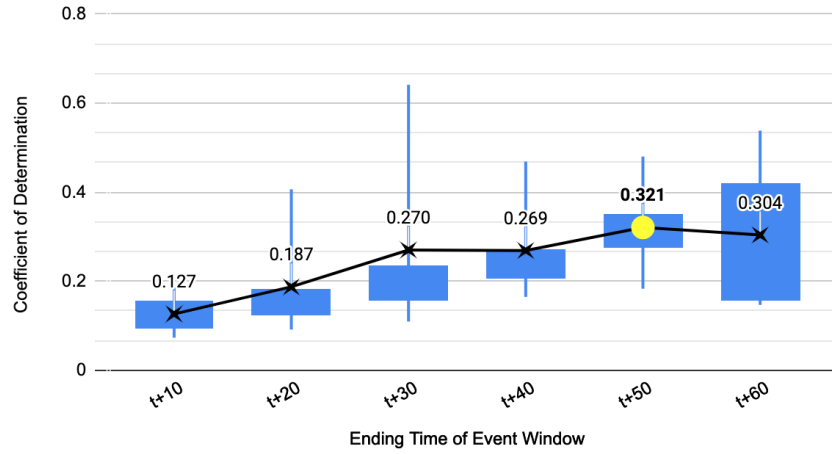
Notes: The horizontal axis of each figure depicts the end time of the considered event window lengths and the vertical axis represents the coefficient of determination, R^2_{OOS} . The cross points represent $\overline{R^2_{OOS}}$ for each event window size, where the solid yellow point represents the event window with the largest $\overline{R^2_{OOS}}$. For each event window, box-and-whisker plots are shown surrounding the corresponding averages. $\overline{R^2_{OOS}}$ measures the average reduction in predictive error achieved by the neural network compared to the naive baseline.

Out-of-sample R^2 for USc1 (Averaged Across Splits)



(a) *USc1*

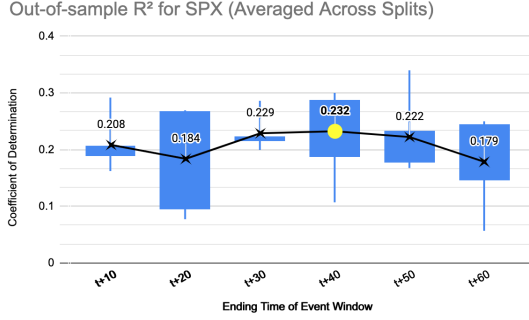
Out-of-sample R^2 for USc2 (Averaged Across Splits)



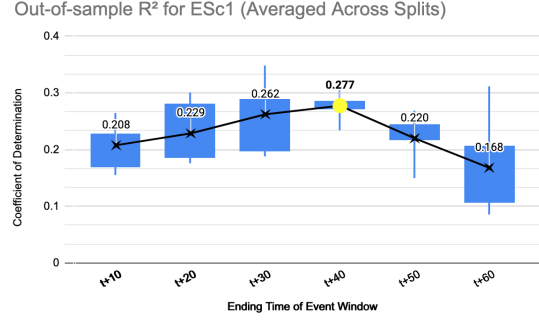
(b) *USc2*

Figure E7: Optimal Event Window Lengths for 30-Year Treasury Futures

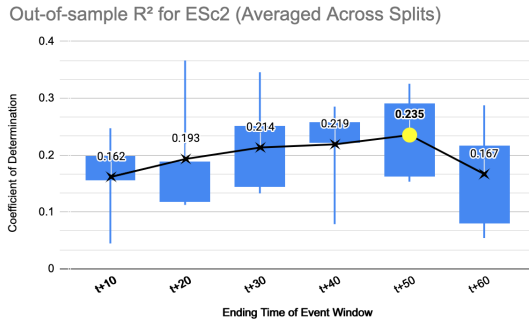
Notes: The horizontal axis of each figure depicts the end time of the considered event window lengths and the vertical axis represents the coefficient of determination, R^2_{OOS} . The cross points represent $\overline{R^2_{OOS}}$ for each event window size, where the solid yellow point represents the event window with the largest $\overline{R^2_{OOS}}$. For each event window, box-and-whisker plots are shown surrounding the corresponding averages. $\overline{R^2_{OOS}}$ measures the average reduction in predictive error achieved by the neural network compared to the naive baseline.



(a) *SPX*

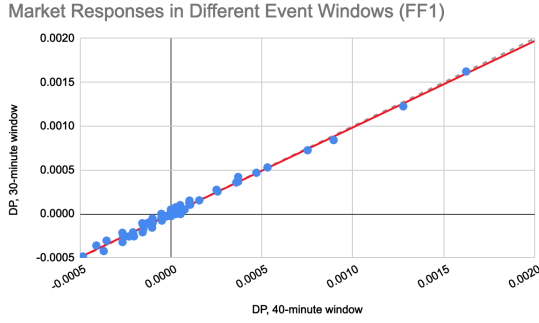


(b) *ESc1*

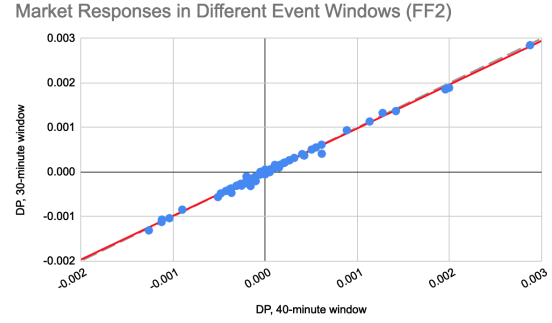


(c) *ESc2*

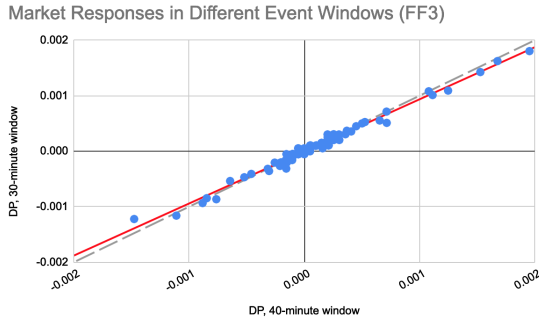
Figure E8: Optimal Event Window Lengths for S&P 500 and E-mini Futures
Notes: The horizontal axis of each figure depicts the end time of the considered event window lengths and the vertical axis represents the coefficient of determination, R^2_{OOS} . The cross points represent $\overline{R^2_{OOS}}$ for each event window size, where the solid yellow point represents the event window with the largest $\overline{R^2_{OOS}}$. For each event window, box-and-whisker plots are shown surrounding the corresponding averages. $\overline{R^2_{OOS}}$ measures the average reduction in predictive error achieved by the neural network compared to the naive baseline.



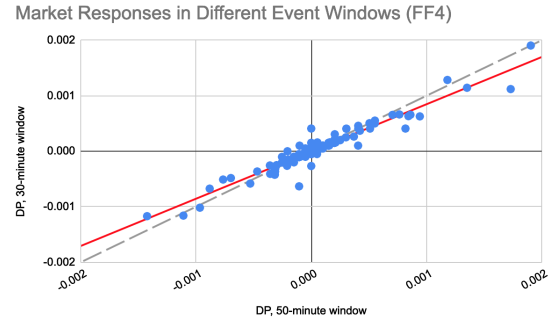
(a) $FF1^{**}$



(b) $FF2^{***}$



(c) $FF3^{***}$

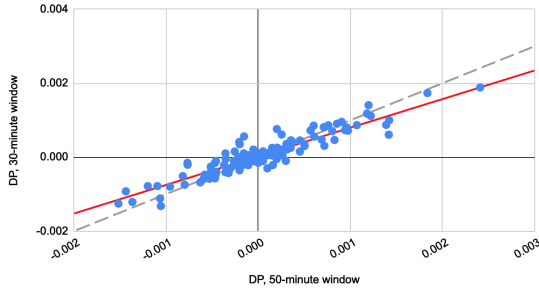


(d) $FF4^{***}$

Figure E9: Comparing Market Responses in Different Event Windows for Federal Funds Futures

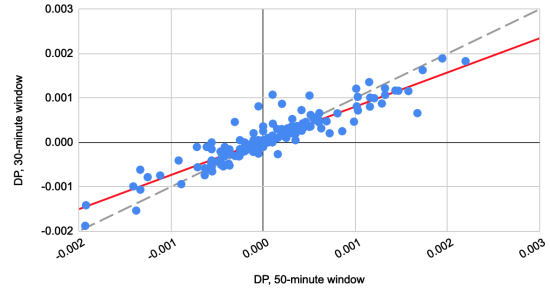
Notes: The horizontal axis depicts the price log-difference within the systematically estimated event window length. The vertical axis represents DP_{t+20} . The 45-degree line is depicted as grey and dashed. The blue dots are market reactions on scheduled FOMC meetings. The price-log differences calculated within the optimal window lengths are regressed on DP_{t+20} through OLS. If the slope coefficient of the regression line is statistically less than 1, it confirms under-reaction to the information content of FOMC statements on release, ex post, and is depicted in red. ** sig. at the 5% level, *** sig. at the 1% level.

Market Responses in Different Event Windows (EDcm2)



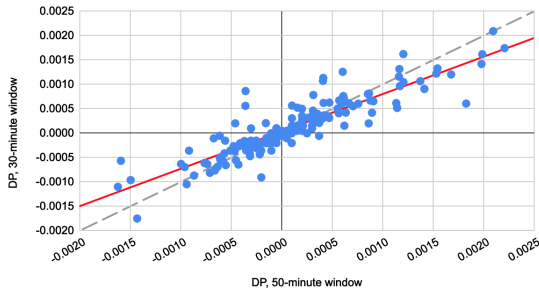
(a) $EDcm2^{***}$

Market Responses in Different Event Windows (EDcm3)



(b) $EDcm3^{***}$

Market Responses in Different Event Windows (EDcm4)

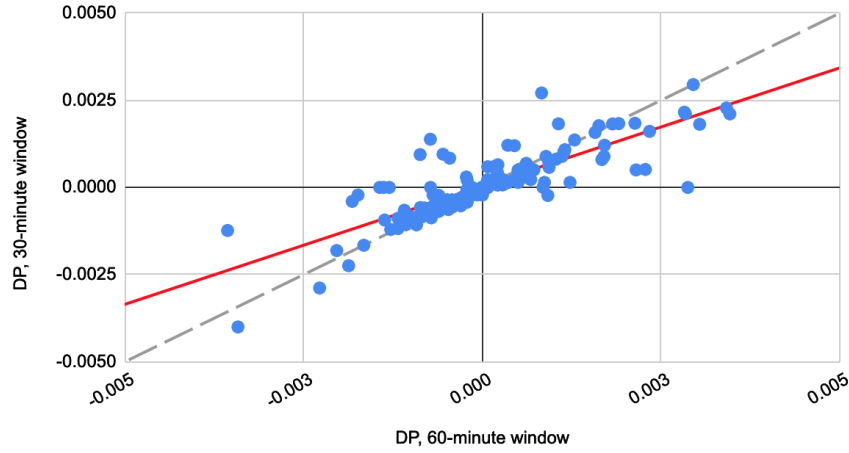


(c) $EDcm4^{***}$

Figure E10: Comparing Market Responses in Different Event Windows for Eurodollar Futures

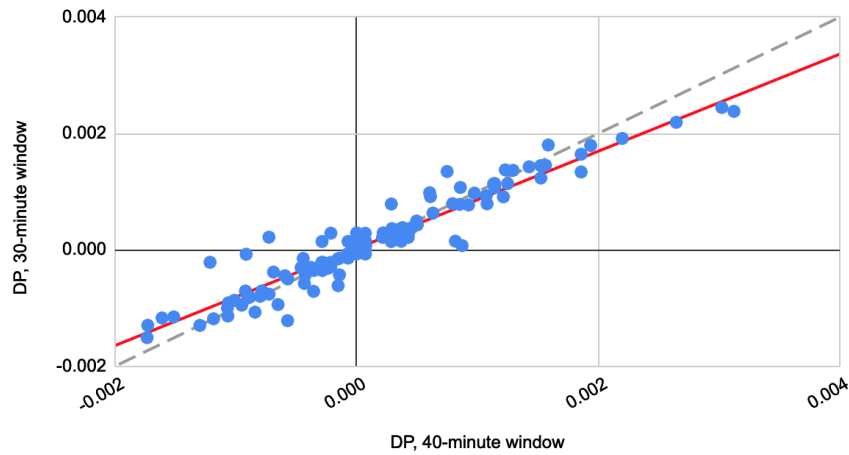
Notes: The horizontal axis depicts the price log-difference within the systematically estimated event window length. The vertical axis represents DP_{t+20} . The 45-degree line is depicted as grey and dashed. The blue dots are market reactions on scheduled FOMC meetings. The price-log differences calculated within the optimal window lengths are regressed on DP_{t+20} through OLS. If the slope coefficient of the regression line is statistically less than 1, it confirms under-reaction to the information content of FOMC statements on release, ex post, and is depicted in red. ** sig. at the 5% level, *** sig. at the 1% level.

Market Responses in Different Event Windows (Tuc1)



(a) $Tuc1^{***}$

Market Responses in Different Event Windows (Tuc2)

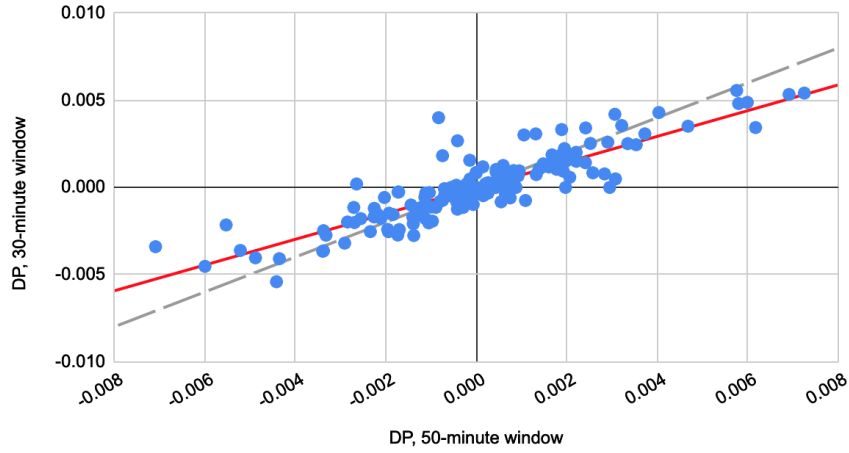


(b) $Tuc2^{***}$

Figure E11: Comparing Market Responses in Different Event Windows for 2-Year Treasury Futures

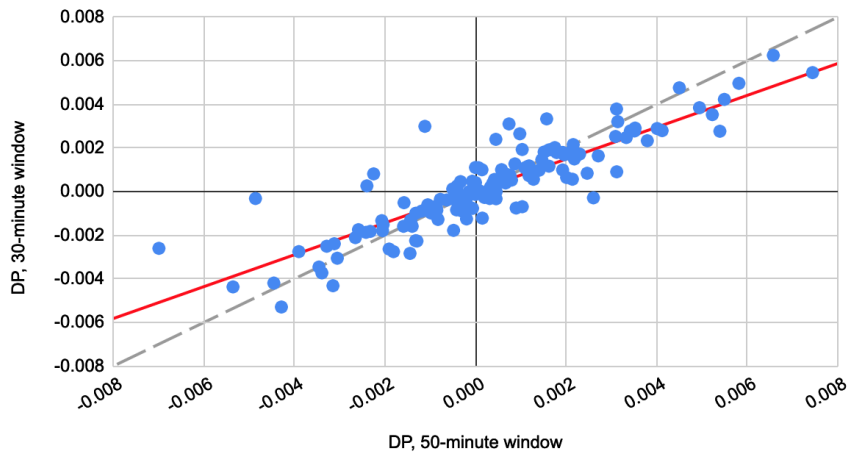
Notes: The horizontal axis depicts the price log-difference within the systematically estimated event window length. The vertical axis represents DP_{t+20} . The 45-degree line is depicted as grey and dashed. The blue dots are market reactions on scheduled FOMC meetings. The price-log differences calculated within the optimal window lengths are regressed on DP_{t+20} through OLS. If the slope coefficient of the regression line is statistically less than 1, it confirms under-reaction to the information content of FOMC statements on release, ex post, and is depicted in red. ** sig. at the 5% level, *** sig. at the 1% level.

Market Responses in Different Event Windows (FVc1)



(a) $FVc1^{***}$

Market Responses in Different Event Windows (FVc2)

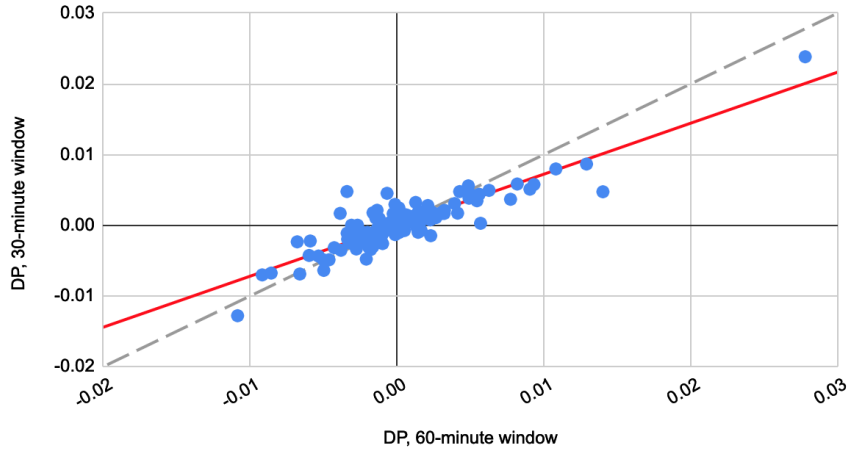


(b) $FVc2^{***}$

Figure E12: Comparing Market Responses in Different Event Windows for 5-Year Treasury Futures

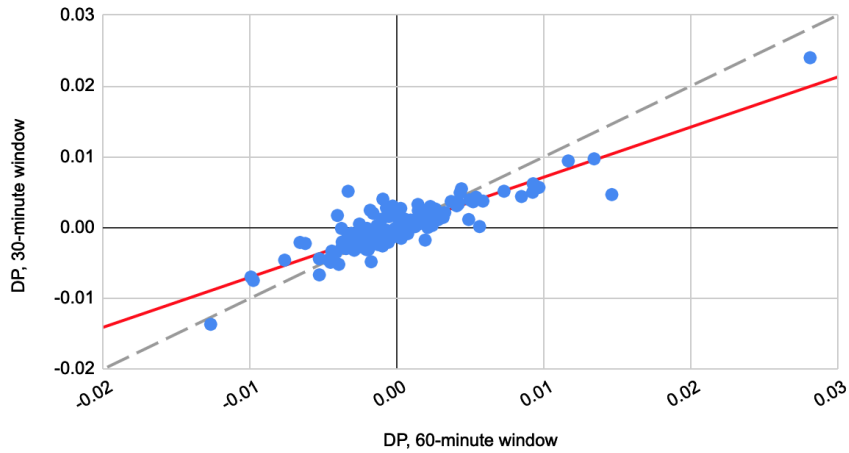
Notes: The horizontal axis depicts the price log-difference within the systematically estimated event window length. The vertical axis represents DP_{t+20} . The 45-degree line is depicted as grey and dashed. The blue dots are market reactions on scheduled FOMC meetings. The price-log differences calculated within the optimal window lengths are regressed on DP_{t+20} through OLS. If the slope coefficient of the regression line is statistically less than 1, it confirms under-reaction to the information content of FOMC statements on release, ex post, and is depicted in red. ** sig. at the 5% level, *** sig. at the 1% level.

Market Responses in Different Event Windows (TYc1)



(a) $TYc1^{***}$

Market Responses in Different Event Windows (TYc2)

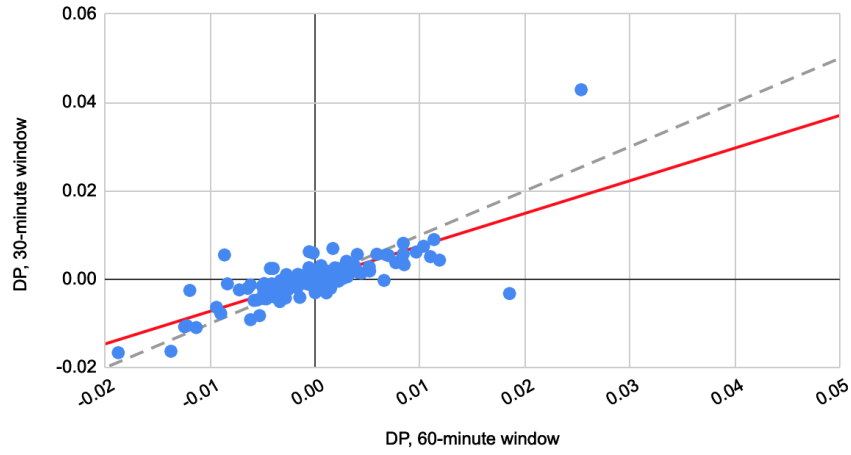


(b) $TYc2^{***}$

Figure E13: Comparing Market Responses in Different Event Windows for 10-Year Treasury Futures

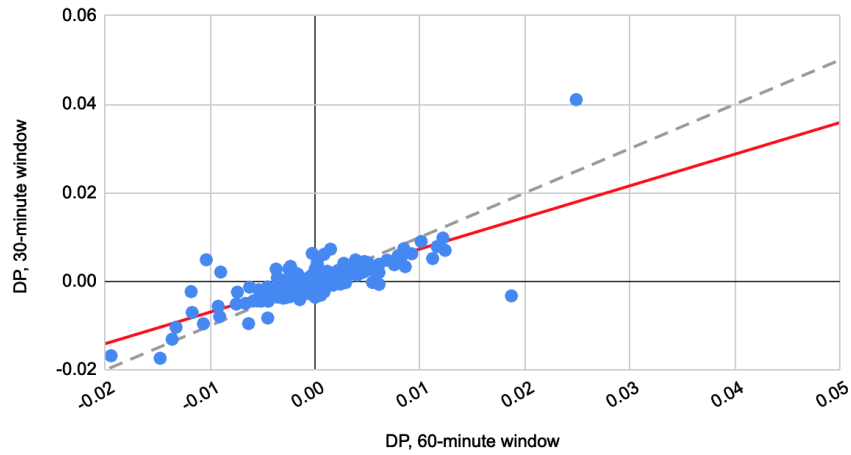
Notes: The horizontal axis depicts the price log-difference within the systematically estimated event window length. The vertical axis represents DP_{t+20} . The 45-degree line is depicted as grey and dashed. The blue dots are market reactions on scheduled FOMC meetings. The price-log differences calculated within the optimal window lengths are regressed on DP_{t+20} through OLS. If the slope coefficient of the regression line is statistically less than 1, it confirms under-reaction to the information content of FOMC statements on release, ex post, and is depicted in red. ** sig. at the 5% level, *** sig. at the 1% level.

Market Responses in Different Event Windows (USc1)



(a) $USc1^{***}$

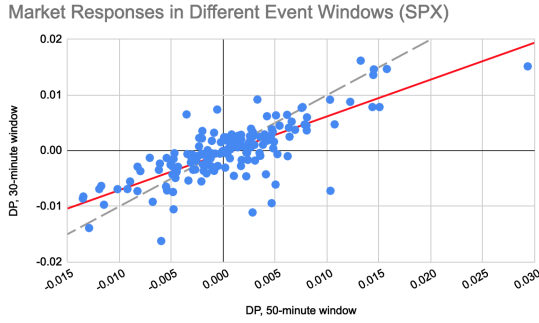
Market Responses in Different Event Windows (USc2)



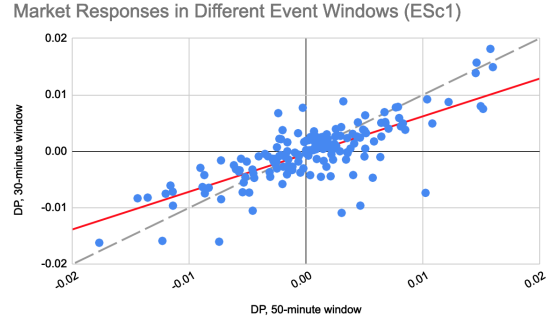
(b) $USc2^{***}$

Figure E14: Comparing Market Responses in Different Event Windows for 30-Year Treasury Futures

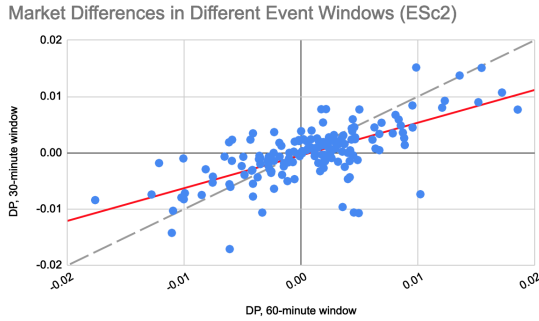
Notes: The horizontal axis depicts the price log-difference within the systematically estimated event window length. The vertical axis represents DP_{t+20} . The 45-degree line is depicted as grey and dashed. The blue dots are market reactions on scheduled FOMC meetings. The price-log differences calculated within the optimal window lengths are regressed on DP_{t+20} through OLS. If the slope coefficient of the regression line is statistically less than 1, it confirms under-reaction to the information content of FOMC statements on release, ex post, and is depicted in red. ** sig. at the 5% level, *** sig. at the 1% level.



(a) SPX^{***}



(b) $ESc1^{***}$



(c) $ESc2^{***}$

Figure E15: Comparing Market Responses in Different Event Windows for S&P 500 and E-mini Futures

Notes: The horizontal axis depicts the price log-difference within the systematically estimated event window length. The vertical axis represents DP_{t+20} . The 45-degree line is depicted as grey and dashed. The blue dots are market reactions on scheduled FOMC meetings. The price-log differences calculated within the optimal window lengths are regressed on DP_{t+20} through OLS. If the slope coefficient of the regression line is statistically less than 1, it confirms under-reaction to the information content of FOMC statements on release, ex post, and is depicted in red. ** sig. at the 5% level, *** sig. at the 1% level.

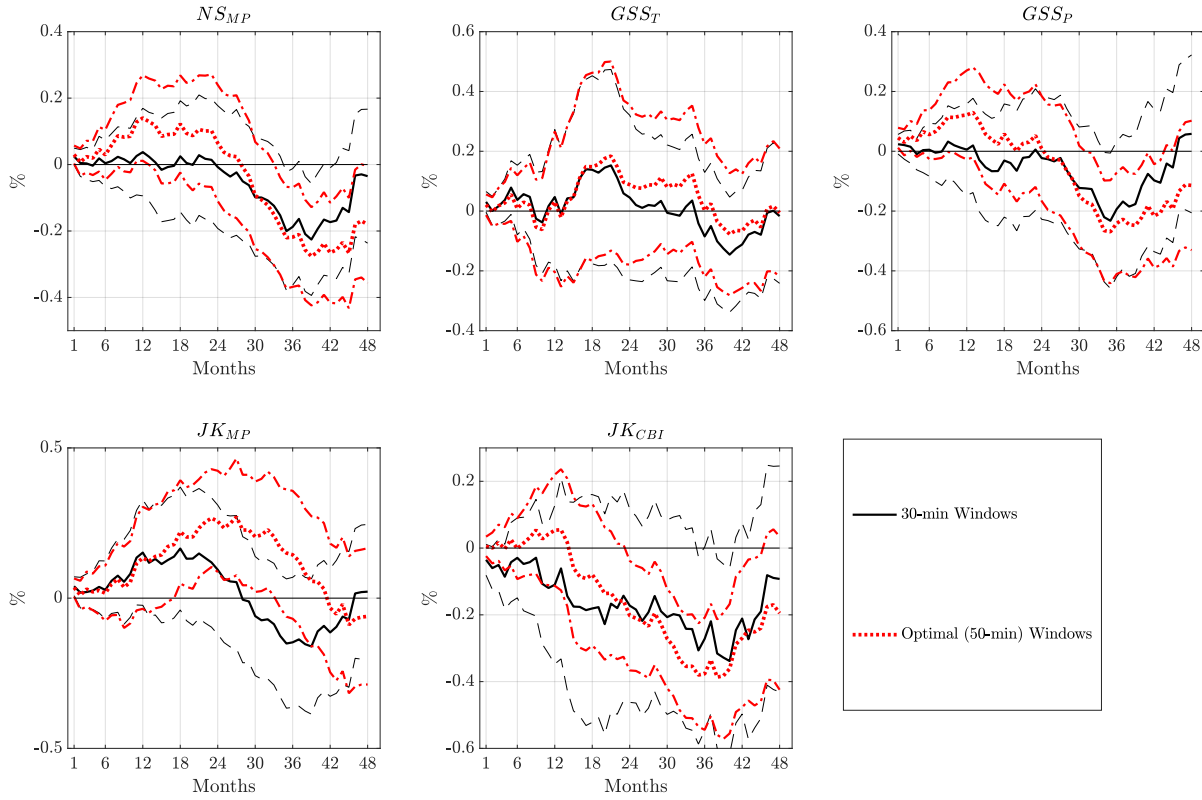


Figure E16: Effects of Event Window Choice on IP Impulse Responses

Notes: Impulse responses are calculated using the lag-augmented local projection from Olea and Plagborg-Møller (2021). Confidence bands are at the 90% level. Standard errors are Eicker-Huber-White heteroscedasticity-robust standard errors. The responses are to a 100 basis point increase in the monetary policy shock series. The monetary policy shocks are constructed using the full set of monetary policy surprise instruments.

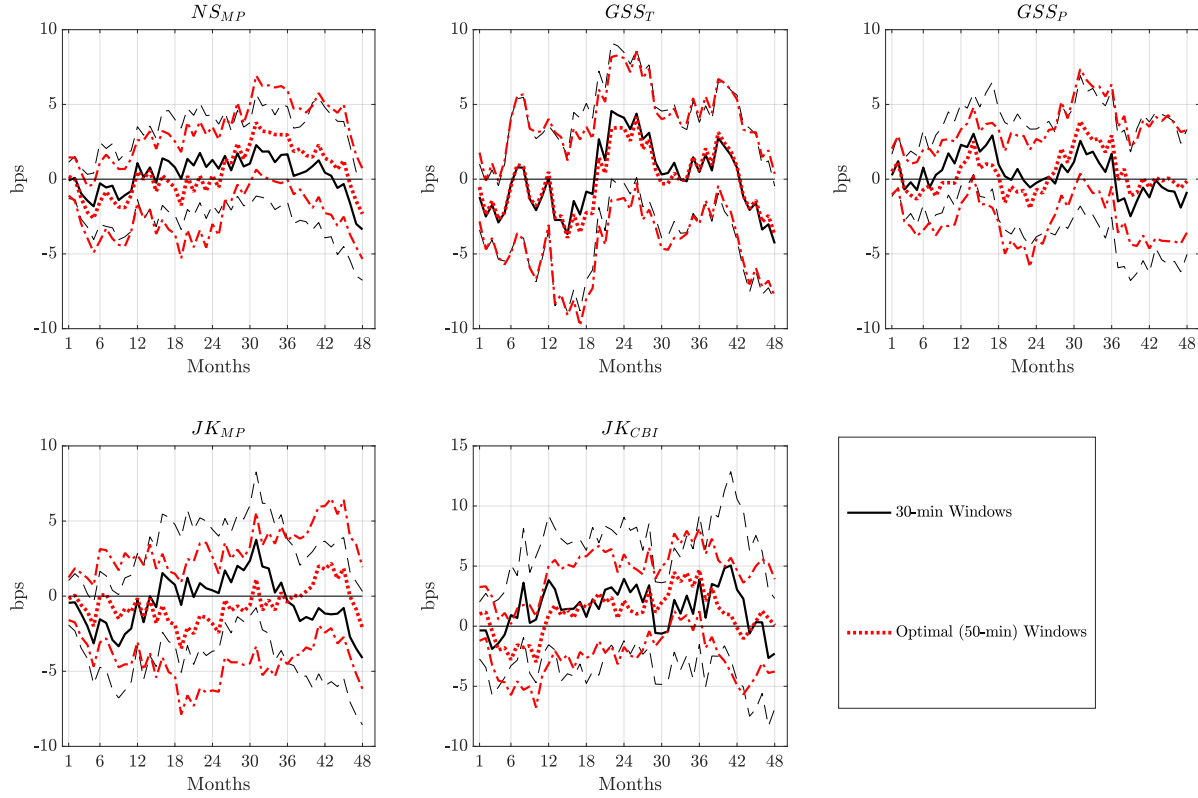


Figure E17: Effects of Event Window Choice on Excess Bond Premium Impulse Responses

Notes: Impulse responses are calculated using the lag-augmented local projection from Olea and Plagborg-Møller (2021). Confidence bands are at the 90% level. Standard errors are Eicker-Huber-White heteroscedasticity-robust standard errors. The responses are to a 100 basis point increase in the monetary policy shock series. The monetary policy shocks are constructed using the full set of monetary policy surprise instruments.

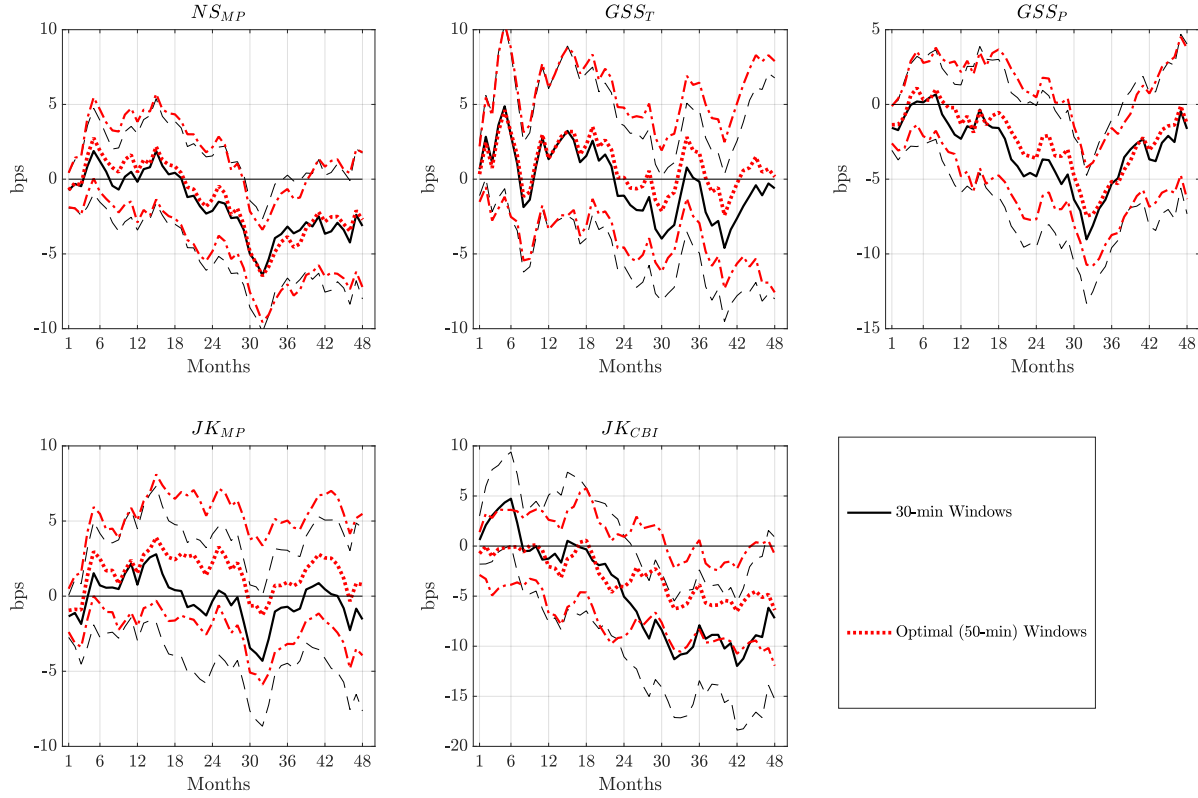


Figure E18: Effects of Event Window Choice on Two-year Treasury Yield Impulse Responses

Notes: Impulse responses are calculated using the lag-augmented local projection from Olea and Plagborg-Møller (2021). Confidence bands are at the 90% level. Standard errors are Eicker-Huber-White heteroscedasticity-robust standard errors. The responses are to a 100 basis point increase in the monetary policy shock series. The monetary policy shocks are constructed using the full set of monetary policy surprise instruments.

Flesch-Kincaid Grade Level Readability of FOMC Statements

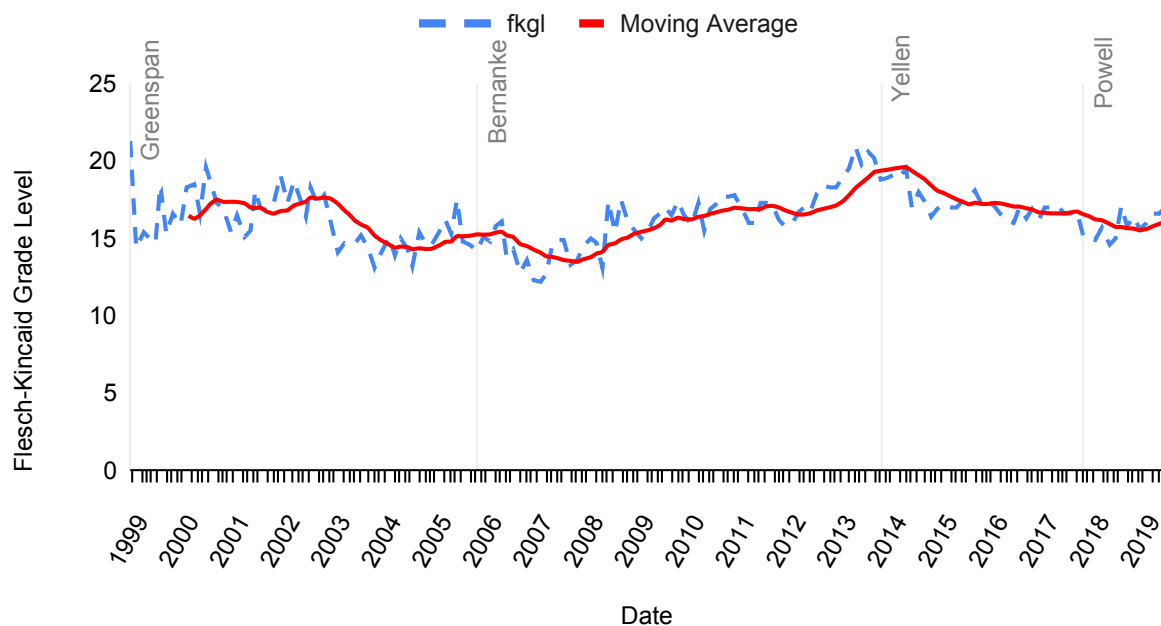


Figure E19: Flesch-Kincaid Grade Level Readability of FOMC Statements

Notes: The complexity of FOMC statements is measured by the Flesch-Kincaid Grade Level, defined as: $0.39 \times \text{average sentence length} + 11.8 \times \text{average number of syllables per word} - 15.59$. From left to right, the vertical grey lines indicate the first FOMC meeting with Greenspan, Bernanke, Yellen, and Powell as Fed Chair. The moving average in solid red is calculated with a period of 10. For a description of U.S. education grade levels, see https://en.wikipedia.org/wiki/Education_in_the_United_States.

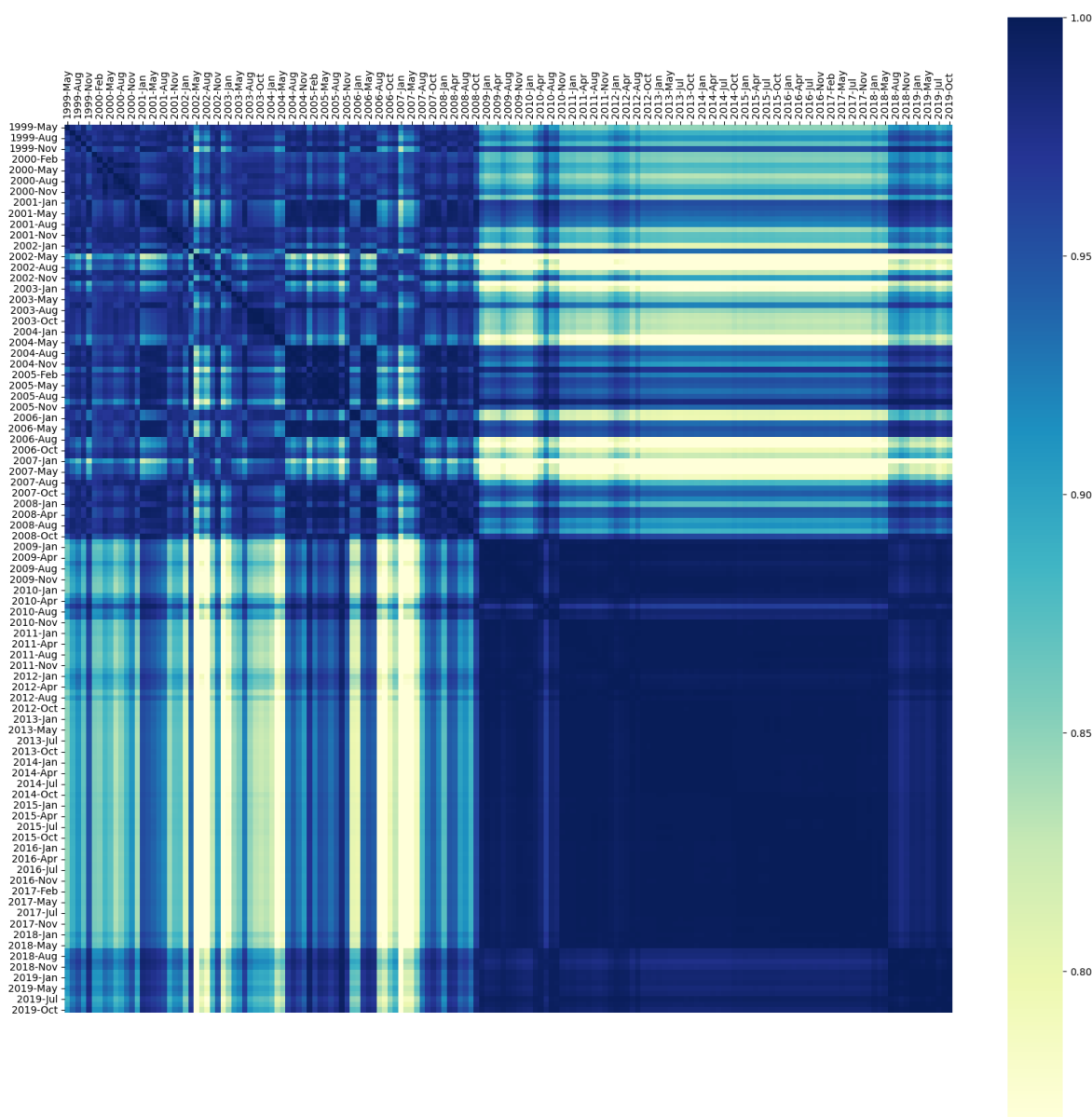


Figure E20: Document Similarity Matrix for FOMC Statements

Notes: Each element in the grid represents the cosine similarity between the dense contextual embeddings of the FOMC statements corresponding to that specific row and column. The resulting metric measures the degree of alignment between the contextual prose and policy concepts of both documents. A darker shade of blue represents document pairs with a high similarity approaching one, signifying near-identical communication states. Conversely, a lighter shade of green highlights pairs that are semantically distinct due to structural shifts or major informational updates. The main diagonal consists entirely of ones, as it reflects the cosine similarity calculated between a given FOMC statement and itself.

Out-of-sample R^2 Using "One Signal" Approach (FFFs)

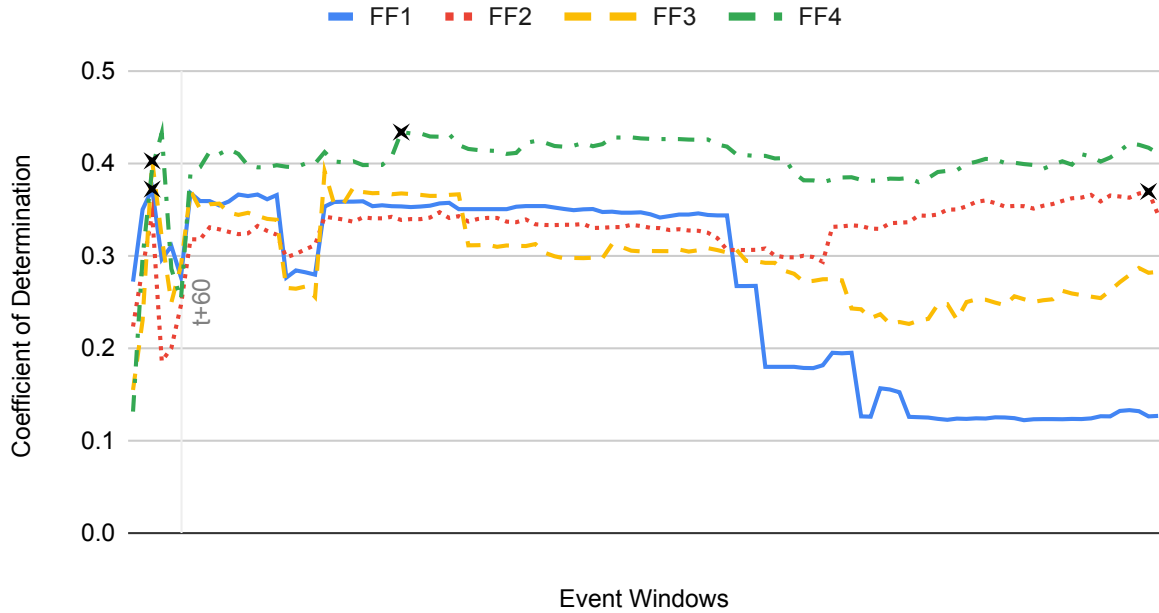


Figure E21: $\overline{R^2_{OOS}}$ Calculated Using the “One Signal” Approach for Federal Funds Futures

Notes: The horizontal axis depicts the event window lengths, starting from a 20-minute event window and ending at an event window starting 10 minutes before and ending 18 hours after FOMC statement release. The cross points represent the event window length associated with the largest $\overline{R^2_{OOS}}$. Estimates to the left of the vertical grey line labelled “ $t + 60$ ” are calculated using the “joint” approach. The 1-month and 3-month-ahead federal funds futures contract are the two exceptions that see the largest $\overline{R^2_{OOS}}$ within a window length than the systematically estimated length. As a robustness check, $\overline{R^2_{OOS}}$ was calculated using systematic estimation for both futures and considered event window lengths, which can be found in the rightmost box-and-whisker plot of Appendix Sub-figures E2b and E2d, respectively.

Out-of-sample R^2 Using "One Signal" Approach (EDs)

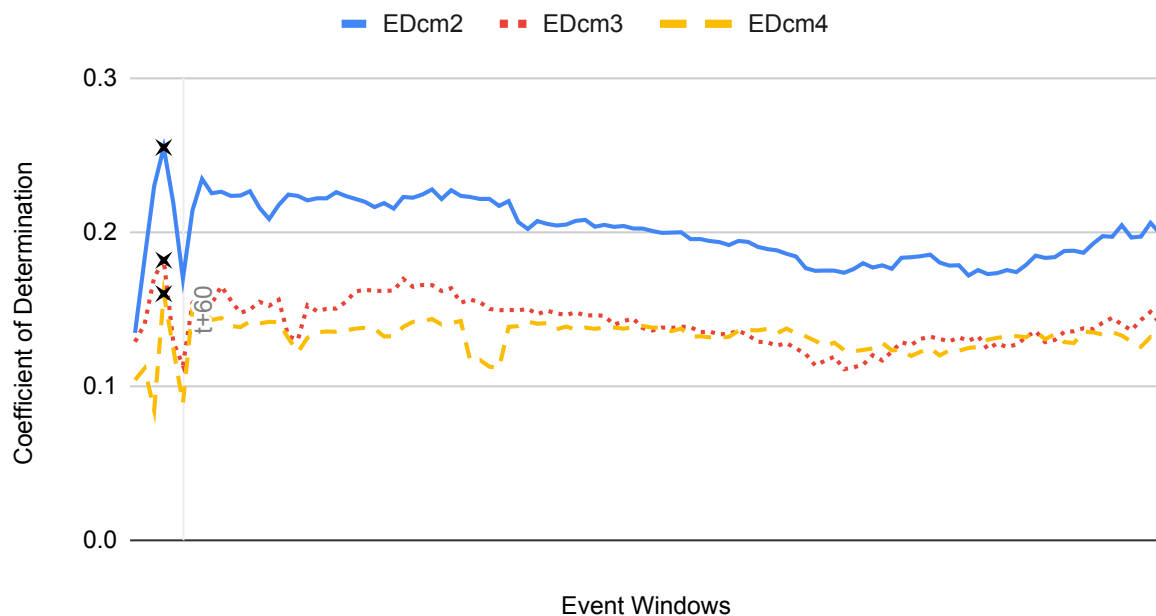


Figure E22: $\overline{R^2_{OOS}}$ Calculated Using the "One Signal" Approach for Eurodollar Futures

Notes: The horizontal axis depicts the event window lengths, starting from a 20-minute event window and ending at an event window starting 10 minutes before and ending 18 hours after FOMC statement release. The cross points represent the event window length associated with the largest $\overline{R^2_{OOS}}$. Estimates to the left of the vertical grey line labelled " $t + 60$ " are calculated using the "joint" approach.

Out-of-sample R^2 Using "One Signal" Approach (TUs)

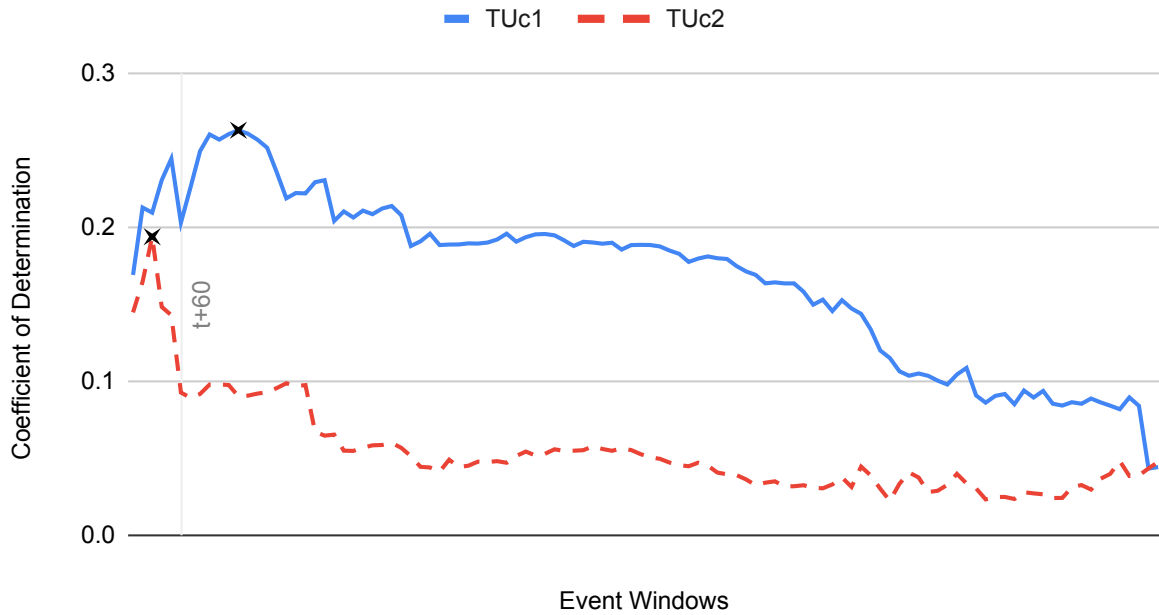


Figure E23: $\overline{R^2_{OOS}}$ Calculated Using the "One Signal" Approach for 2-year Treasury Futures

Notes: The horizontal axis depicts the event window lengths, starting from the systematically estimated event window length and ending at an event window starting 10 minutes before and ending 18 hours after FOMC statement release. The cross points represent the event window length associated with the largest $\overline{R^2_{OOS}}$. Estimates to the left of the vertical grey line labelled " $t + 60$ " are calculated using the "joint" approach. The front-month 2-year Treasury futures contract is the one exception that sees the largest $\overline{R^2_{OOS}}$ within a window length than the systematically estimated length. As a robustness check, $\overline{R^2_{OOS}}$ was calculated with systematic estimation for the futures contract and considered event window length, which can be found in the rightmost box-and-whisker plot of Appendix Sub-figure E4a.

Out-of-sample R^2 Using "One Signal" Approach (FVs)

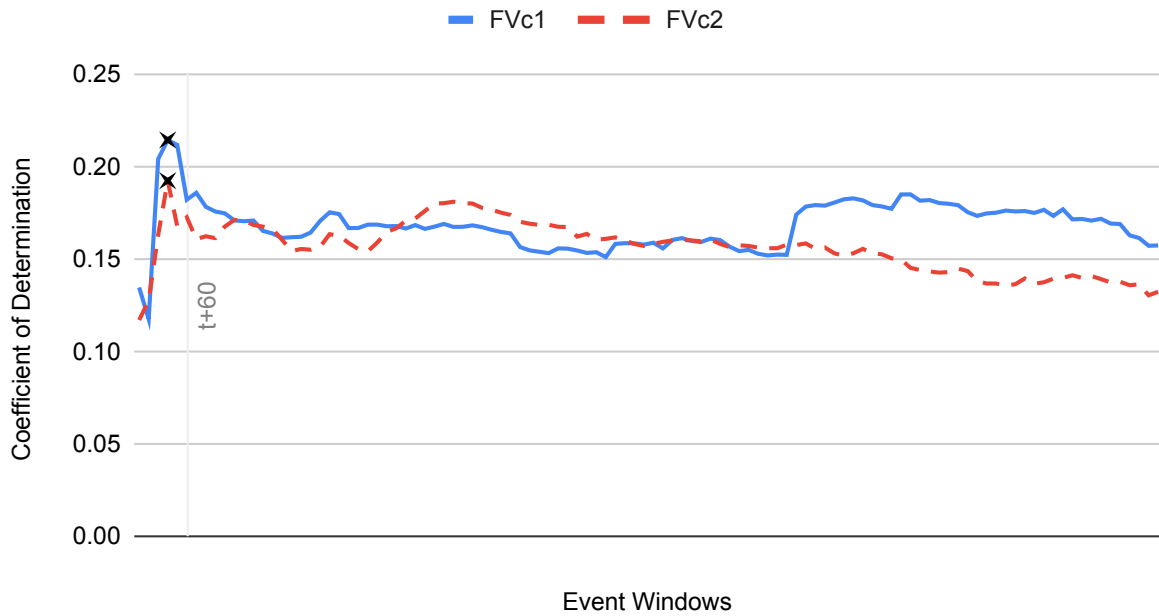


Figure E24: $\overline{R^2_{OOS}}$ Calculated Using the "One Signal" Approach for 5-year Treasury Futures

Notes: The horizontal axis depicts the event window lengths, starting from the systematically estimated event window length and ending at an event window starting 10 minutes before and ending 18 hours after FOMC statement release. The cross points represent the event window length associated with the largest $\overline{R^2_{OOS}}$. Estimates to the left of the vertical grey line labelled " $t + 60$ " are calculated using the "joint" approach.

Out-of-sample R^2 Using "One Signal" Approach (TYs)

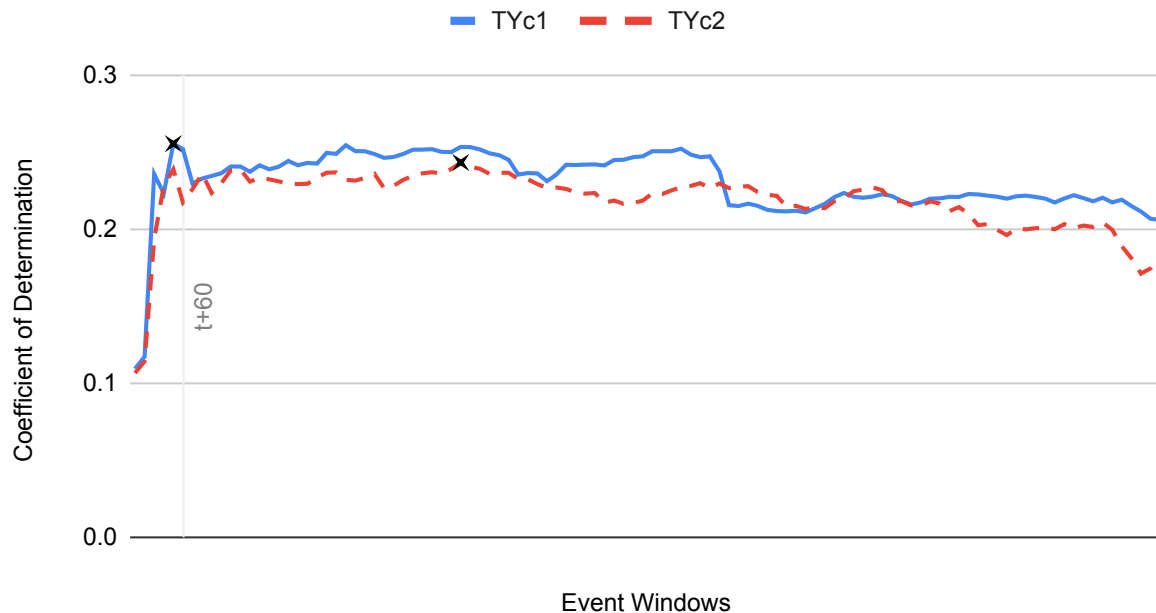


Figure E25: $\overline{R^2_{OOS}}$ Calculated Using the "One Signal" Approach for 10-year Treasury Futures

Notes: The horizontal axis depicts the event window lengths, starting from the systematically estimated event window length and ending at an event window starting 10 minutes before and ending 18 hours after FOMC statement release. The cross points represent the event window length associated with the largest $\overline{R^2_{OOS}}$. Estimates to the left of the vertical grey line labelled " $t + 60$ " are calculated using the "joint" approach. The second-month 10-year Treasury futures contract is the one exception that sees the largest $\overline{R^2_{OOS}}$ within a window length than the systematically estimated length. As a robustness check, $\overline{R^2_{OOS}}$ was calculated with systematic estimation for the futures contract and considered event window length, which can be found in the rightmost box-and-whisker plot of Appendix Sub-figure E6b.

Out-of-sample R^2 Using "One Signal" Approach (USs)

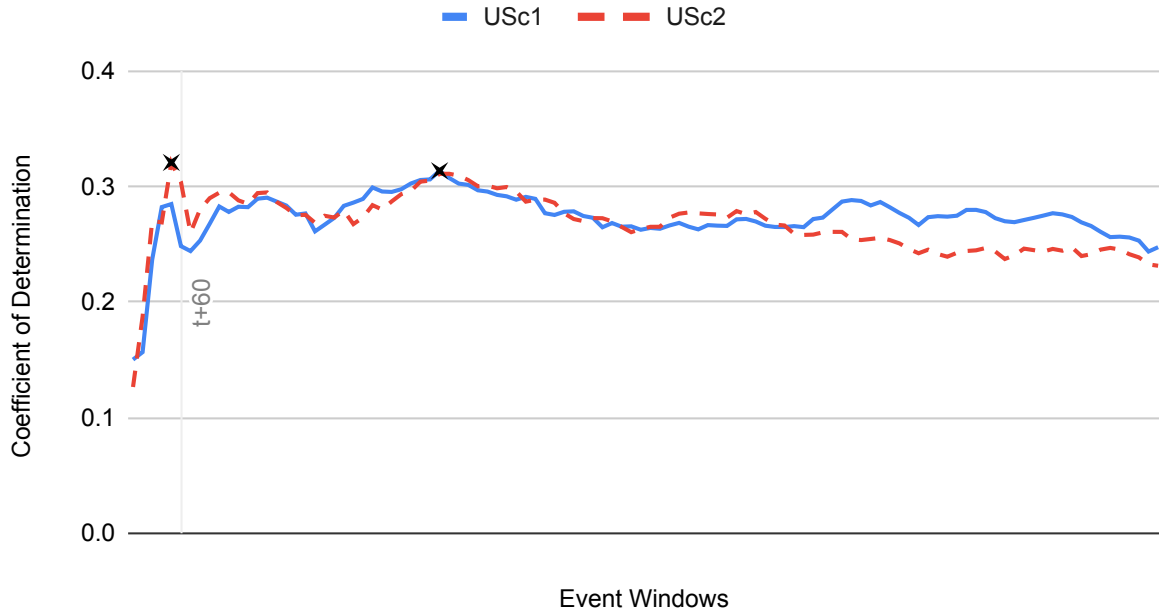


Figure E26: $\overline{R^2_{OOS}}$ Calculated Using the "One Signal" Approach for 30-year Treasury Futures

Notes: The horizontal axis depicts the event window lengths, starting from the systematically estimated event window length and ending at an event window starting 10 minutes before and ending 18 hours after FOMC statement release. The cross points represent the event window length associated with the largest $\overline{R^2_{OOS}}$. Estimates to the left of the vertical grey line labelled " $t + 60$ " are calculated using the "joint" approach. The front-month 30-year Treasury futures contract is the one exception that sees the largest $\overline{R^2_{OOS}}$ within a window length than the systematically estimated length. As a robustness check, $\overline{R^2_{OOS}}$ was calculated with systematic estimation for the futures contract and considered event window length, which can be found in the rightmost box-and-whisker plot of Appendix Sub-figure E7a.

Out-of-sample R^2 Using "One Signal" Approach (S&P 500)

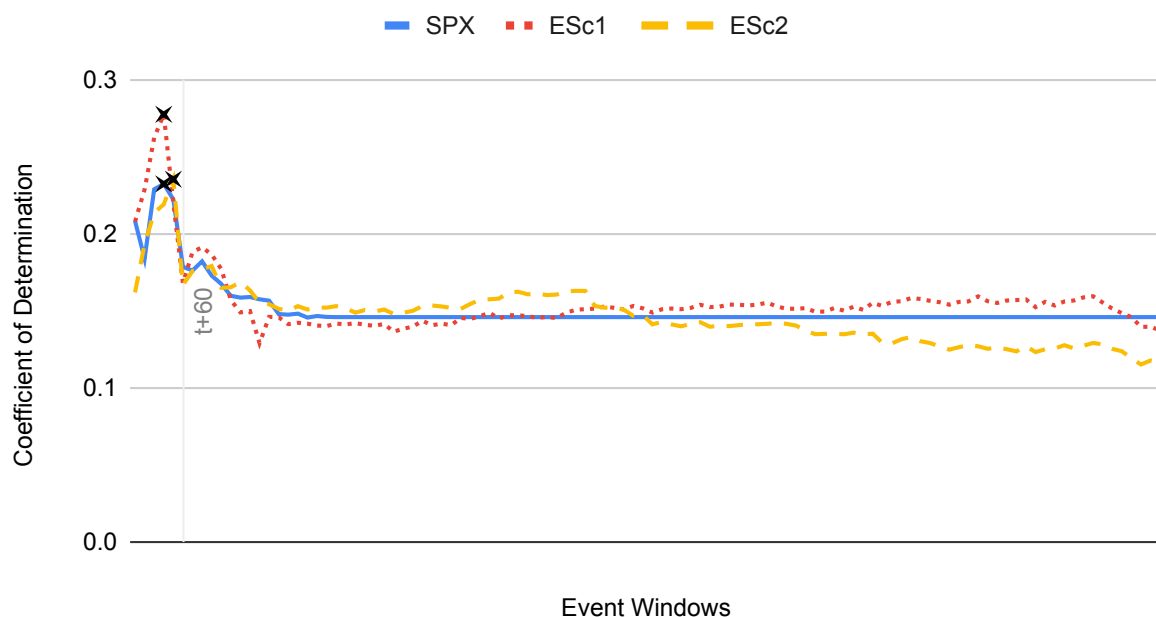


Figure E27: $\overline{R_{OOS}^2}$ Calculated Using the "One Signal" Approach for S&P 500 and E-mini Futures

Notes: The horizontal axis depicts the event window lengths, starting from the systematically estimated event window length and ending at an event window starting 10 minutes before and ending 18 hours after FOMC statement release. The cross points represent the event window length associated with the largest $\overline{R_{OOS}^2}$. Estimates to the left of the vertical grey line labelled " $t + 60$ " are calculated using the "joint" approach.

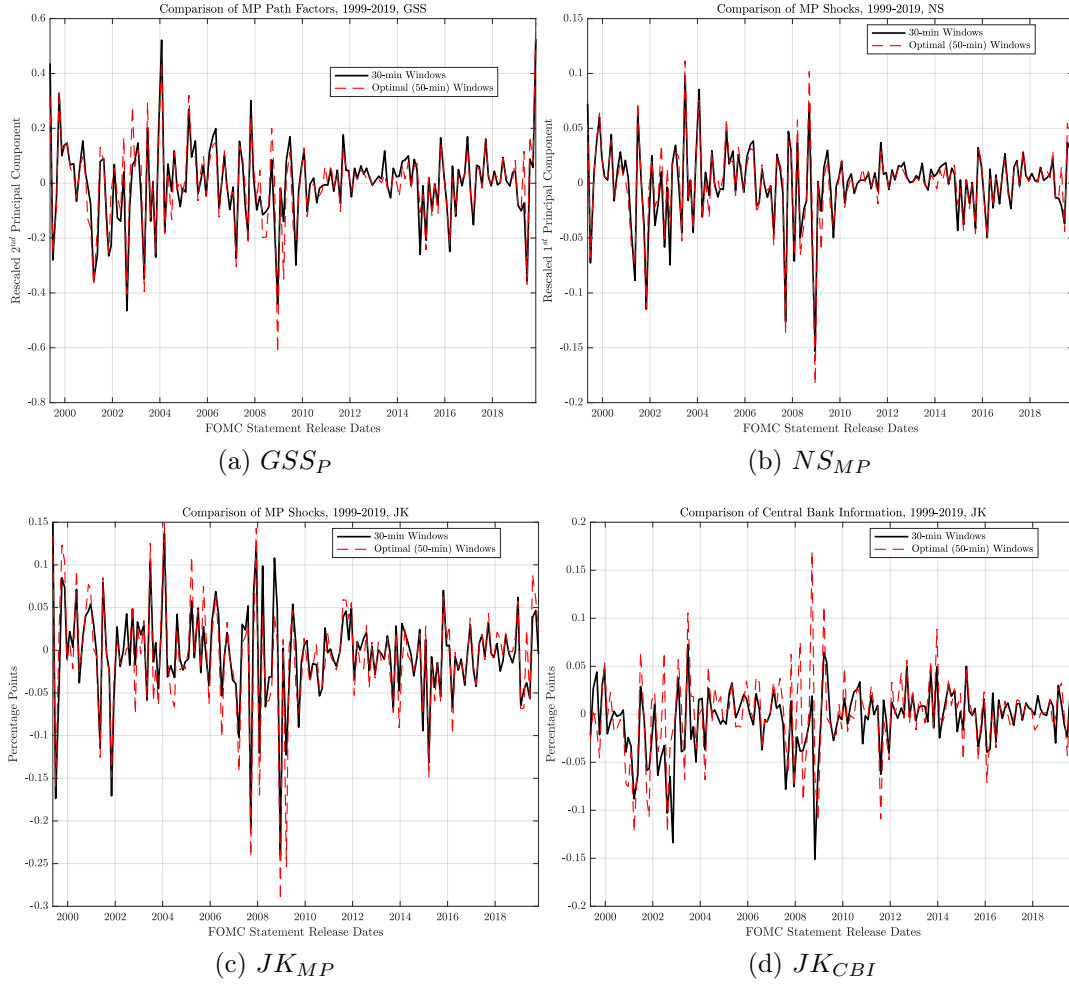


Figure E28: Comparing Monetary Policy Shock Series Derived from Optimal Window Length v. 30 Minutes

Notes: For each sub-figure, the horizontal axis represents FOMC statement release dates. The vertical axis depicts the principal components scaled according to the original specification of the authors. For all construction methods, the black-solid and red-dotted lines represent the shocks derived from surprises measured within 30 minutes and the median optimal event window length of 50 minutes, respectively.

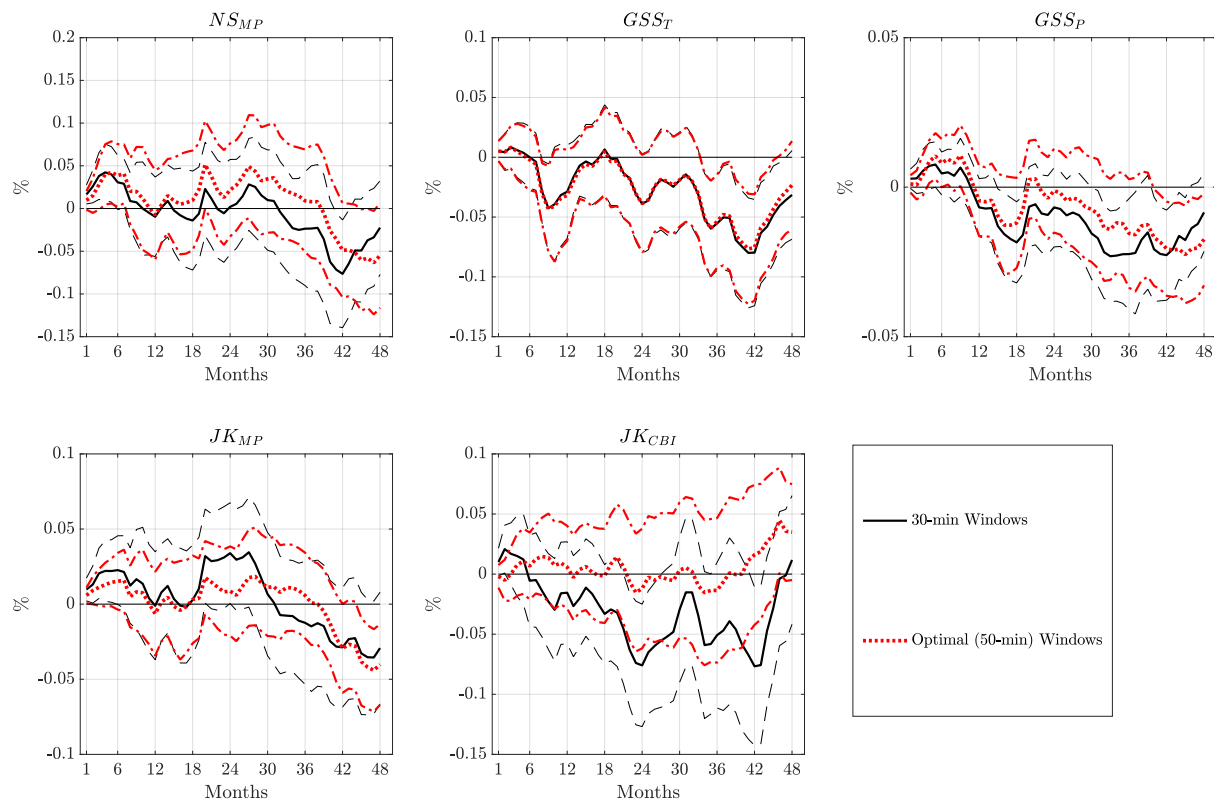


Figure E29: Effects of Event Window Choice on CPI Impulse Responses

Notes: Impulse responses are calculated using the lag-augmented local projection from Olea and Plagborg-Møller (2021). Confidence bands are at the 90% level. Standard errors are Eicker-Huber-White heteroscedasticity-robust standard errors. The responses are to a 100 basis point increase in the monetary policy shock series. The monetary policy shocks are constructed using the original author set of monetary policy surprise instruments.

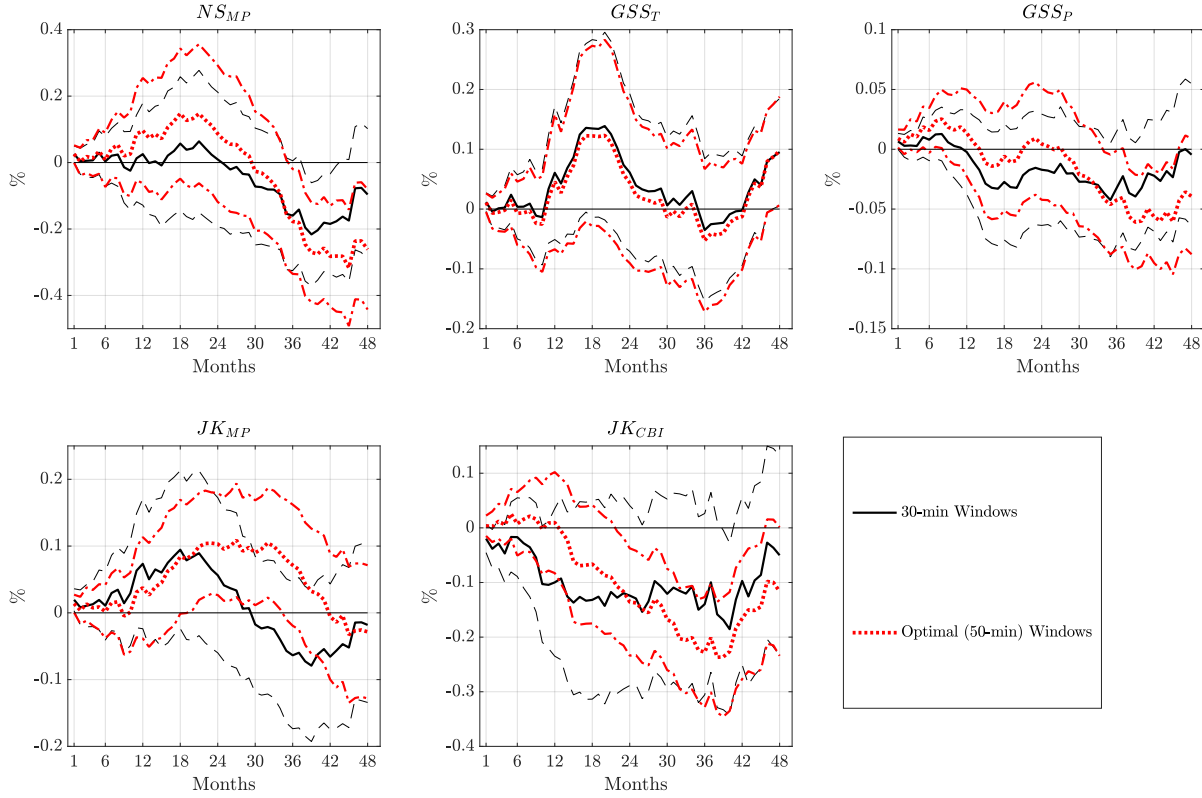


Figure E30: Effects of Event Window Choice on IP Impulse Responses

Notes: Impulse responses are calculated using the lag-augmented local projection from Olea and Plagborg-Møller (2021). Confidence bands are at the 90% level. Standard errors are Eicker-Huber-White heteroscedasticity-robust standard errors. The responses are to a 100 basis point increase in the monetary policy shock series. The monetary policy shocks are constructed using the original author set of monetary policy surprise instruments.

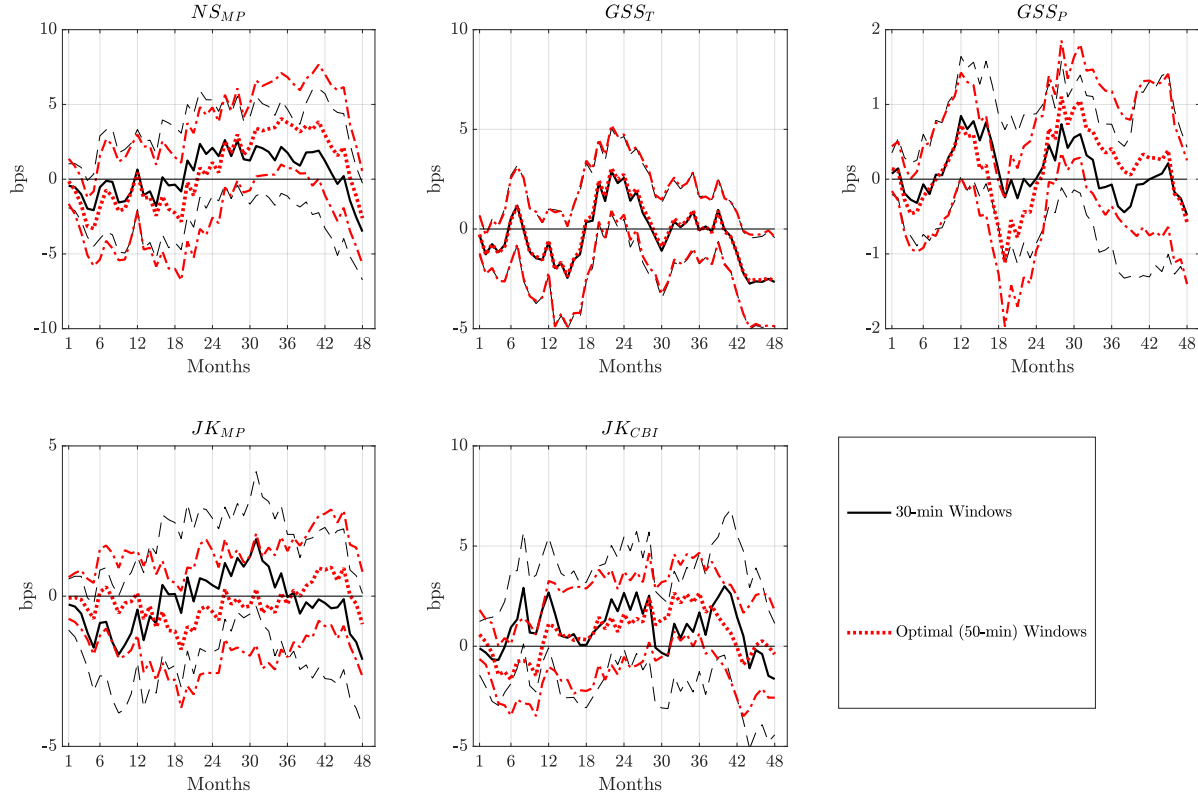


Figure E31: Effects of Event Window Choice on Excess Bond Premium Impulse Responses

Notes: Impulse responses are calculated using the lag-augmented local projection from Olea and Plagborg-Møller (2021). Confidence bands are at the 90% level. Standard errors are Eicker-Huber-White heteroscedasticity-robust standard errors. The responses are to a 100 basis point increase in the monetary policy shock series. The monetary policy shocks are constructed using the original author set of monetary policy surprise instruments.

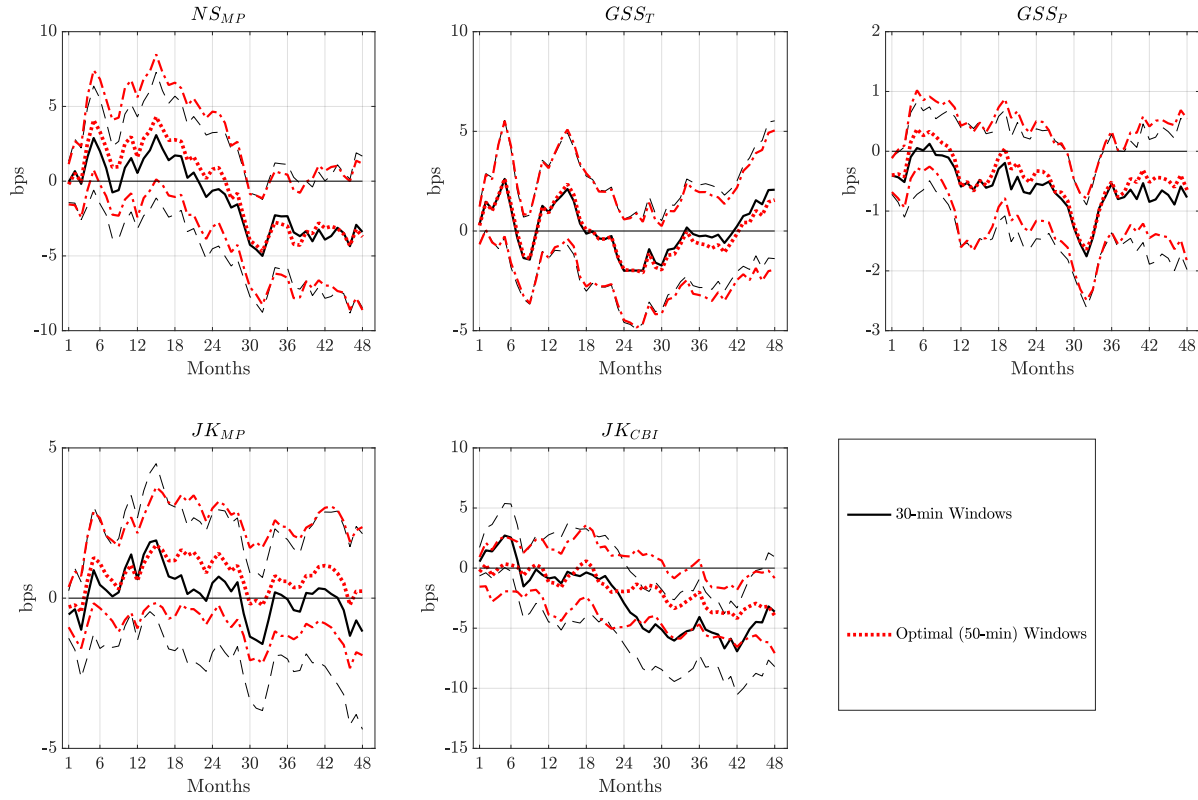


Figure E32: Effects of Event Window Choice on Two-year Treasury Yield Impulse Responses

Notes: Impulse responses are calculated using the lag-augmented local projection from Olea and Plagborg-Møller (2021). Confidence bands are at the 90% level. Standard errors are Eicker-Huber-White heteroscedasticity-robust standard errors. The responses are to a 100 basis point increase in the monetary policy shock series. The monetary policy shocks are constructed using the original author set of monetary policy surprise instruments.

	Scenario 1	Scenario 2	Scenario 3
<i>Framework Simulation Parameters</i>			
P_i^f	$\in [-100, 100]$	$\in [-100, 100]$	$\in [-100, 100]$
$\varepsilon_{i,0}^c$	$\in [-100, 100]$	$\in [-100, 100]$	$\in [-100, 100]$
$\varepsilon_{i,0}^n$	0	0	0
σ_c	100	1	50
$\mathcal{D}$	0.5	2	0.75
σ_n	0.1	1.13	1
ρ_c	0.47	0.47	0.47
σ_s	$\in \mathbb{R}$	$\in \mathbb{R}$	$\in \mathbb{R}$
<i>Simulation Results</i>			
t^*	16	5	10
$\hat{t}$	15	5	10

Table E1: Framework Simulation Parameters and Results for Different Asset Market Scenarios

Notes: The results are from 10,000 simulations of the asset price framework for the three asset market scenarios considered. t^* and $\hat{t}$ are defined as the time at which the asset market fully reacts to news on average, where the former is calculated using the fundamental price component and the latter with the observed signal according to the motivating framework.

Name	Maturity	Ticker	Sample	Observations
<i>Inputs</i>				
FOMC Statements	N/A	N/A	1999–2019	165
<i>Outputs</i>				
Federal Funds Rate Futures	Front-month	<i>FFc1</i>	1999–2019	165
Federal Funds Rate Futures	1-month-ahead	<i>FFc2</i>	1999–2019	165
Federal Funds Rate Futures	2-month-ahead	<i>FFc3</i>	1999–2019	165
Federal Funds Rate Futures	3-month-ahead	<i>FFc4</i>	1999–2019	165
Eurodollar Futures	2-quarter	<i>EDcm2</i>	1999–2019	165
Eurodollar Futures	3-quarter	<i>EDcm3</i>	1999–2019	165
Eurodollar Futures	4-quarter	<i>EDcm4</i>	1999–2019	165
2-year Treasury Futures	Front-month	<i>TUc1</i>	1999–2019	165
2-year Treasury Futures	Second-month	<i>TUc2</i>	1999–2019	165
5-year Treasury Futures	Front-month	<i>FVc1</i>	1999–2019	165
5-year Treasury Futures	Second-month	<i>FVc2</i>	1999–2019	165
10-year Treasury Futures	Front-month	<i>TYc1</i>	1999–2019	165
10-year Treasury Futures	Second-month	<i>TYc2</i>	1999–2019	165
30-year Treasury Futures	Front-month	<i>USc1</i>	1999–2019	165
30-year Treasury Futures	Second-month	<i>USc2</i>	1999–2019	165
S&P 500 Index	N/A	<i>SPX</i>	1999–2019	165
S&P 500 E-mini Futures	Front-month	<i>ESc1</i>	1999–2019	165
S&P 500 E-mini Futures	Second-month	<i>ESc2</i>	1999–2019	165

Table E2: Independent and Dependent Variables for Systematic Estimation of Appropriate Event Window Lengths by Neural Networks

Notes: The table shows the FOMC statements and financial market asset prices considered as independent and dependent variables, respectively, in my analysis. The statements are collected from the Board of Governors of the Federal Reserve System website and the asset prices are from the Thomson Reuters Tick History database. All series begin in May 1999 and end in October 2019. *Ticker* refers to the Reuters Instrument Code, which uniquely identifies the financial instrument. The letters *c* and *cm* standard for continuous futures contracts.

Hyperparameter	Value
Number of Layers	12
Hidden Size	768
Number of Attention Heads	12
Attention Head Size	64
Hidden Dropout	0.1
Attention Dropout	0.1
Max Sequence Length	512
Vocab Size	32,000
Training Batch Size	8
Evaluation Batch Size	8
Max Num of Steps	2040
Max Num of Epochs	120
Manual Random Seed	47
Learning Rate	*
Warmup Ratio	0.06
Adam Epsilon	1e-8
Learning Rate Decay	Linear

Table E3: XLNet-Base Hyperparameters for Fine-tuning

Notes: * denotes the hyperparameter that undergoes tuning during the fine-tuning process. The random seed is set as a constant to ensure replicability by always initialising random components (e.g., weights and biases) to the same initial values.

Asset	$\overline{R_{OOS}^2}$, 30-min	$\overline{R_{OOS}^2}$, Optimal	Difference
<i>FF1</i>	35.0%	37.2%	+2.2 p.p.
<i>FF2</i>	28.7%	34.5%	+5.8 p.p.
<i>FF3</i>	23.0%	40.2%	+17.2 p.p.
<i>FF4</i>	29.8%	43.3%	+13.5 p.p.
<i>EDcm2</i>	18.3%	23.3%	+5 p.p.
<i>EDcm3</i>	14.0%	18.2%	+4.2 p.p.
<i>EDcm4</i>	11.2%	16.0%	+4.8 p.p.
<i>TUc1</i>	21.3%	24.4%	+3.1 p.p.
<i>TUc2</i>	16.5%	19.4%	+2.9 p.p.
<i>FVc1</i>	11.7%	21.4%	+9.7 p.p.
<i>FVc2</i>	12.7%	19.2%	+6.5 p.p.
<i>TYc1</i>	11.7%	25.5%	+13.8 p.p.
<i>TYc2</i>	11.4%	23.9%	+12.5 p.p.
<i>USc1</i>	15.7%	28.5%	+12.8 p.p.
<i>USc2</i>	18.7%	32.1%	+13.4 p.p.
<i>SPX</i>	18.4%	23.2%	+4.8 p.p.
<i>ESc1</i>	22.9%	27.7%	+4.8 p.p.
<i>ESc2</i>	19.3%	23.5%	+4.2 p.p.

Table E4: Differences of $\overline{R_{OOS}^2}$ between 30-minute and Optimal Event Windows

Series Name	Notation	Description
Gürkaynak, Sack, and Swanson (2005) Target Shocks	GSS_T	First principal component of monetary policy surprises that is rotated such that it drives unexpected changes in the surprise in the current federal funds rate. The authors re-scale the shock originally to be one-for-one with the surprise in the current federal funds rate.
Gürkaynak, Sack, and Swanson (2005) Path Shocks	GSS_P	Second principal component of monetary policy surprises that is rotated such that on average, it has no effect on the surprise in the current federal funds rate. The authors re-scale the shock originally to have equal effect with GSS_T on the surprise in the four-quarter Eurodollar futures.
Nakamura and Steinsson (2018) Shocks	NS_{MP}	First principal component of monetary policy surprises. The authors re-scale the shock originally to be one-for-one with the daily change in the zero-coupon, nominal one-year Treasury yield.
Jarociński and Karadi (2020) Shocks	JK_{MP}	First principal component of monetary policy surprises that have negative comovement with stock market changes. The authors make the first principal component to have equal standard deviation with that of the surprise in the four-quarter Eurodollar futures.
Jarociński and Karadi (2020) Central Bank Information	JK_{CBI}	First principal component of monetary policy surprises that have positive comovement with stock market changes. The authors make the first principal component to have equal standard deviation with that of the surprise in the four-quarter Eurodollar futures.

Table E5: Monetary Policy Shock Series

Notes: All monetary policy shocks are re-scaled to be one-for-one with the daily change in the zero-coupon, nominal one-year Treasury yield for easy interpretation and comparison. Before applying sign restrictions to obtain JK_{MP} and JK_{CBI} , I follow Jarociński and Karadi (2020) by first re-scaling the first principal component of monetary policy surprises in terms of the standard deviation of the zero-coupon, nominal one-year Treasury yield.

Metric	GSS_T	GSS_P	NS_{MP}	JK_{MP}	JK_{CBI}
Count	165 (165)	165 (165)	165 (165)	165 (165)	165 (165)
Mean	0 (0)	0 (0)	0 (0)	-0.0023 (-0.0035)	-0.0028 (-0.0004)
SD	0.0219 (0.0216)	0.0282 (0.0309)	0.0356 (0.0375)	0.0305 (0.0297)	0.0180 (0.0233)
Max	0.0756 (0.0652)	0.1223 (0.1079)	0.1333 (0.1135)	0.0968 (0.0840)	0.0388 (0.0984)
75 th	0.0076 (0.0084)	0.0134 (0.0167)	0.0193 (0.0203)	0.0175 (0.0124)	0.0069 (0.0112)
Median	0.0011 (0.0023)	-0.0010 (0.0019)	0.0033 (0.0003)	-0.0017 (-0.0009)	-0.0003 (0.0022)
25 th	-0.0069 (-0.0077)	-0.0124 (-0.0127)	-0.0187 (-0.0164)	-0.0153 (-0.0138)	-0.0086 (-0.0103)
Min	-0.1038 (-0.1025)	-0.1611 (-0.1618)	-0.1403 (-0.1722)	-0.1222 (-0.1491)	-0.0812 (-0.0828)

Table E6: Descriptive Statistics for Monetary Policy Shock Series, FOMC Statement Frequency for 1999–2019

Notes: Numbers in parentheses are summary statistics for each shock series derived from monetary policy surprises calculated within the median optimal event window length of 50 minutes. All shocks series have been re-scaled to be one-for-one with the daily change in the zero-coupon, nominal one-year Treasury yield.

Metric	$\ln(CPI)$	$\ln(IP)$	EBP	TY_2
Count	246	246	246	246
Mean	5.356	4.597	0.110	2.210
SD	0.126	0.057	0.715	1.794
Max	5.551	4.706	3.283	6.650
75 th	5.466	4.642	0.341	3.569
Median	5.381	4.605	-0.108	1.673
25 th	5.241	4.548	-0.334	0.662
Min	5.112	4.467	-1.140	0.188

Table E7: Descriptive Statistics for Impulse Response Variables, Monthly for 1999–2019
Notes: All logs are natural logarithms. Consumer Price Index (CPI) and Industrial Production (IP) are sourced from the Federal Reserve Economic Data. The Excess Bond Premium (EBP) is from Gilchrist and Zakrajšek (2012) and is expressed in percentage points. The two-year Treasury yield (TY_2) is from Gürkaynak, Sack, and Swanson (2005).

Metric	GSS_T	GSS_P	NS_{MP}	JK_{MP}	JK_{CBI}
Count	165 (165)	165 (165)	165 (165)	165 (165)	165 (165)
Mean	0 (0)	0 (0)	0 (0)	-0.0015 (-0.0024)	-0.0019 (-0.0002)
SD	0.0180 (0.0177)	0.0230 (0.0253)	0.0291 (0.0307)	0.0250 (0.0243)	0.0148 (0.0190)
Max	0.0756 (0.0652)	0.1223 (0.1079)	0.1333 (0.1135)	0.0968 (0.0840)	0.0388 (0.0984)
75 th	0.0042 (0.0042)	0.0078 (0.0089)	0.0114 (0.0110)	0.0055 (0.0058)	0.0031 (0.0075)
Median	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
25 th	-0.0021 (-0.0011)	-0.0064 (-0.0052)	-0.0036 (-0.0039)	-0.0073 (-0.0070)	-0.0037 (-0.0035)
Min	-0.1038 (-0.1025)	-0.1611 (-0.1618)	-0.1403 (-0.1722)	-0.1222 (-0.1491)	-0.0812 (-0.0828)

Table E8: Descriptive Statistics for Monetary Policy Shock Series, Monthly for 1999–2019
Notes: Numbers in parentheses are summary statistics for each shock series derived from monetary policy surprises calculated within the median optimal event window length of 50 minutes. All shocks series have been re-scaled to be one-for-one with the daily change in the zero-coupon, nominal one-year Treasury yield. All shocks are zero for any month that does not have an FOMC meeting.

Metric	FKGL	S^1
Count	165	164
Mean	16.361	0.991
SD	1.715	0.017
Max	21.3	0.999
75 th	17.3	0.999
Median	16.5	0.998
25 th	15.1	0.987
Min	12.2	0.856

Table E9: Descriptive Statistics for Heterogeneity Analyses
Notes: The complexity of FOMC statements is measured by the Flesch-Kincaid Grade Level (FKGL), defined as: $0.39 \times \text{average sentence length} + 11.8 \times \text{average number of syllables per word} - 15.59$. S^1 is the cosine similarity measure between sequential FOMC statements.

	30-minute Window					Optimal Window					Difference		
	ΔTY_1	ΔTY_2	ΔTY_5	ΔTY_{10}		ΔTY_1	ΔTY_2	ΔTY_5	ΔTY_{10}		ΔTY_2	ΔTY_5	ΔTY_{10}
GSS_T	0.36*** (0.11)	0.28** (0.13)	0.15 (0.12)	0.05 (0.11)		0.35*** (0.11)	0.27** (0.13)	0.13 (0.11)	0.03 (0.10)		-0.01	-0.01	-0.02
GSS_P	0.21*** (0.02)	0.25*** (0.03)	0.24*** (0.04)	0.15*** (0.04)		0.21*** (0.03)	0.28*** (0.04)	0.27*** (0.05)	0.19*** (0.06)		+0.01	+0.02	+0.03
NS_{MP}	1.00*** (0.09)	1.11*** (0.14)	0.95*** (0.18)	0.57*** (0.19)		1.00*** (0.11)	1.16*** (0.15)	1.04*** (0.20)	0.67*** (0.22)		+0.05	+0.09	+0.10
JK_{MP}	0.53*** (0.08)	0.64*** (0.11)	0.61*** (0.16)	0.39** (0.18)		0.44*** (0.08)	0.55*** (0.11)	0.56*** (0.18)	0.39* (0.21)		-0.09	-0.09	-0.06
JK_{CBI}	0.51*** (0.16)	0.39** (0.19)	0.15 (0.23)	0.03 (0.24)		0.52*** (0.15)	0.50*** (0.19)	0.29 (0.25)	0.12 (0.24)		+0.01	+0.11	+0.15

Table E10: Differences in Responses of Nominal Interest Rates to Shocks from Event Window Choice

Notes: Each estimate comes from a separate OLS regression. For each regression, the dependent variable is the one-day change in end-of-day Treasury yields, represented by columns. The independent variable is the change in monetary policy shock series over a given event window around the time of scheduled FOMC announcements. Starting from the left, columns 2–5 represent the OLS regressions performed on shock series constructed within 30-minute windows. Columns 6–9 are equivalent, but for shocks constructed within the median optimal window length of 50 minutes. The sample period consists of all scheduled FOMC statement release dates from May 1999 through October 2019, resulting in 165 observations. Newey and West (1987) standard errors are reported in parentheses. A constant is included in all specifications but not reported. Columns 10–12 represent the differences between the coefficients of the regressions using the optimal windows v. those using 30-minute windows. ** sig. at the 5% level. *** sig. at the 1% level.

	30-minute Window			Optimal Window			Difference	
	$\Delta TIPS_2$	$\Delta TIPS_5$	$\Delta TIPS_{10}$	$\Delta TIPS_2$	$\Delta TIPS_5$	$\Delta TIPS_{10}$	$\Delta TIPS_2$	$\Delta TIPS_5$
GSS_T	-0.43 (0.75)	0.21 (0.21)	0.15 (0.15)	-0.40 (0.71)	0.21 (0.20)	0.14 (0.14)	+0.03	-0.00
GSS_P	0.25** (0.12)	0.21*** (0.06)	0.16*** (0.05)	0.28** (0.13)	0.26*** (0.08)	0.21*** (0.07)	+0.02	+0.06
NS_{MP}	0.56 (0.93)	0.89*** (0.30)	0.69*** (0.23)	0.68 (0.93)	1.07*** (0.32)	0.83*** (0.27)	+0.12	+0.18
JK_{MP}	0.53 (0.52)	0.60** (0.24)	0.48** (0.20)	0.54 (0.38)	0.60** (0.25)	0.49** (0.22)	+0.01	-0.00
JK_{CBI}	-0.39 (0.56)	0.07 (0.25)	0.03 (0.23)	-0.23 (0.71)	0.25 (0.29)	0.11 (0.25)	+0.16	+0.17

Table E11: Differences in Responses of Real Interest Rates to Shocks from Event Window Choice

Notes: Each estimate comes from a separate OLS regression. For each regression, the dependent variable is the one-day change in end-of-day TIPS yields, represented by columns. The independent variable is the change in monetary policy shock series over a given event window around the time of scheduled FOMC announcements. Starting from the left, columns 2–4 represent the OLS regressions performed on shock series constructed within 30-minute windows. Columns 5–7 are equivalent, but for shocks constructed within the median optimal window length of 50 minutes. The sample period consists of all scheduled FOMC statement release dates from May 1999 to October 2019, resulting in 165 observations. Newey and West (1987) standard errors are reported in parentheses. A constant is included in all specifications but not reported. Columns 8–10 represent the differences between the coefficients of the regressions using the optimal windows v. those using 30-minute windows. ** sig. at the 5% level. *** sig. at the 1% level.

	30-minute Window			Optimal Window			Difference		
	ΔBEI_2	ΔBEI_5	ΔBEI_{10}	ΔBEI_2	ΔBEI_5	ΔBEI_{10}	ΔBEI_2	ΔBEI_5	ΔBEI_{10}
GSS_T	0.71 (0.75)	-0.06 (0.13)	-0.10 (0.09)	0.67 (0.71)	-0.07 (0.13)	-0.11 (0.09)	-0.04	-0.02	-0.01
GSS_P	0.00 (0.11)	0.03 (0.05)	-0.01 (0.03)	-0.00 (0.12)	0.01 (0.05)	-0.02 (0.03)	-0.00	-0.02	-0.01
NS_{MP}	0.55 (0.90)	0.06 (0.24)	-0.13 (0.13)	0.48 (0.92)	-0.03 (0.24)	-0.15 (0.12)	-0.07	-0.09	-0.03
JK_{MP}	0.11 (0.50)	0.02 (0.16)	-0.09 (0.07)	0.01 (0.36)	-0.04 (0.12)	-0.10* (0.05)	-0.10	-0.05	-0.01
JK_{CBI}	0.78 (0.55)	0.07 (0.17)	-0.00 (0.11)	0.73 (0.67)	0.04 (0.18)	0.01 (0.13)	-0.05	-0.03	+0.01

Table E12: Differences in Responses of Break-even Inflation to Shocks from Event Window Choice

Notes: Each estimate comes from a separate OLS regression. For each regression, the dependent variable is the one-day change in end-of-day break-even inflation, represented by columns. The independent variable is the change in monetary policy shock series over a given event window around the time of scheduled FOMC announcements. Starting from the left, columns 2–4 represent the OLS regressions performed on shock series constructed within 30-minute windows. Columns 5–7 are equivalent, but for shocks constructed within the median optimal window length of 50 minutes. The sample period consists of all scheduled FOMC statement release dates from May 1999 to October 2019, resulting in 165 observations. Newey and West (1987) standard errors are reported in parentheses. A constant is included in all specifications but not reported. Columns 8–10 represent the differences between the coefficients of the regressions using the optimal windows v. those using 30-minute windows. * sig. at the 10% level. ** sig. at the 5% level. *** sig. at the 1% level.

	30-minute Window			Optimal Window			Percentage Difference		
	DP_{SPX}	DP_{ESc1}	DP_{ESc2}	DP_{SPX}	DP_{ESc1}	DP_{ESc2}	DP_{SPX}	DP_{ESc1}	DP_{ESc2}
GSS_T	-3.46* (1.36)	-3.54** (1.39)	-2.68* (1.40)	-3.24*** (1.23)	-3.25*** (1.25)	-2.67** (1.20)	-6.52%	-8.15%	-0.57%
GSS_P	-1.39*** (0.39)	-1.47*** (0.40)	-1.39*** (0.36)	-1.43*** (0.51)	-1.45*** (0.53)	-1.69*** (0.48)	+2.52%	-1.12%	+21.24%
NS_{MP}	-7.57*** (1.52)	-7.88*** (1.58)	-6.96*** (1.57)	-7.33*** (1.81)	-7.41*** (1.87)	-7.82*** (1.82)	-3.32%	-6.00%	+12.30%
JK_{MP}	-8.16*** (0.43)	-8.35*** (0.47)	-7.53*** (0.57)	-8.01*** (0.58)	-8.15*** (0.60)	-7.91*** (0.62)	-1.75%	-2.35%	+5.07%
JK_{CBI}	8.27*** (1.25)	8.20*** (1.29)	7.68*** (1.24)	8.04*** (1.44)	8.25*** (1.46)	7.00*** (1.47)	-2.84%	+0.66%	-8.97%

Table E13: Differences in Responses of Stock Prices to Shocks from Event Window Choice

Notes: Each estimate comes from a separate OLS regression. For each regression, the dependent variable is the price log-difference of the S&P 500 Index or E-mini futures, represented by columns. The independent variable is the change in monetary policy shock series over a given event window around the time of scheduled FOMC announcements. Starting from the left, columns 2–4 represent the OLS regressions performed on shock series constructed within 30-minute windows. Columns 5–7 are equivalent, but for shocks constructed within the median optimal window length of 50 minutes. The sample period consists of all scheduled FOMC statement release dates from May 1999 to October 2019, resulting in 165 observations. Newey and West (1987) standard errors are reported in parentheses. A constant is included in all specifications but not reported. Columns 8–10 represent the percentage differences between the coefficients of the regressions using the optimal windows v. those using 30-minute windows, where positive (negative) values represent a stronger (weaker) effect in the same direction. *** sig. at the 1% level.